



(12) **United States Patent**
Kawasumi et al.

(10) **Patent No.:** **US 12,399,446 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **IMAGE FORMING APPARATUS INCLUDING DUCT STRUCTURE THAT SUPPORTS LIGHT EMITTING ELEMENTS FOR EXPOSING A PHOTOCONDUCTOR**

(58) **Field of Classification Search**
CPC G03G 15/011; G03G 15/0178; G03G 15/04054; G03G 15/04063;
(Continued)

(71) Applicant: **CANON KABUSHIKI KAISHA**,
Tokyo (JP)

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(72) Inventors: **Ryoichi Kawasumi**, Ibaraki (JP);
Takehiro Ishidate, Tokyo (JP); **Hiroki Takano**, Saitama (JP)

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(73) Assignee: **CANON KABUSHIKI KAISHA**,
Tokyo (JP)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/756,140**

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(22) Filed: **Jun. 27, 2024**

Extended European Search Report issued in European Appl. No. 23159019.1, mailed Jul. 21, 2023.

(65) **Prior Publication Data**

US 2024/0345502 A1 Oct. 17, 2024

(Continued)

Related U.S. Application Data

(63) Continuation of application No. 18/175,736, filed on Feb. 28, 2023, now Pat. No. 12,066,766.

Primary Examiner — Robert B Beatty

(74) *Attorney, Agent, or Firm* — ROSSI, KIMMS & McDOWELL LLP

(30) **Foreign Application Priority Data**

Mar. 10, 2022 (JP) 2022-037622

(57) **ABSTRACT**

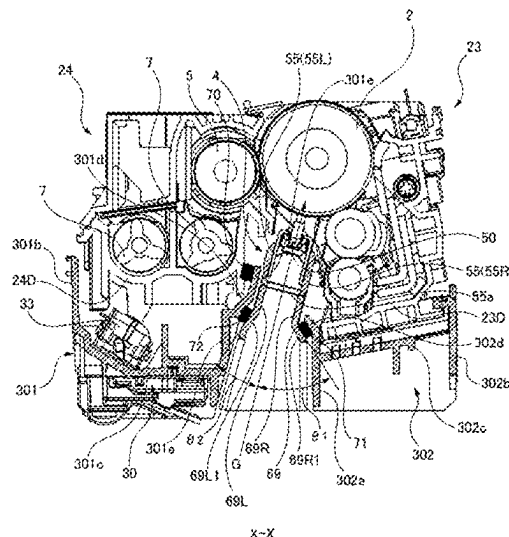
An image forming apparatus includes: a rotatable photoconductor; an exposure portion including a substrate on which a plurality of chips each including a plurality of light emitting portions that emit light for exposing the photoconductor is mounted, and a support portion configured to support the substrate; a fan configured to generate an airflow for cooling the exposure portion; and a duct configured to communicate with the fan and to communicate with the support portion to guide the airflow generated by the fan to the substrate.

(51) **Int. Cl.**
G03G 15/04 (2006.01)
B41J 2/00 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **G03G 15/04054** (2013.01); **B41J 2/4473** (2013.01); **B41J 29/377** (2013.01);
(Continued)

14 Claims, 69 Drawing Sheets



- (51) **Int. Cl.**
B41J 2/447 (2006.01)
B41J 29/377 (2006.01)
G03G 21/20 (2006.01)
- (52) **U.S. Cl.**
CPC **G03G 15/04063** (2013.01); **G03G 21/206**
(2013.01); **G03G 2215/0407** (2013.01); **G03G**
2215/0412 (2013.01); **G03G 2221/1645**
(2013.01)
- (58) **Field of Classification Search**
CPC G03G 21/1666; G03G 21/206; G03G
2215/0119; G03G 2215/0125; G03G
2215/0407; G03G 2215/0409; G03G
2215/0412; G03G 2215/0451; G03G
2215/0453; G03G 2215/0456; G03G
2221/1636; G03G 2221/1645; G06K
15/1247; G06K 15/1261; B41J 2/447;
B41J 2/4473; B41J 2/45; B41J 29/377
USPC 399/92, 118, 220; 347/117, 118, 129,
347/130, 152, 238, 242, 256, 257
See application file for complete search history.
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Apr. 24, 2024.
Extended European Search Report issued in European Appln. No.
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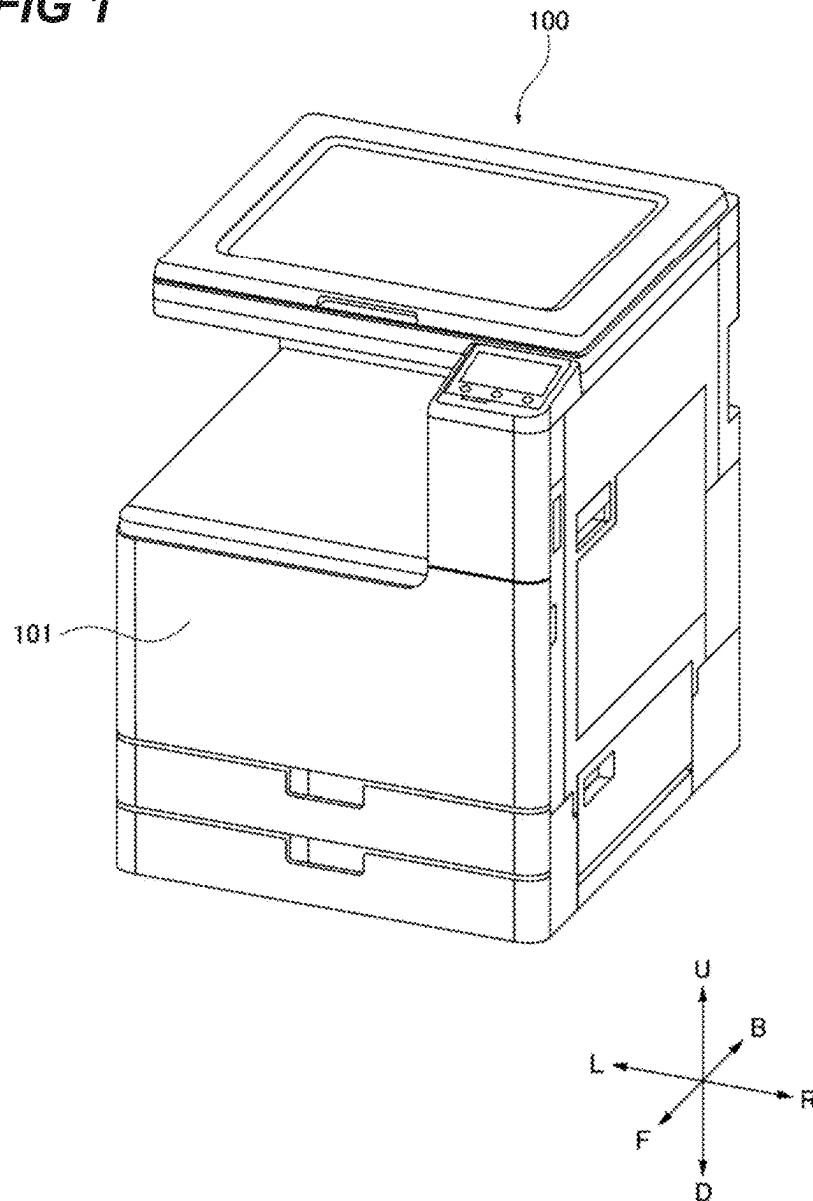
FIG 1

FIG 2

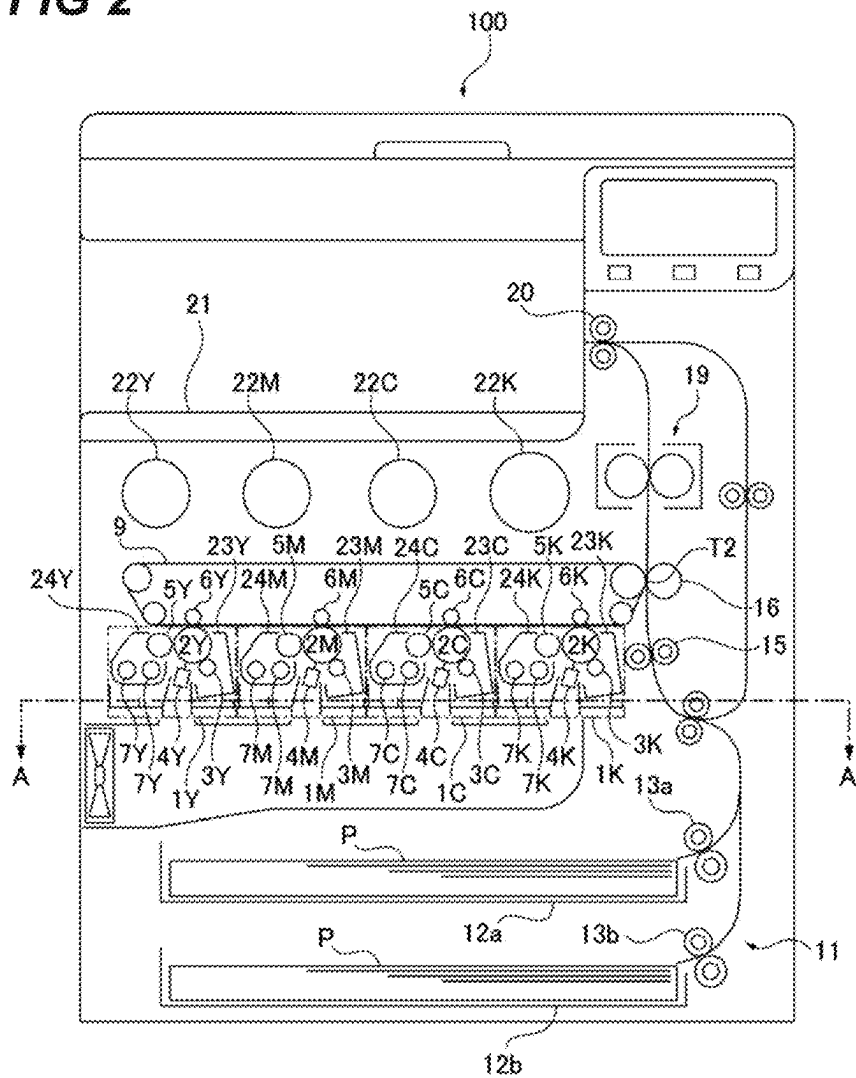


FIG 3

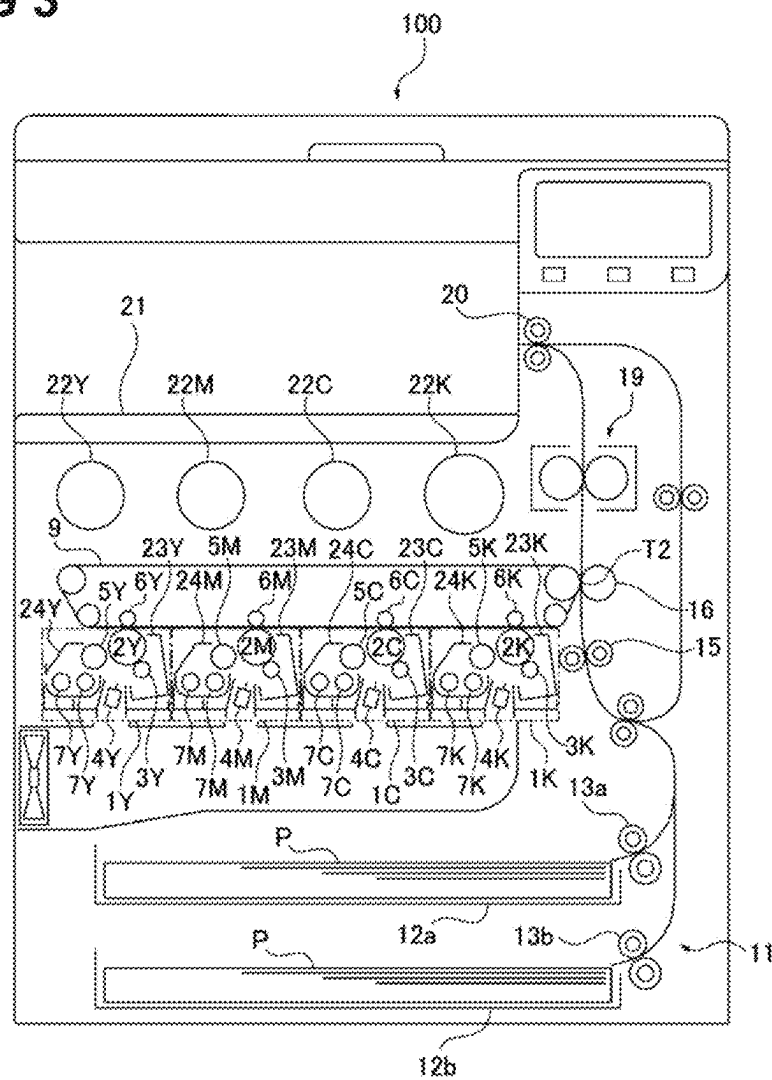


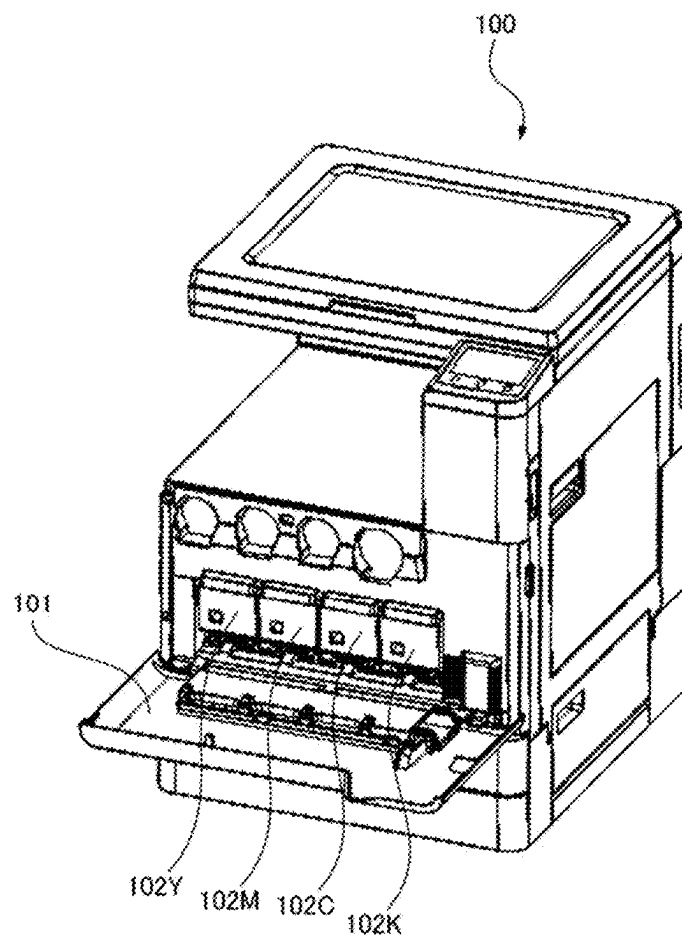
FIG 4

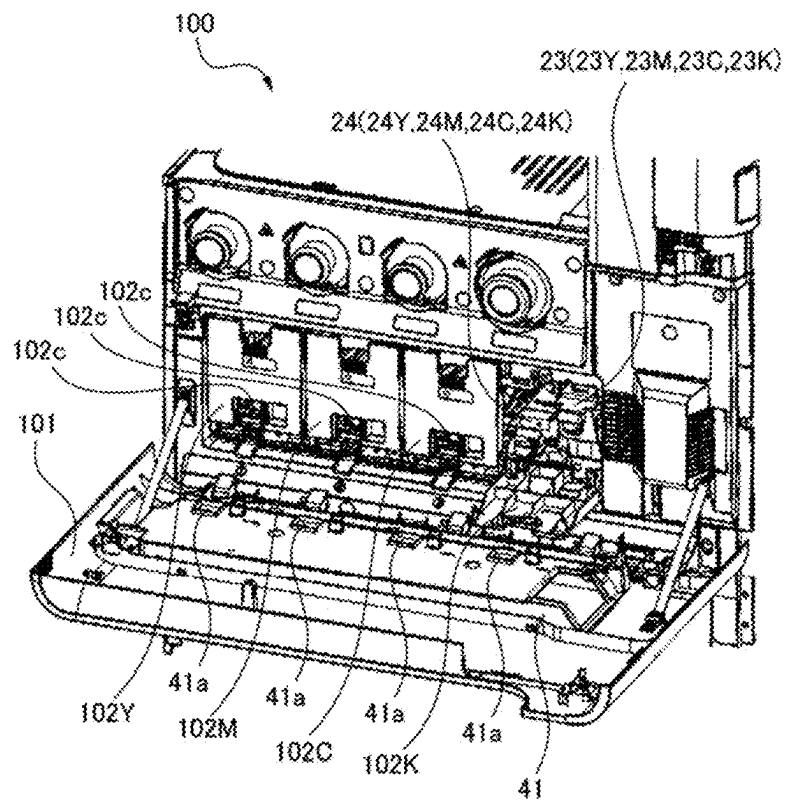
FIG 5

FIG 6

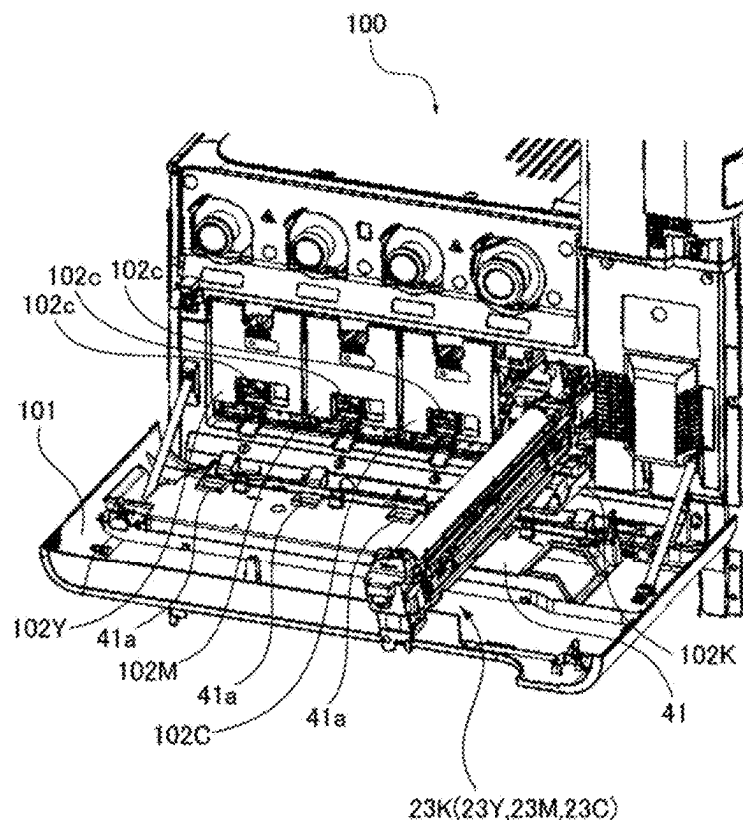


FIG 7

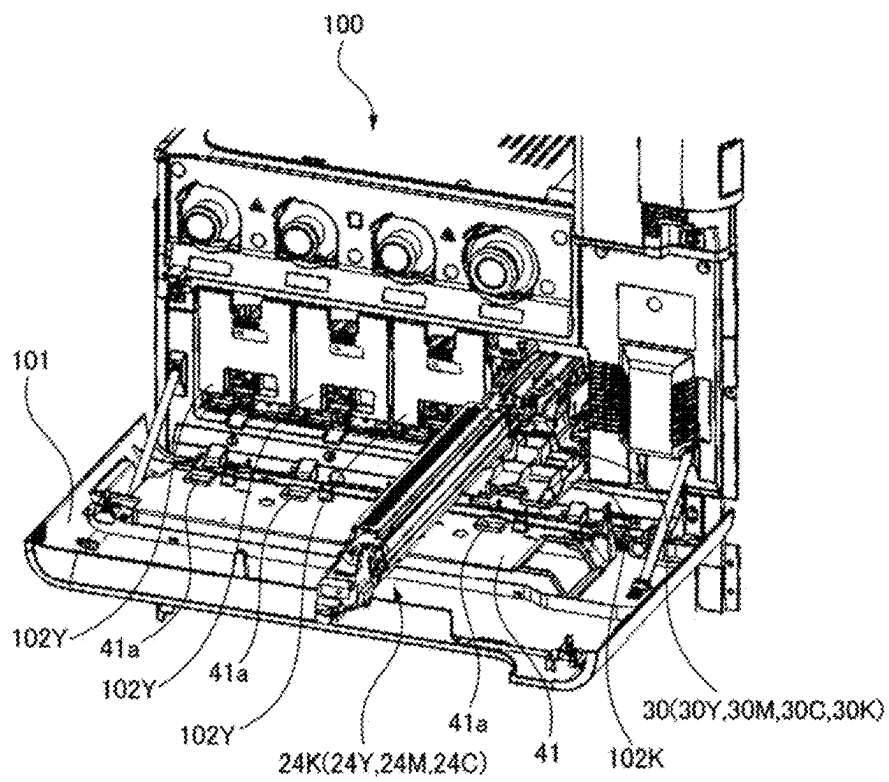


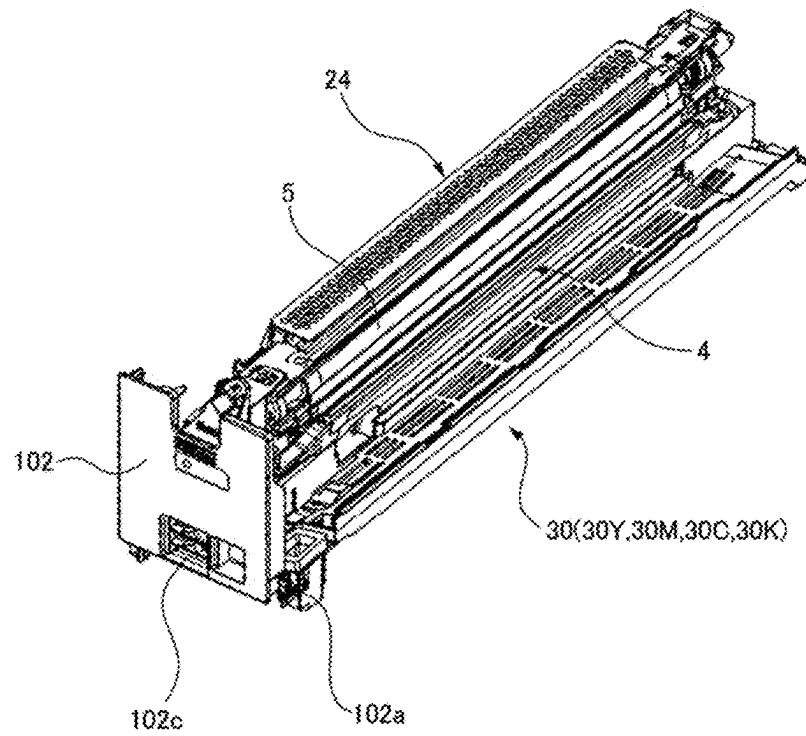
FIG 8

FIG 9

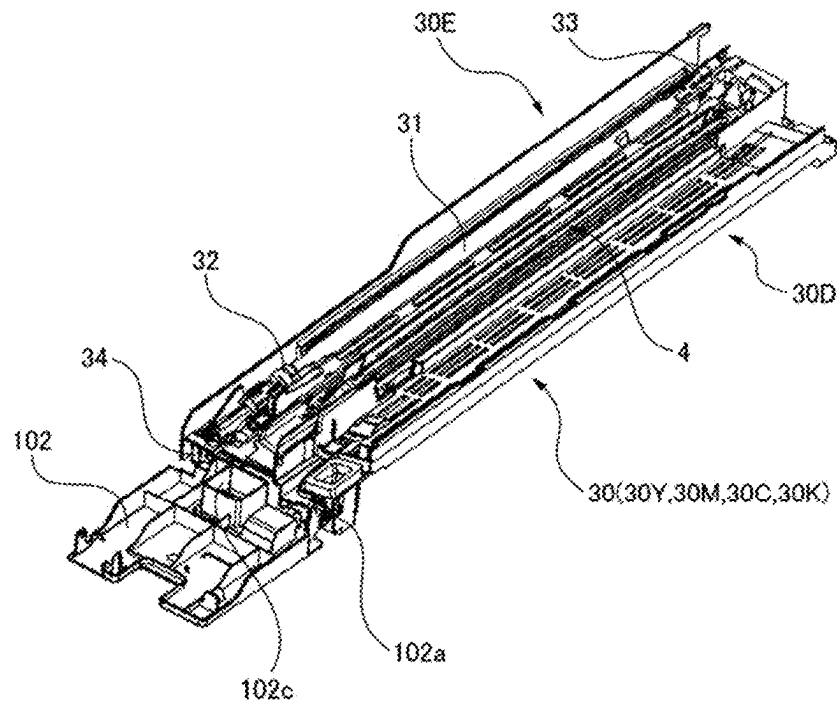


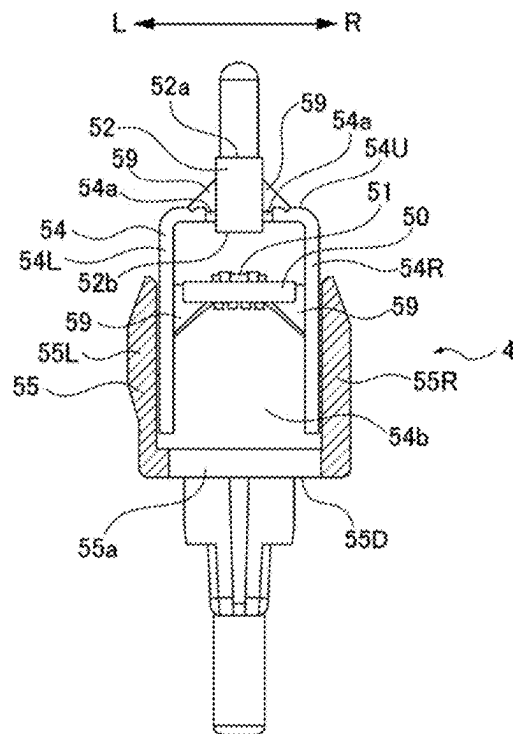
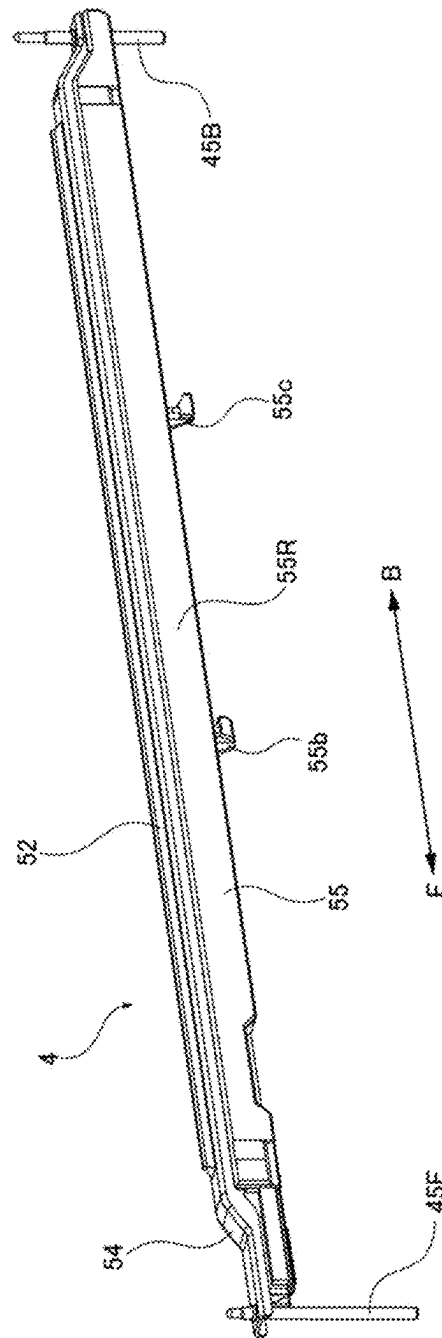
FIG 10

FIG 11



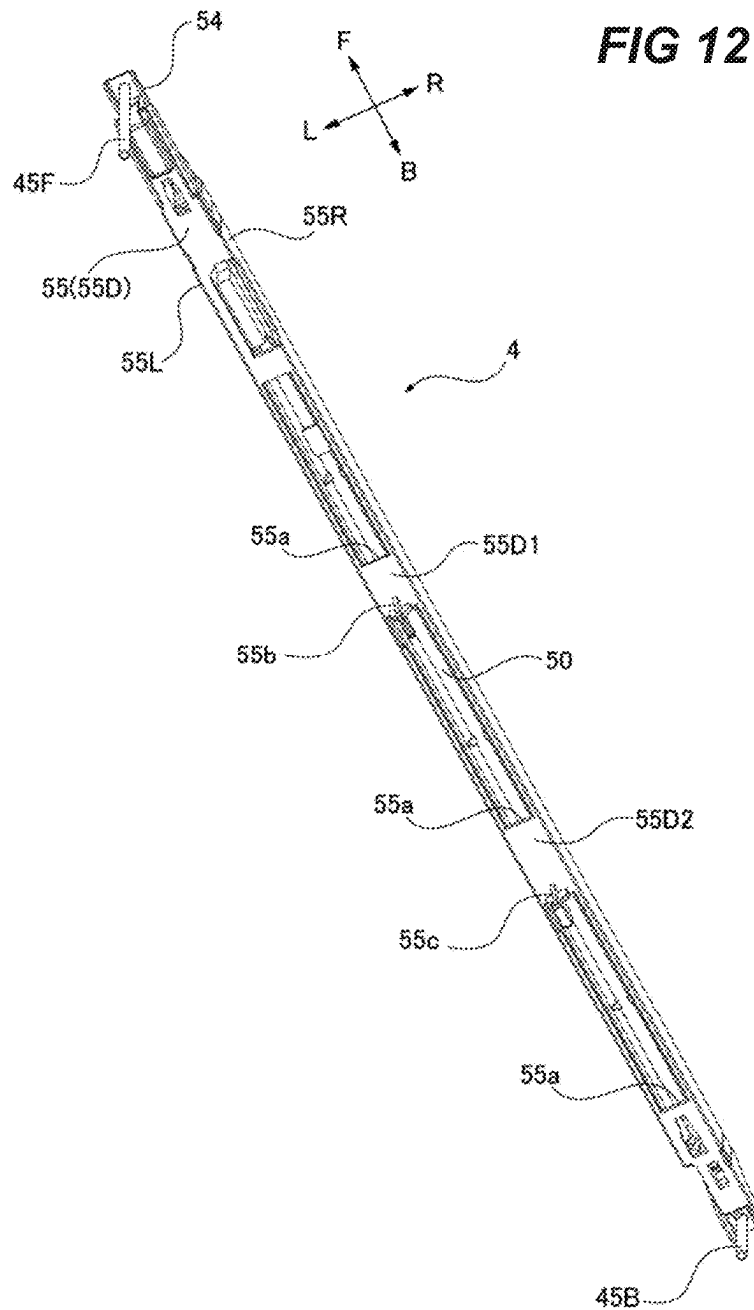


FIG 13A

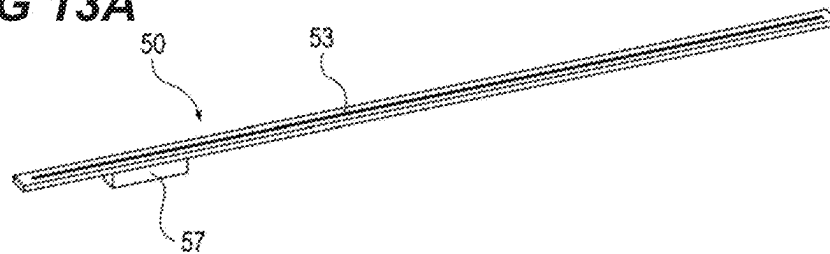


FIG 13B

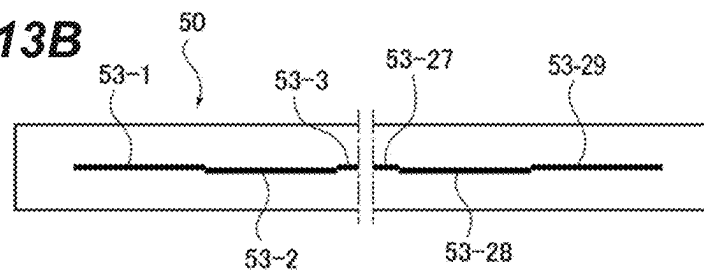


FIG 13C

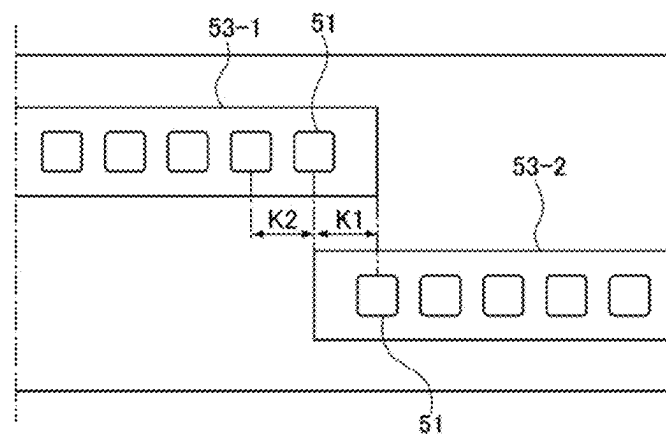


FIG 13D

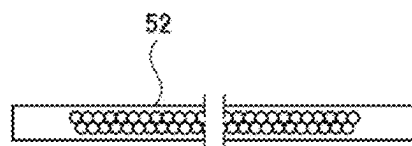
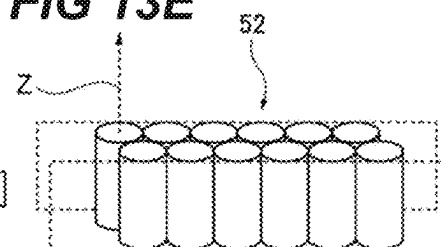


FIG 13E



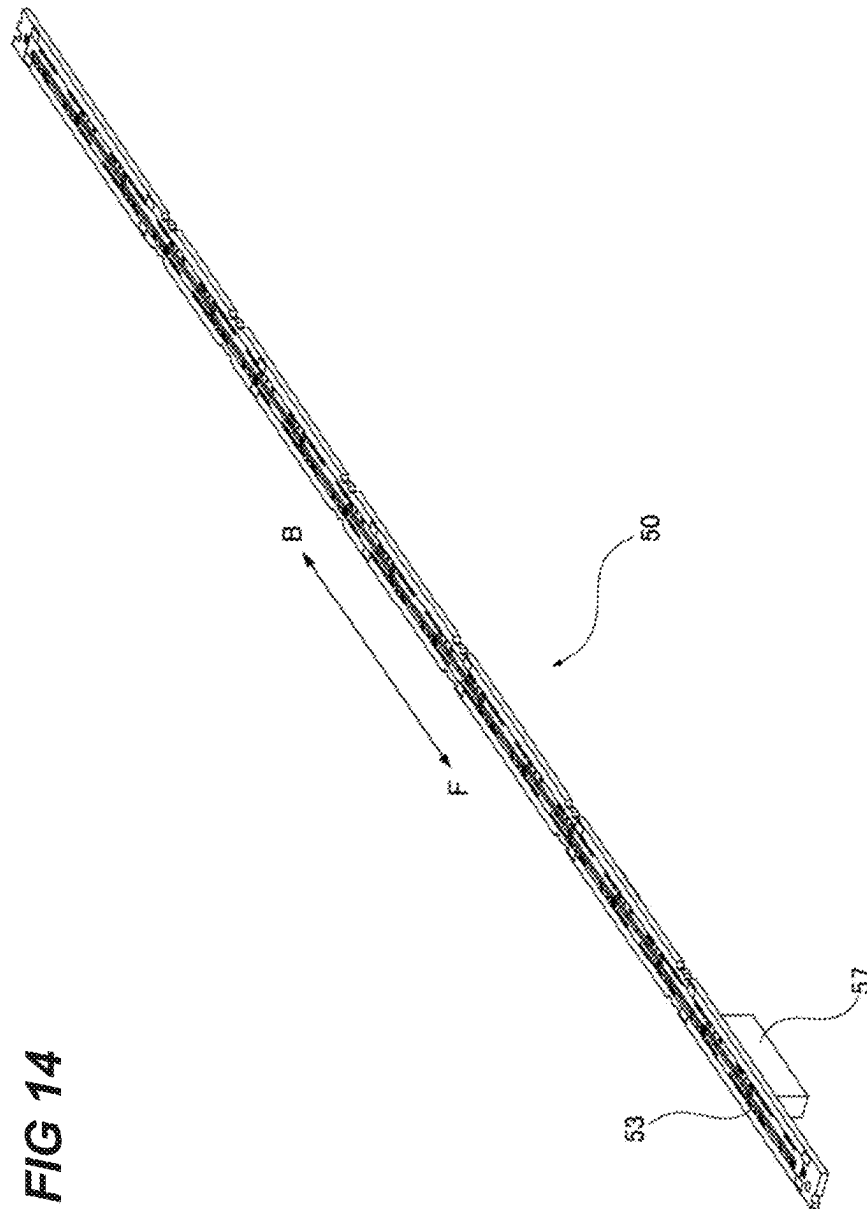


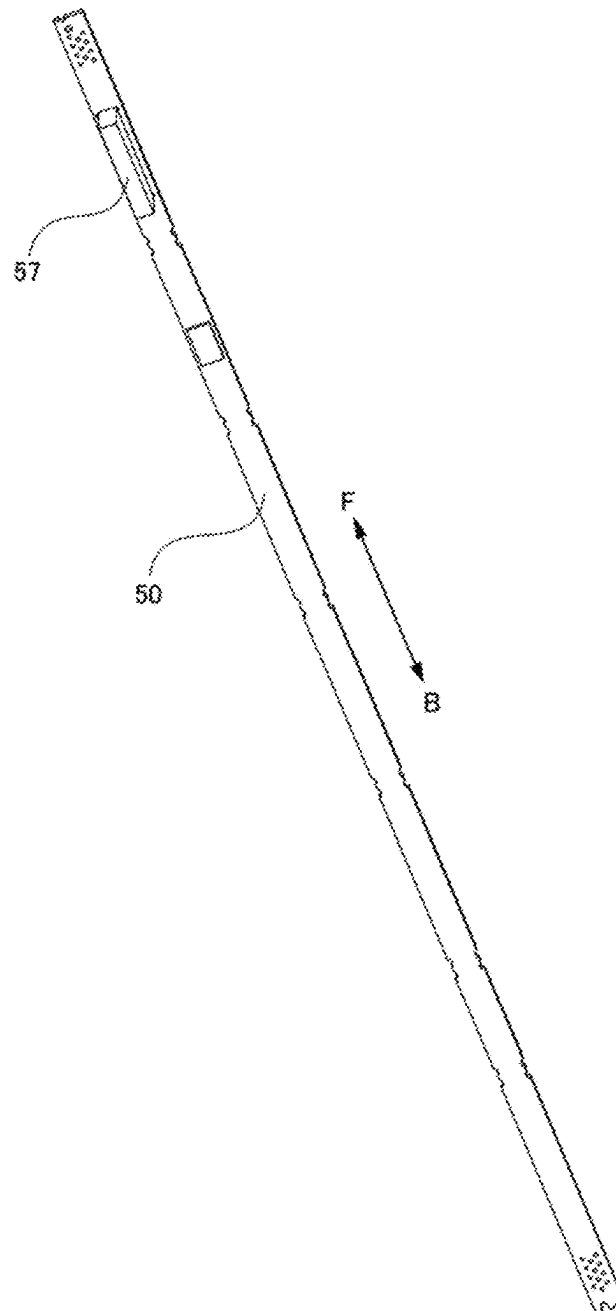
FIG 15

FIG 16

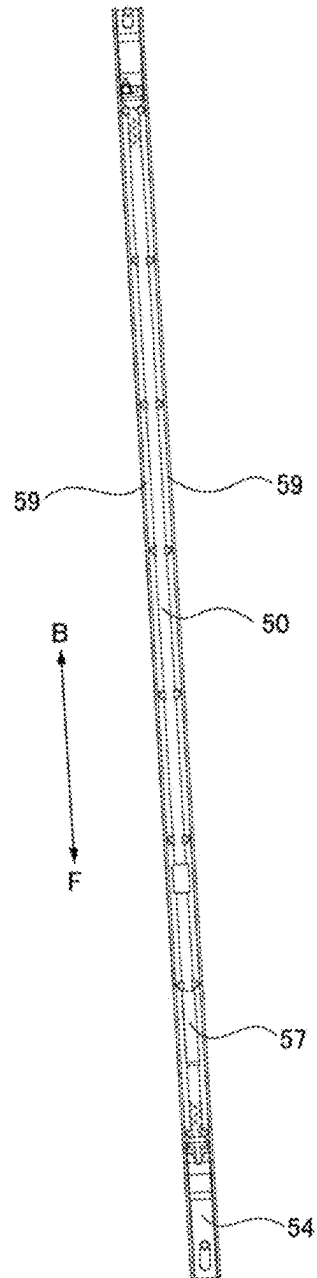


FIG 17

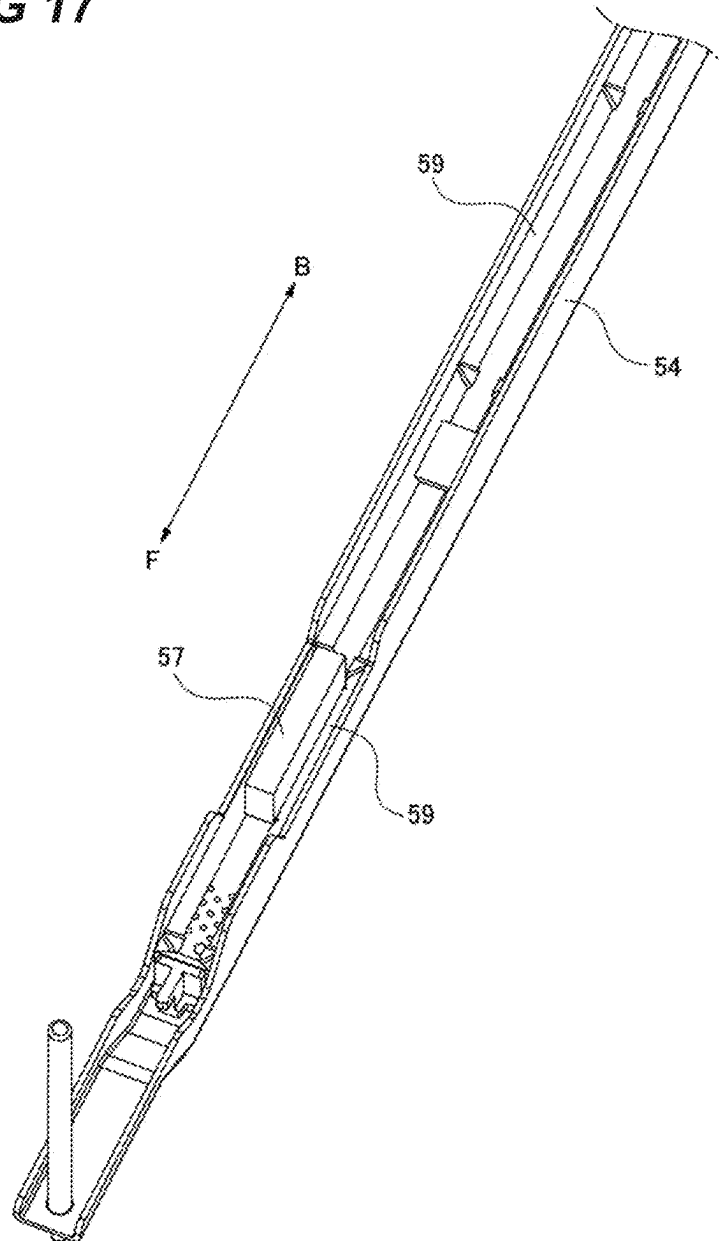


FIG 18

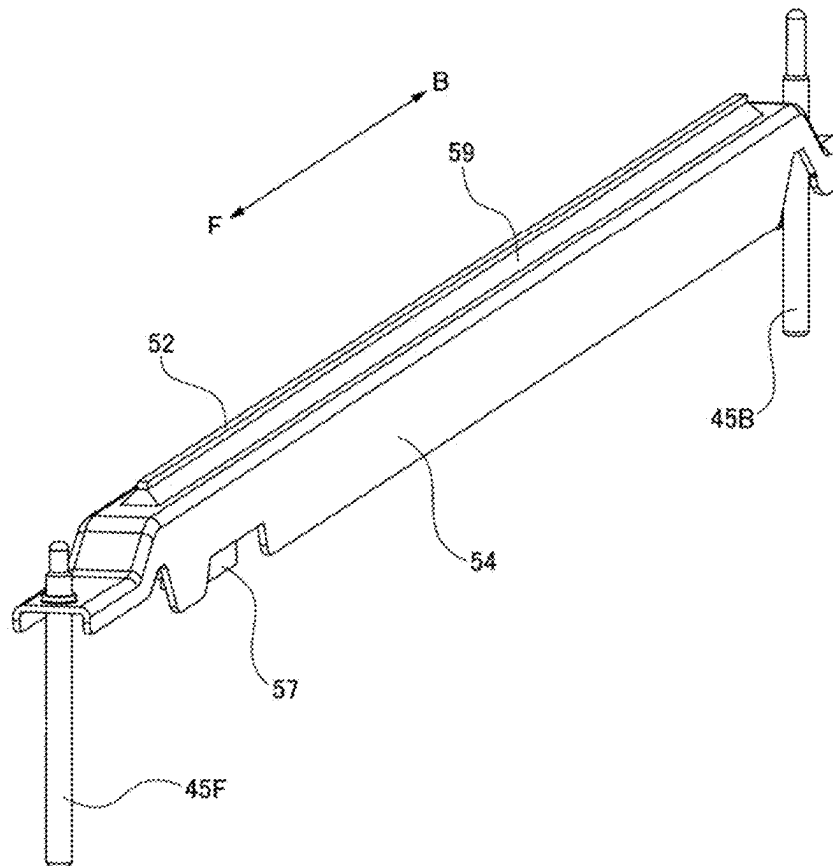


FIG 19

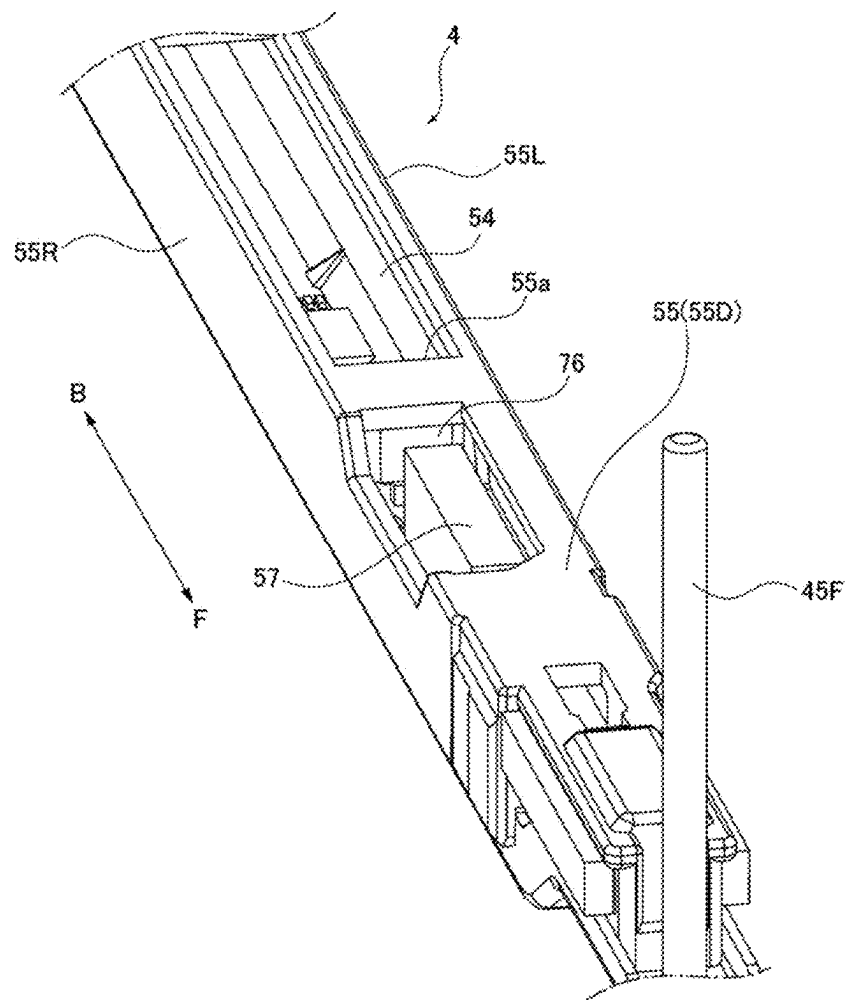


FIG 20

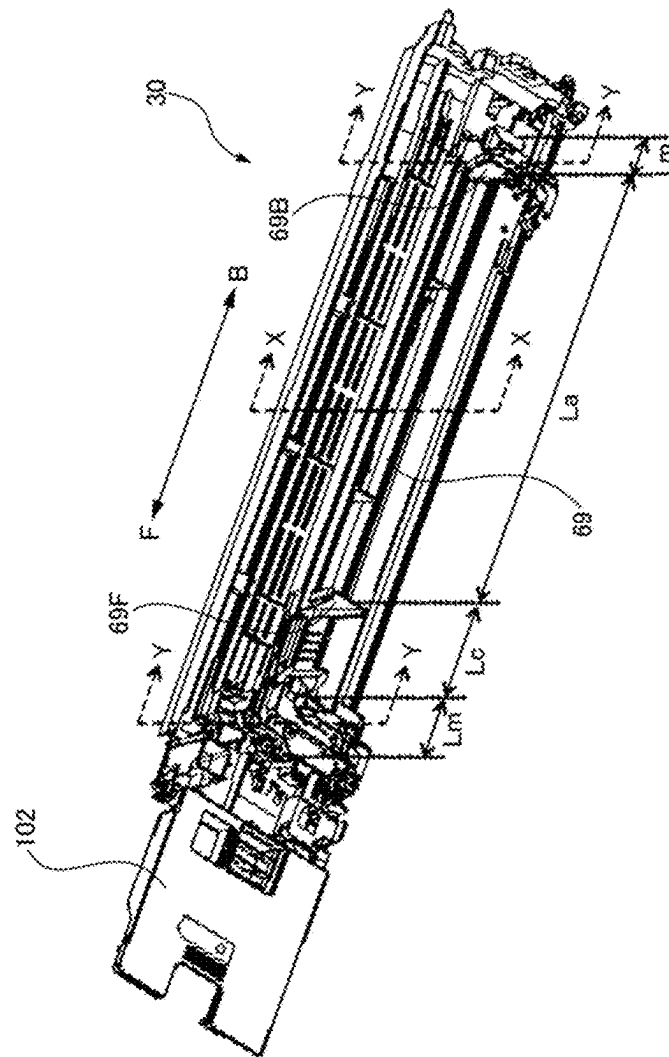


FIG 21

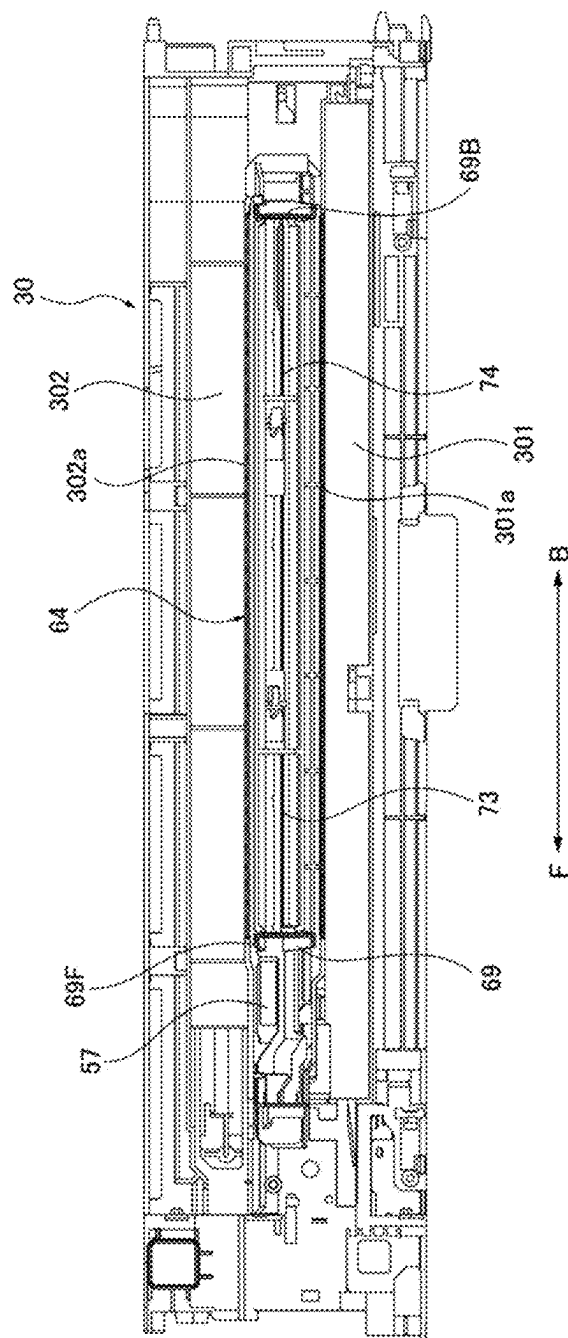


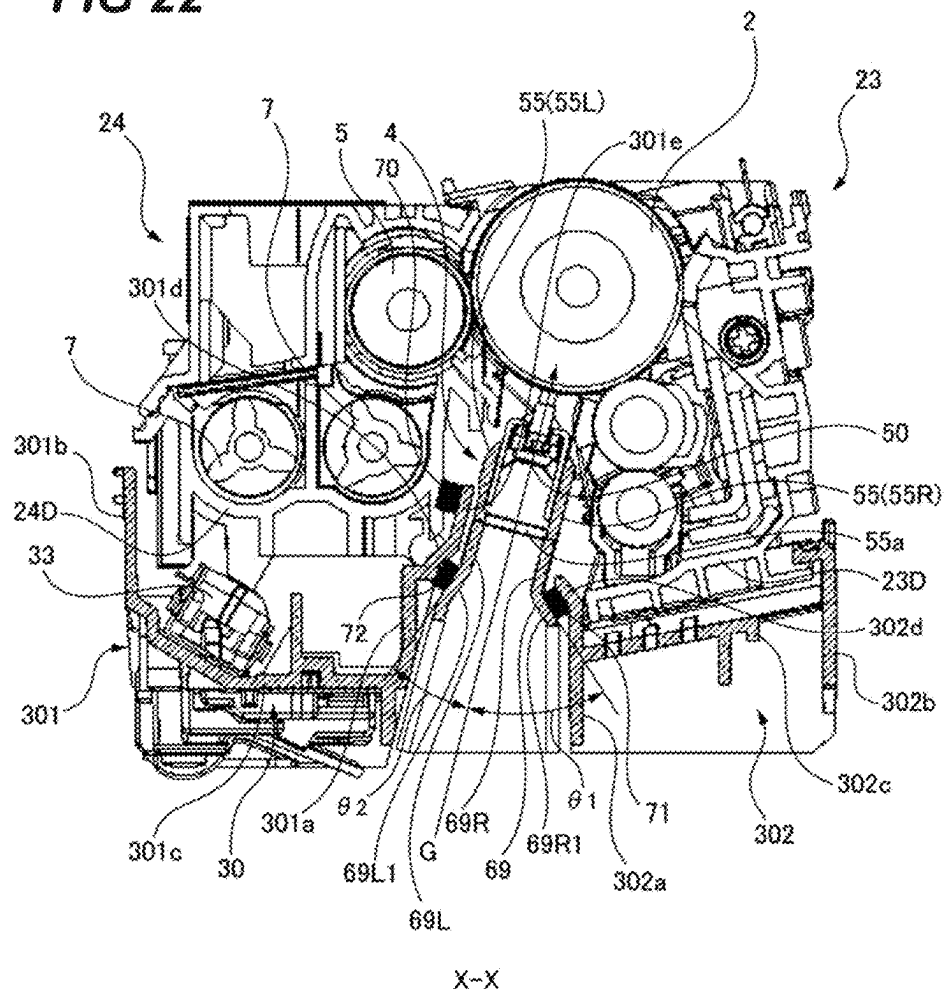
FIG 22

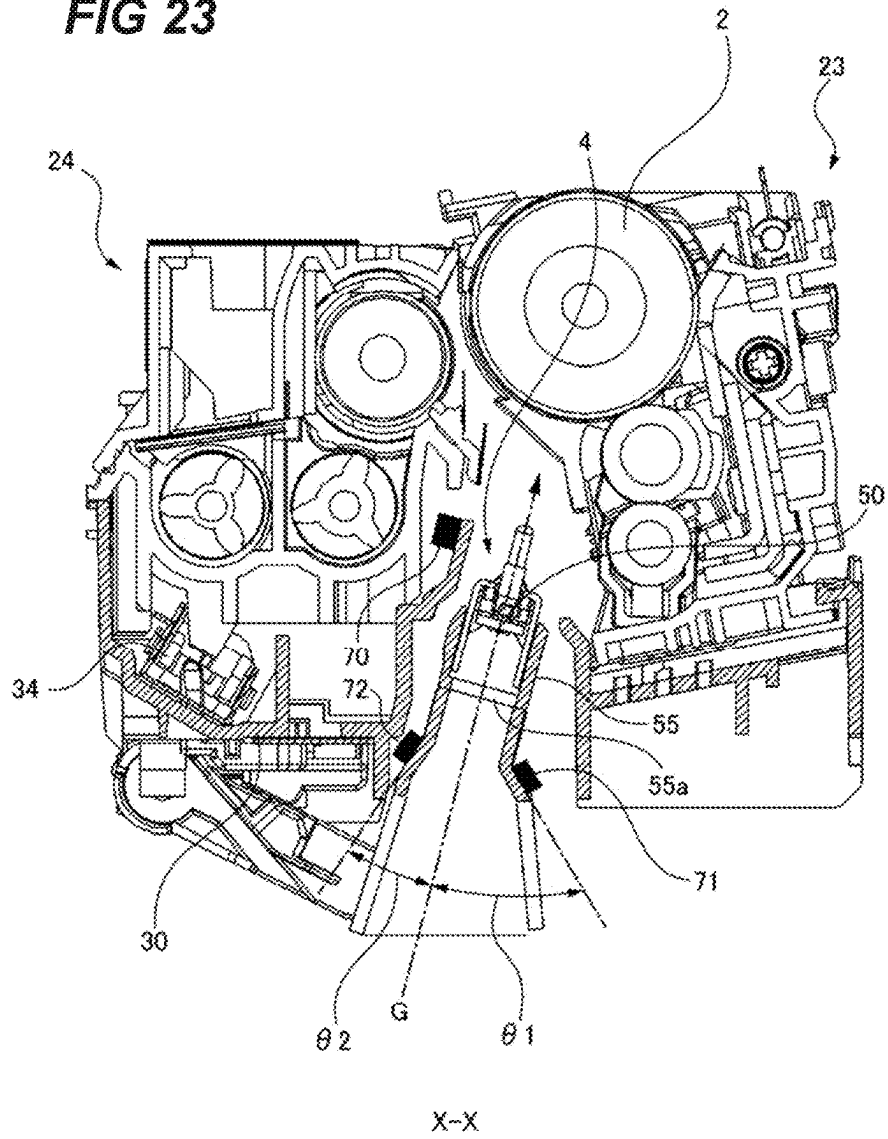
FIG 23

FIG 24

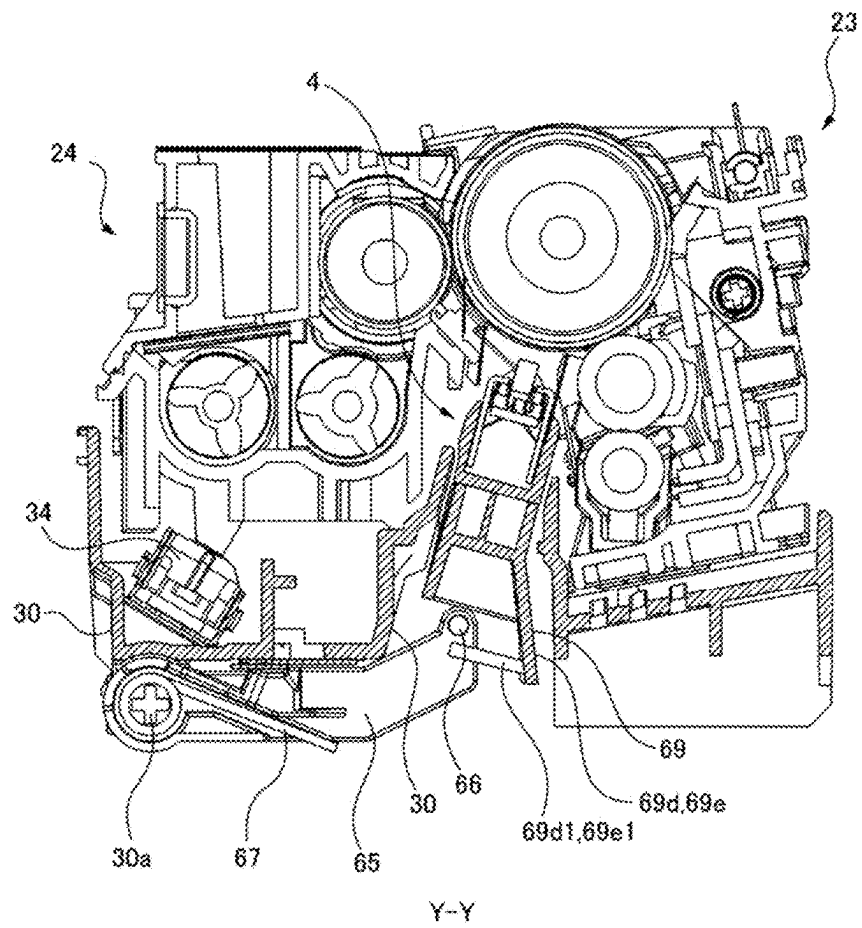


FIG 25

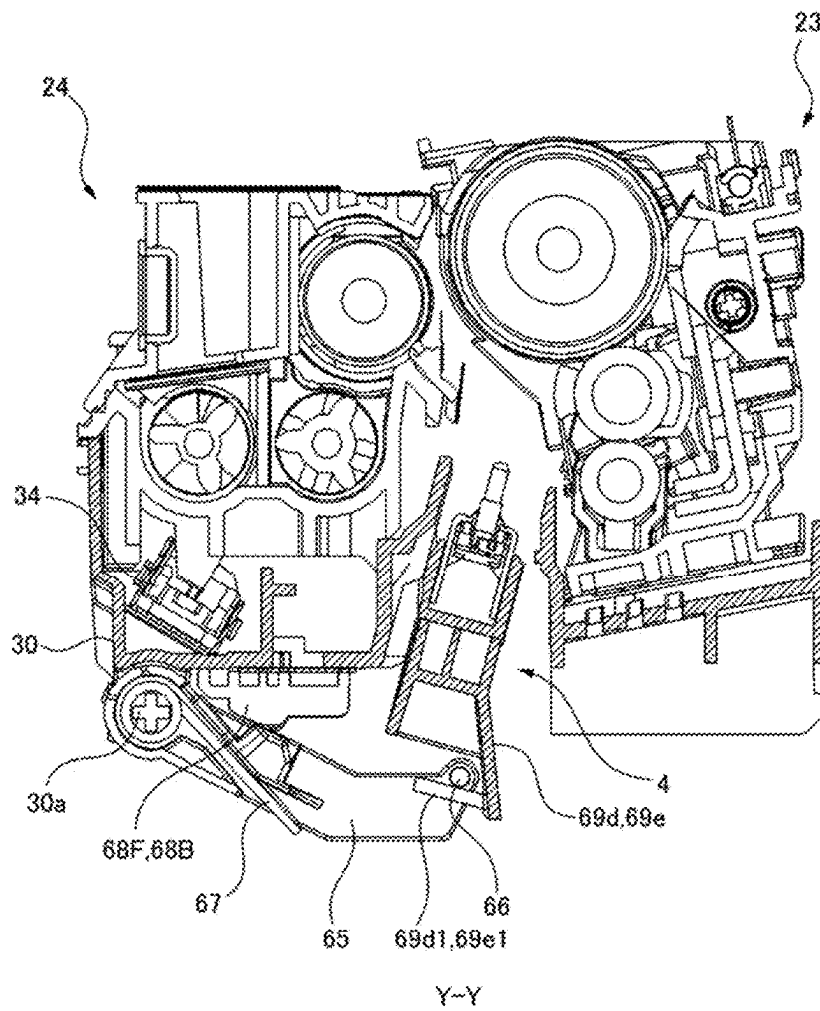


FIG 26

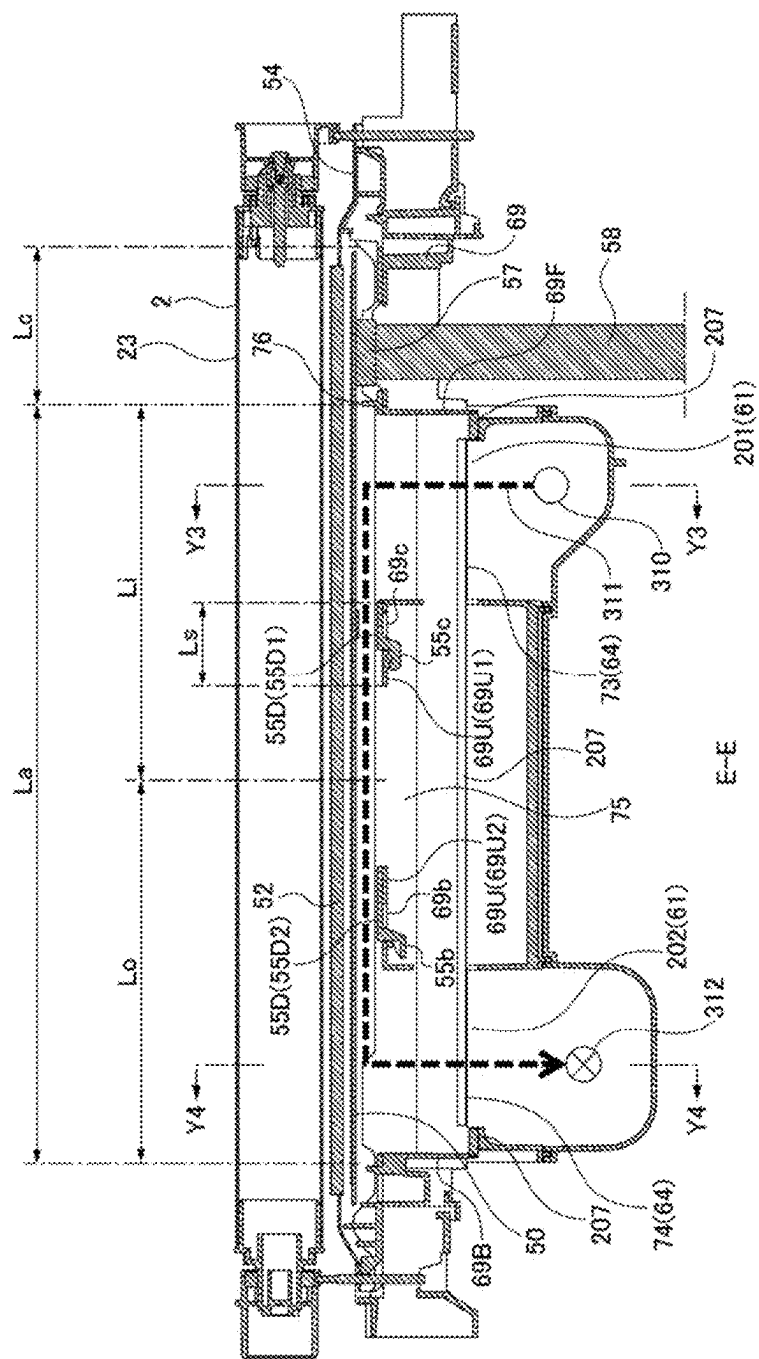


FIG 27

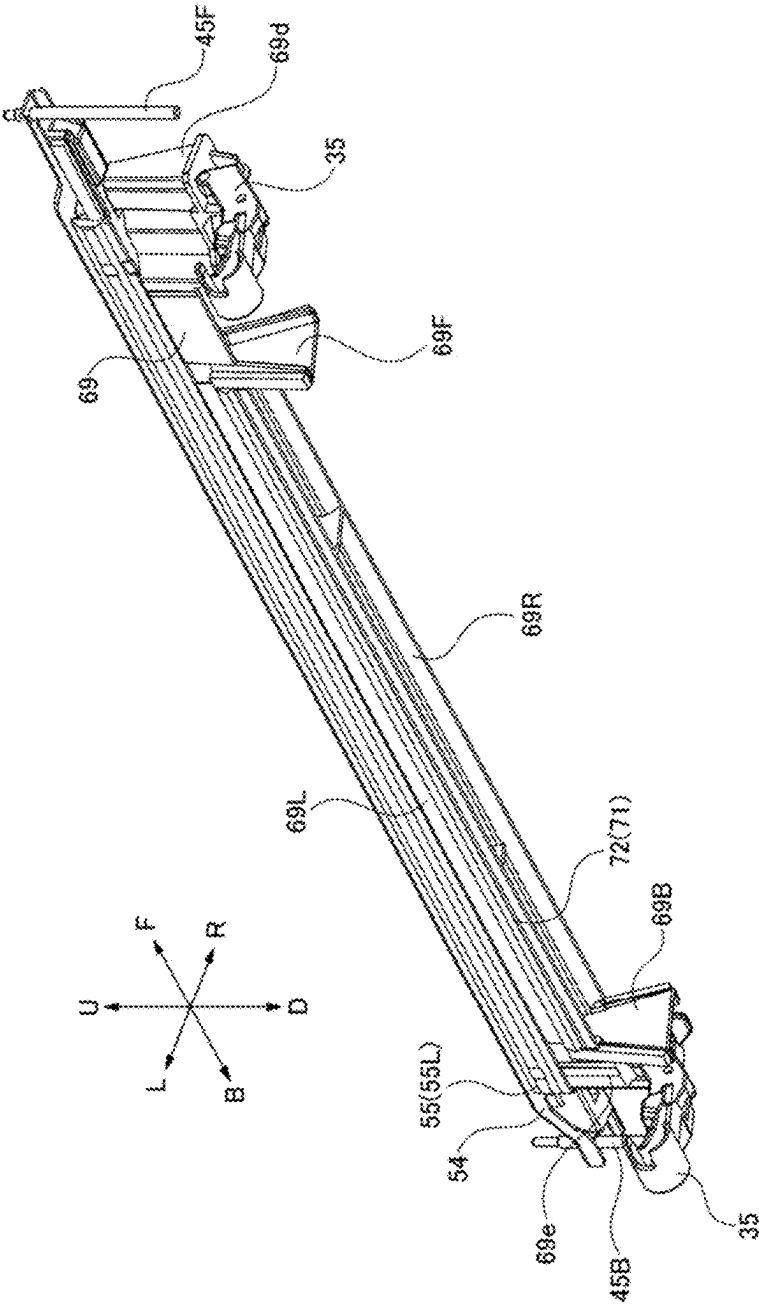


FIG 28

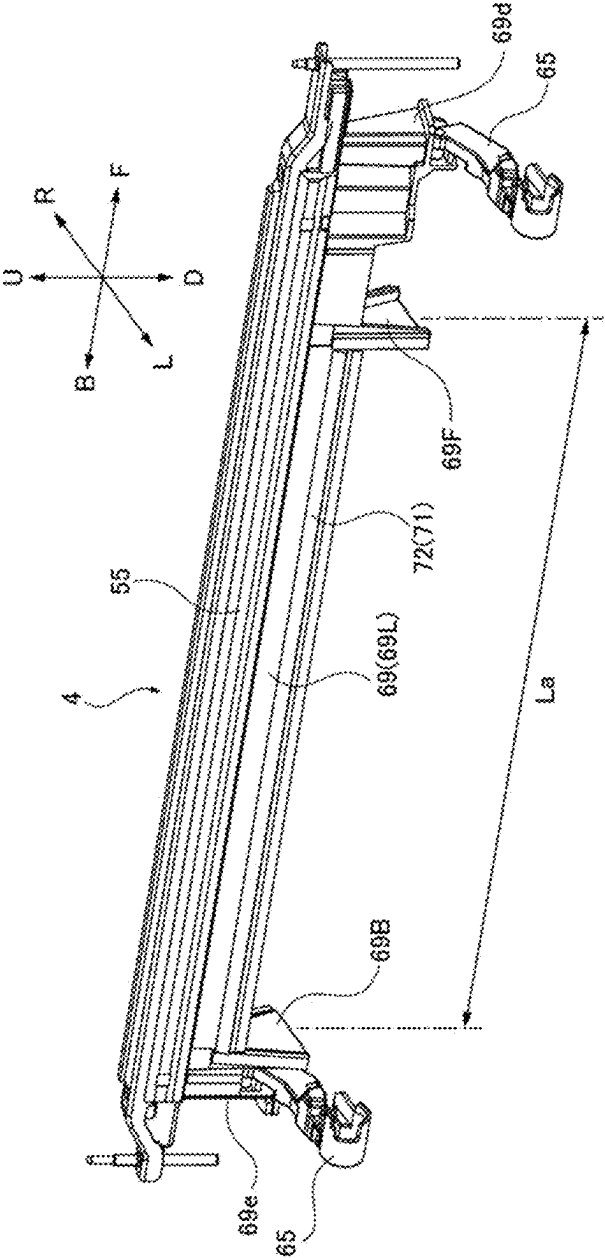


FIG 29

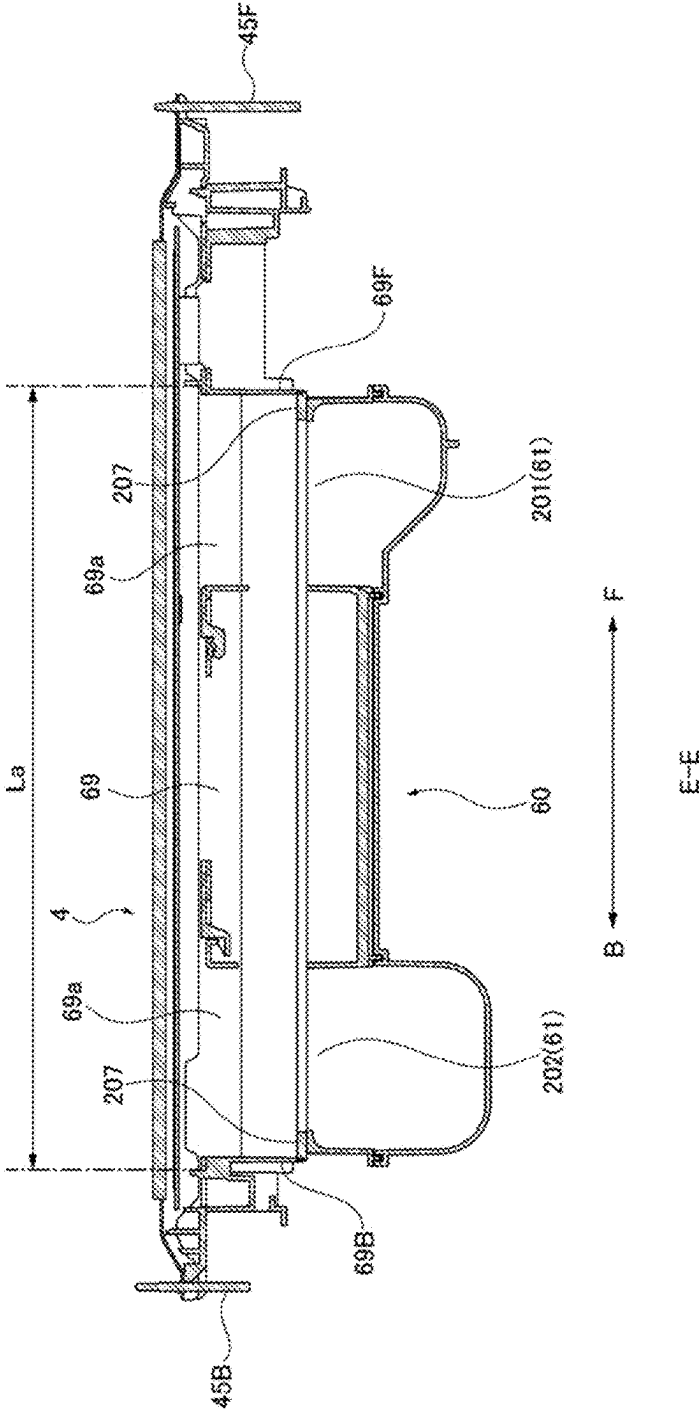


FIG 30

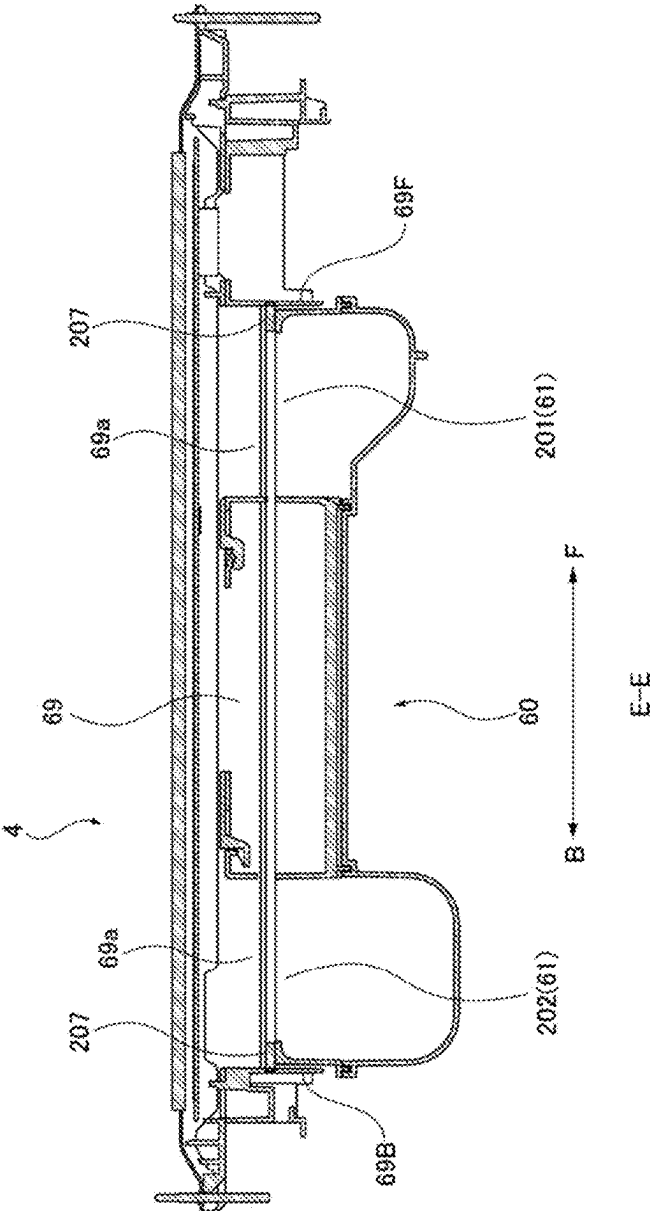


FIG 31

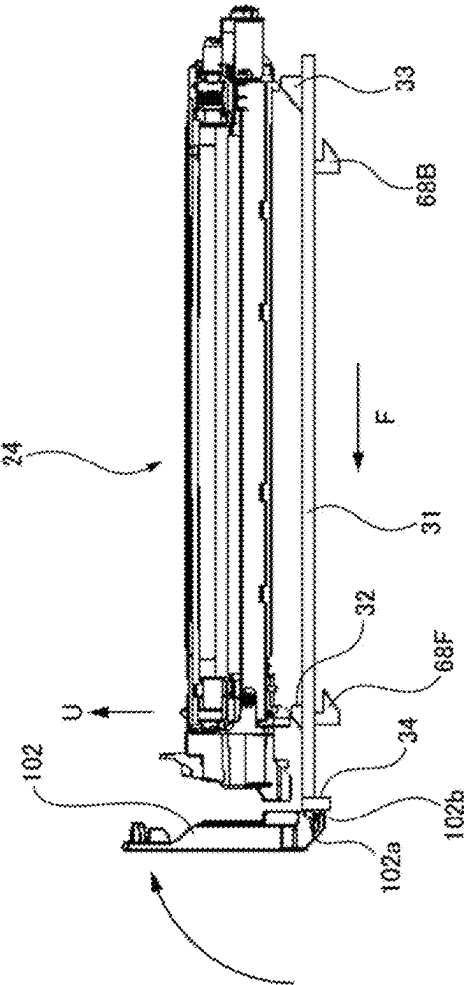
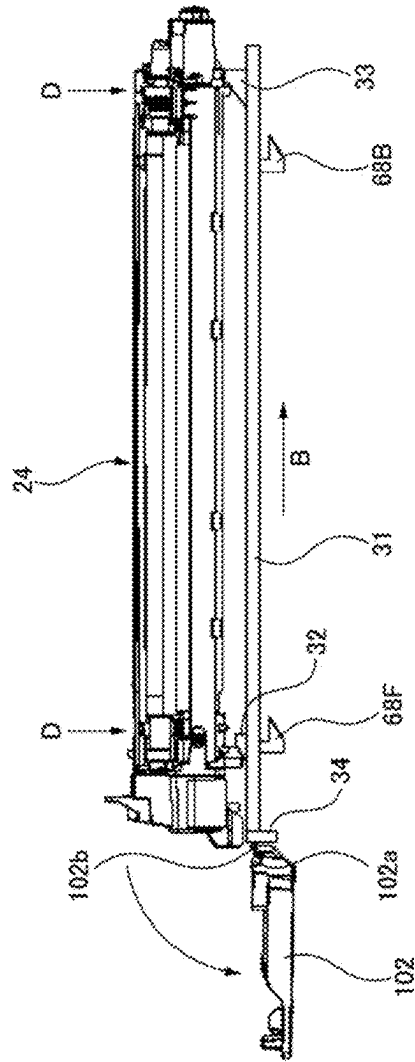


FIG 32



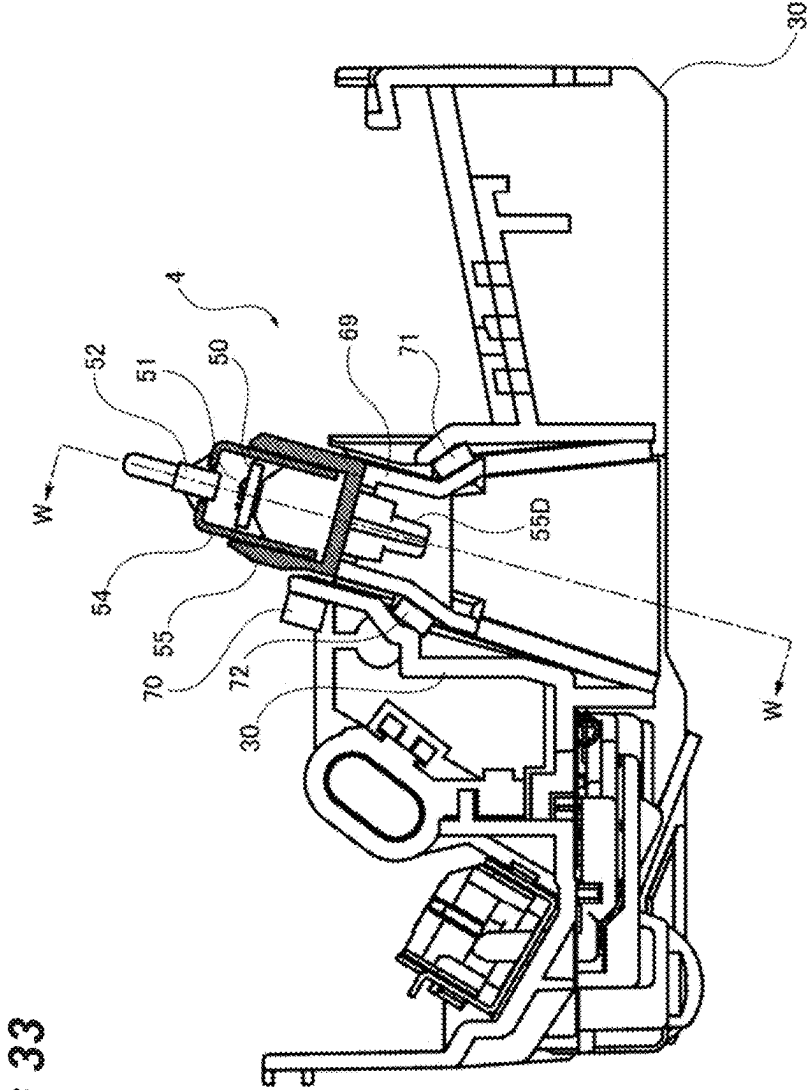
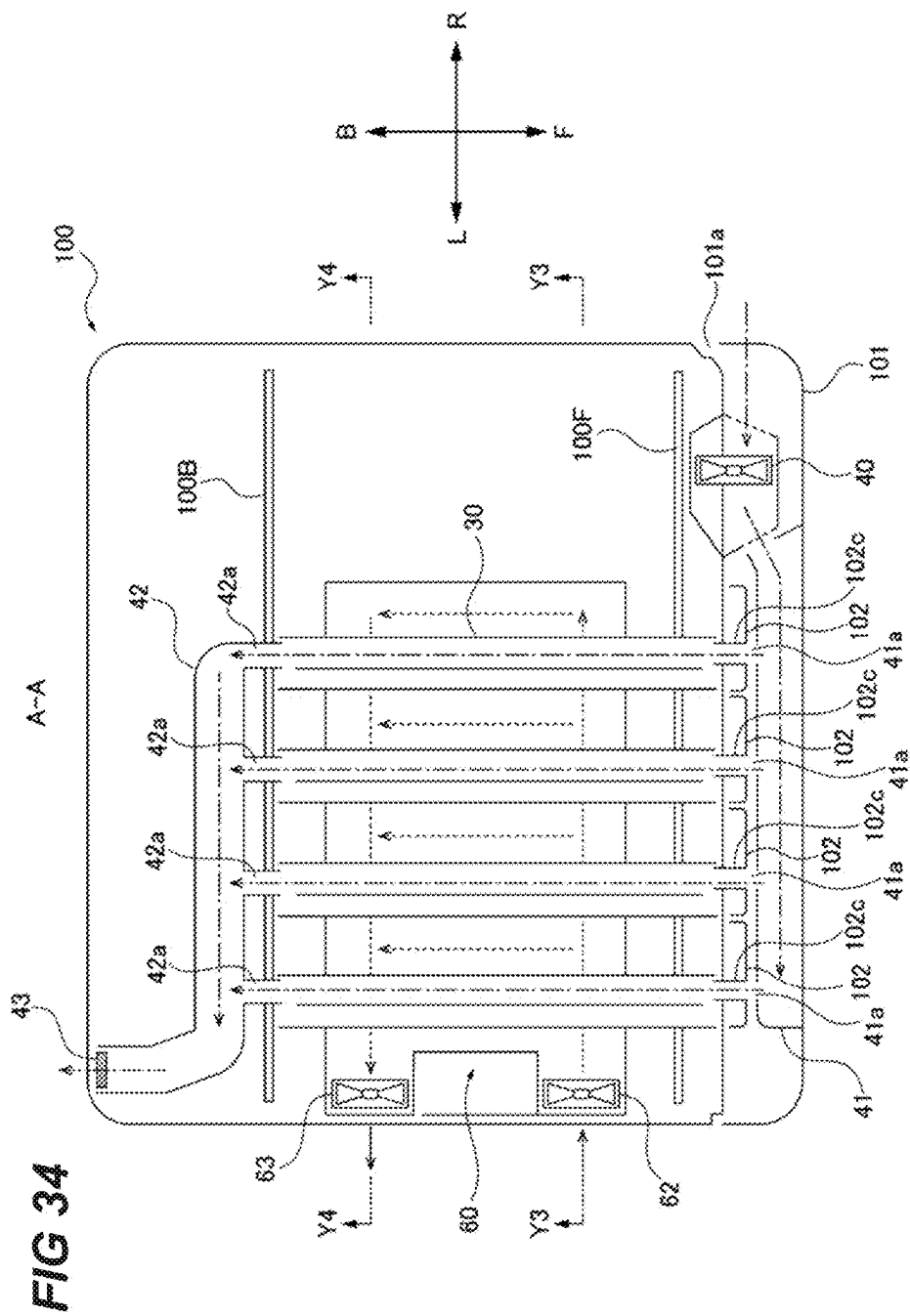
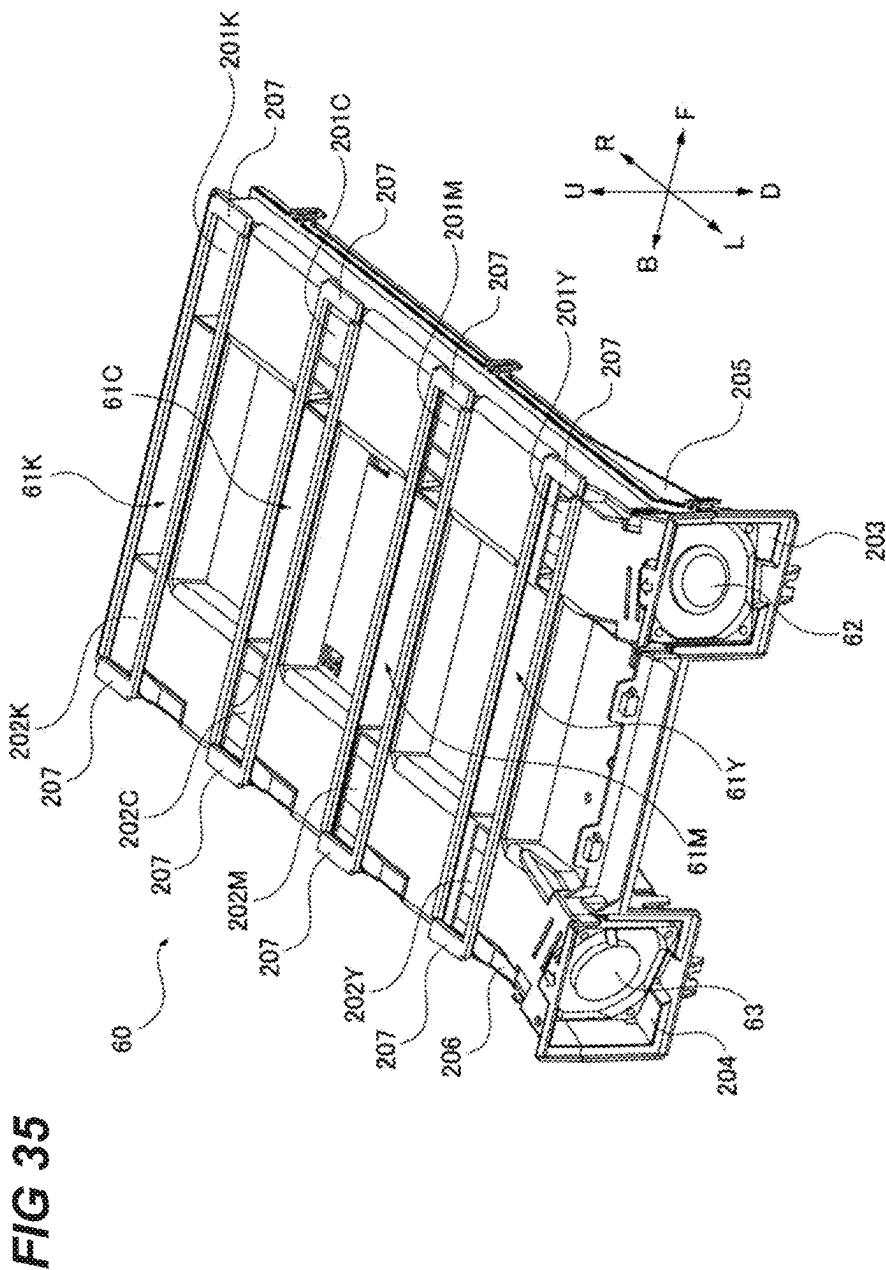
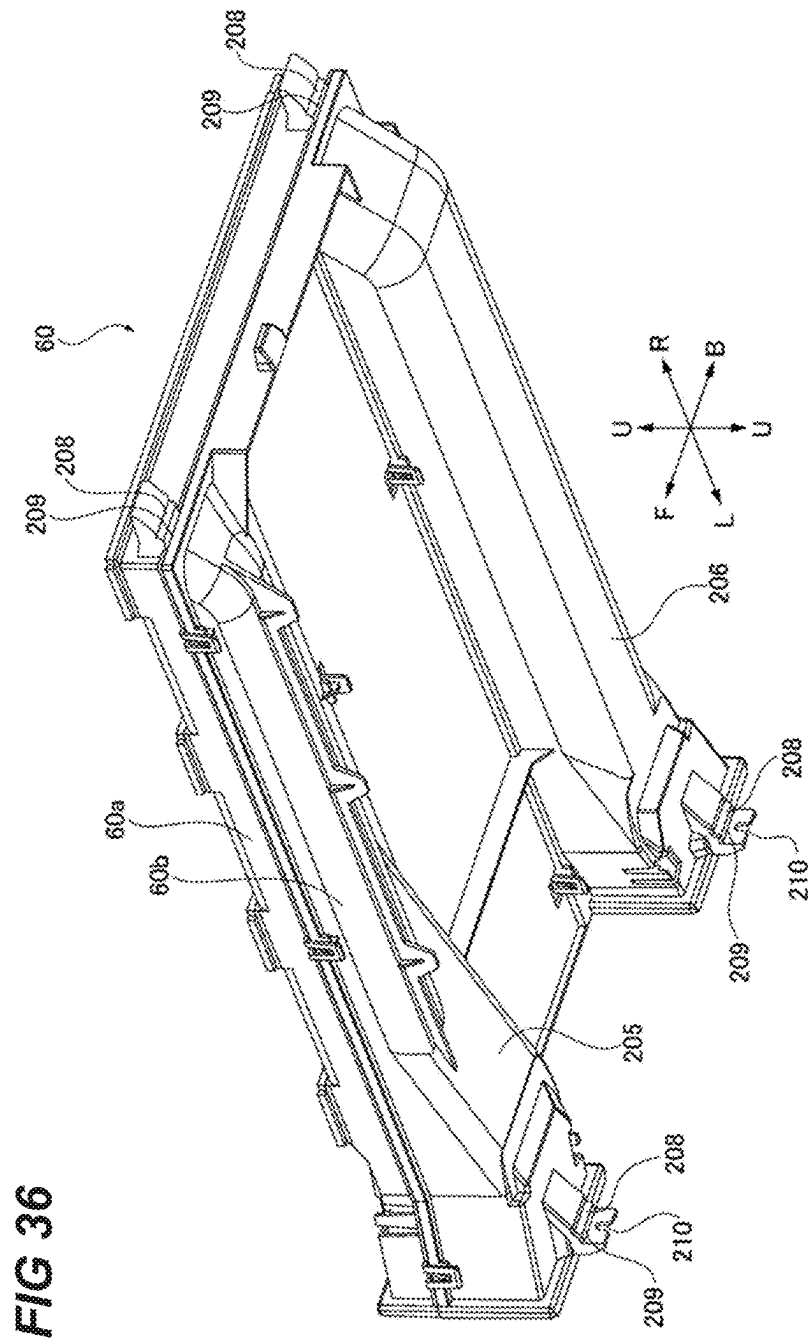


FIG 33







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FIG 37

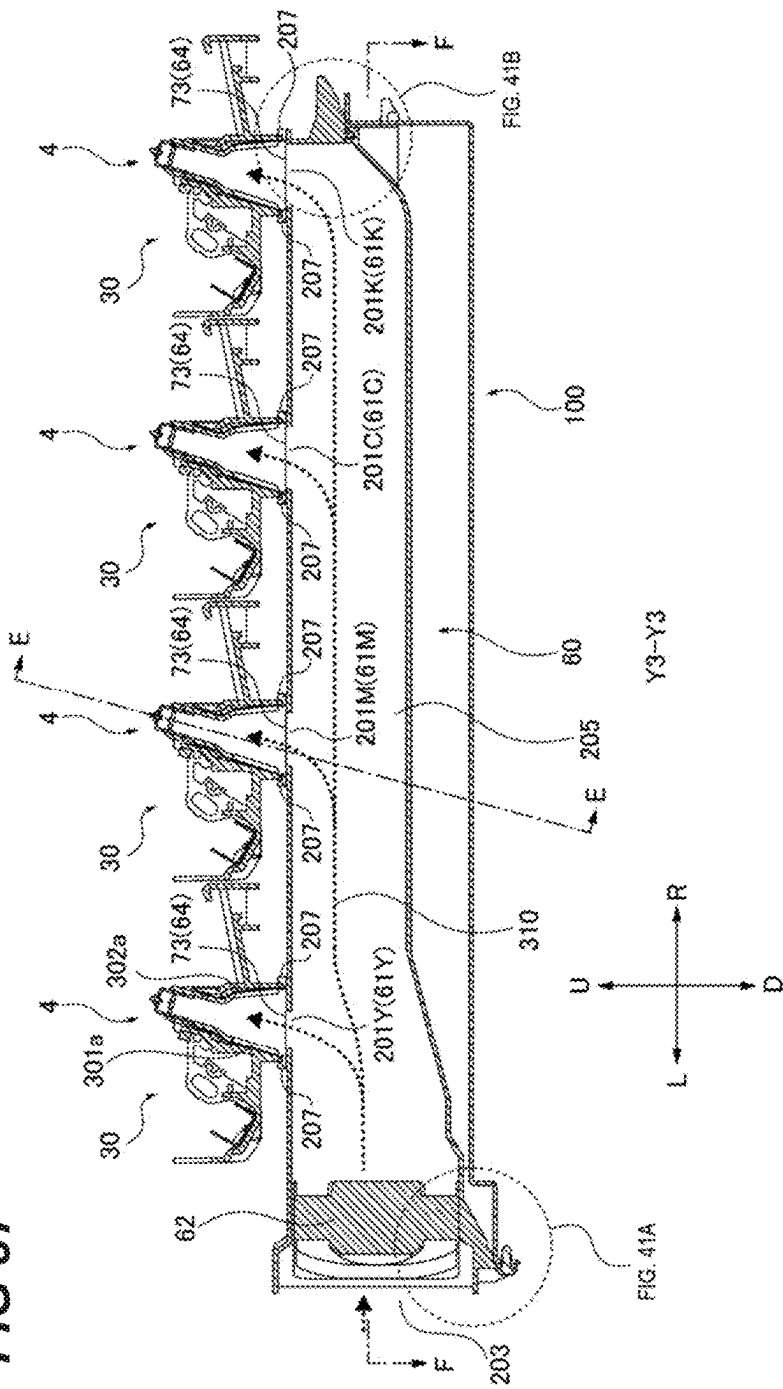


FIG 38

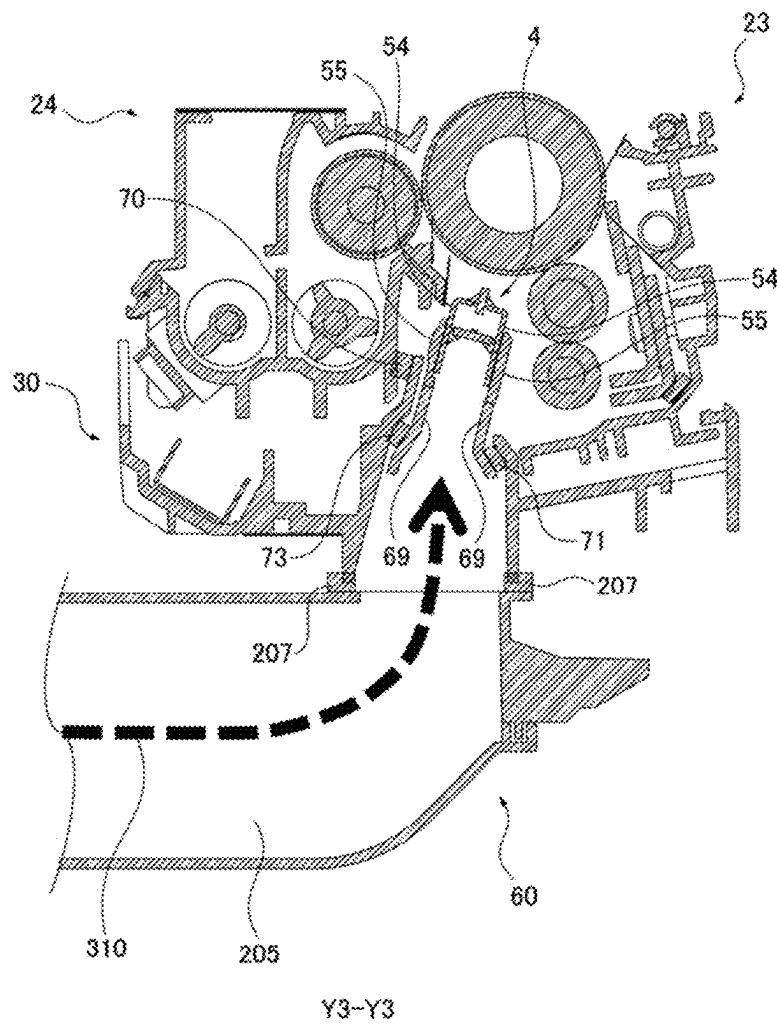


FIG 39

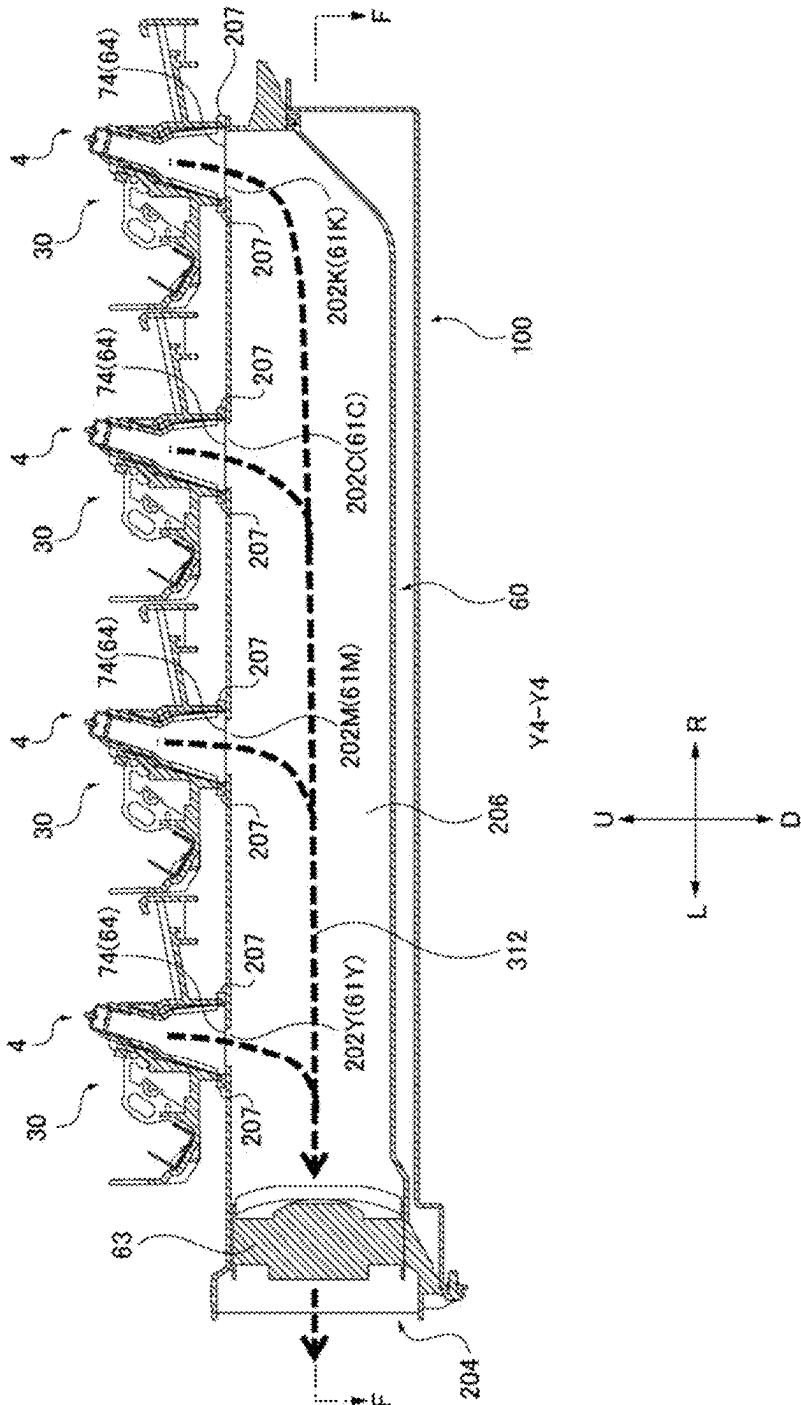


FIG 40

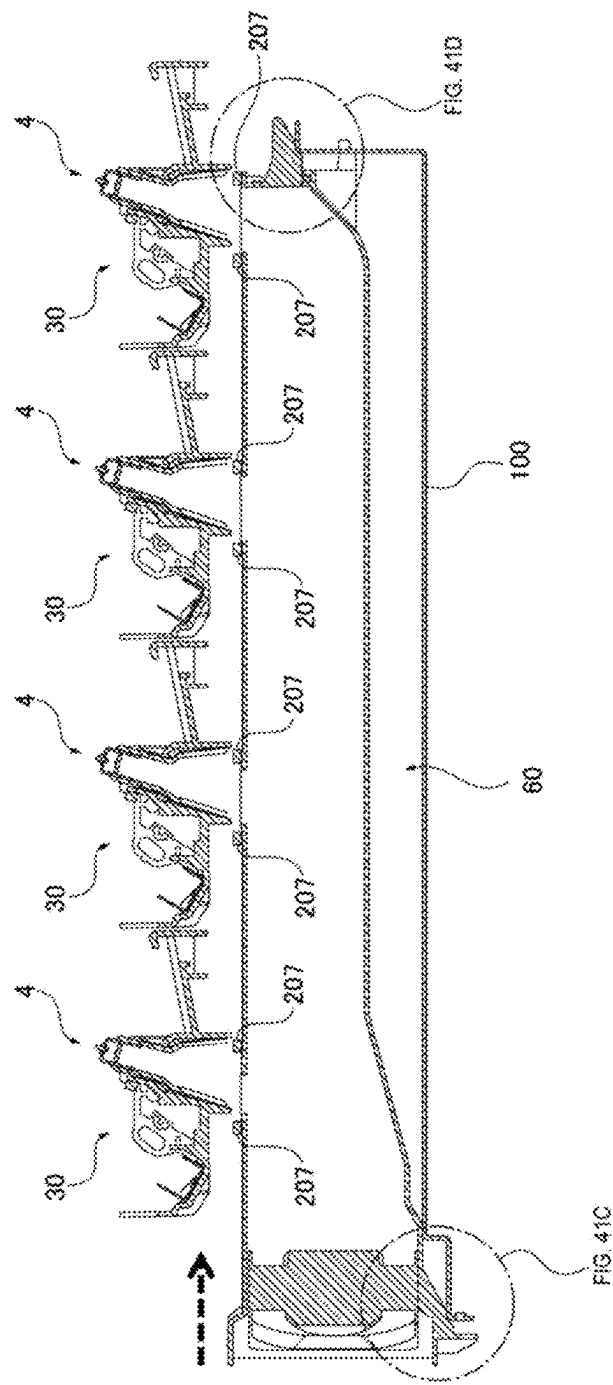


FIG 41A

LEFT SIDE IN FIG. 37

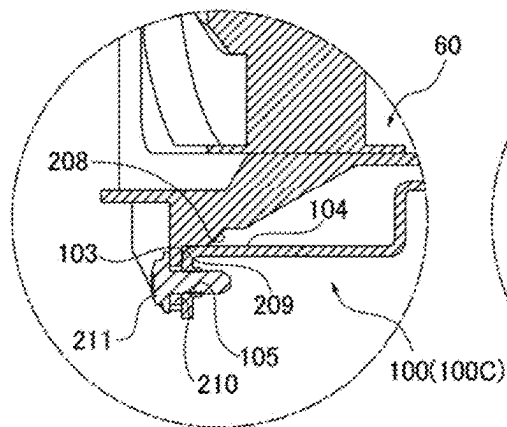


FIG 41B

RIGHT SIDE IN FIG. 37

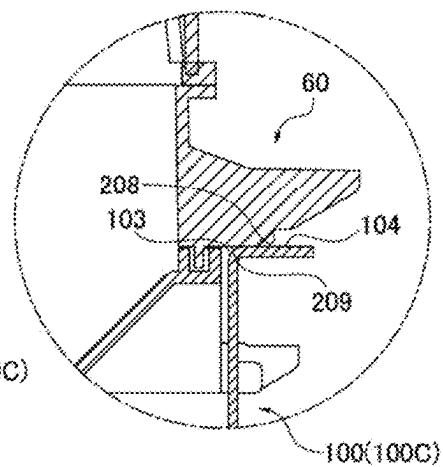


FIG 41C

LEFT SIDE IN FIG. 40

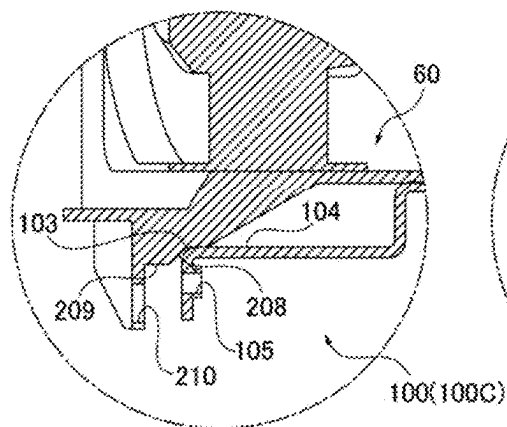


FIG 41D

RIGHT SIDE IN FIG. 40

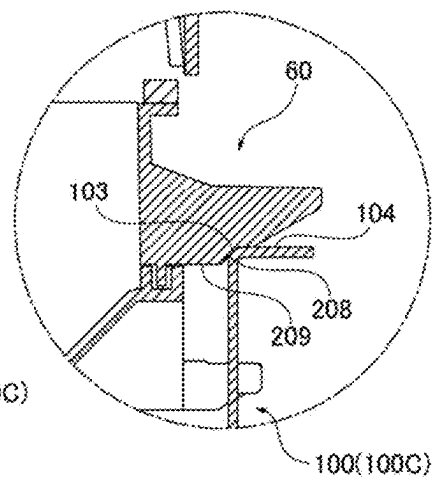


FIG 42

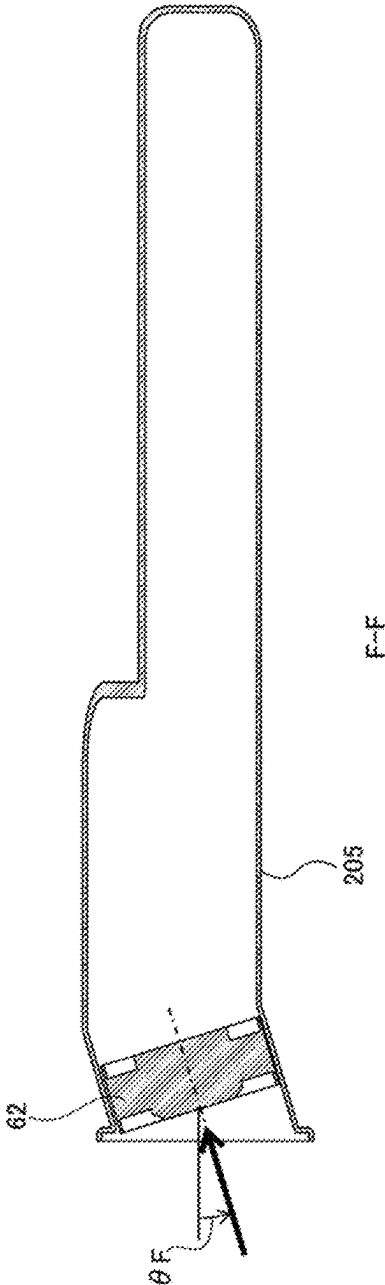
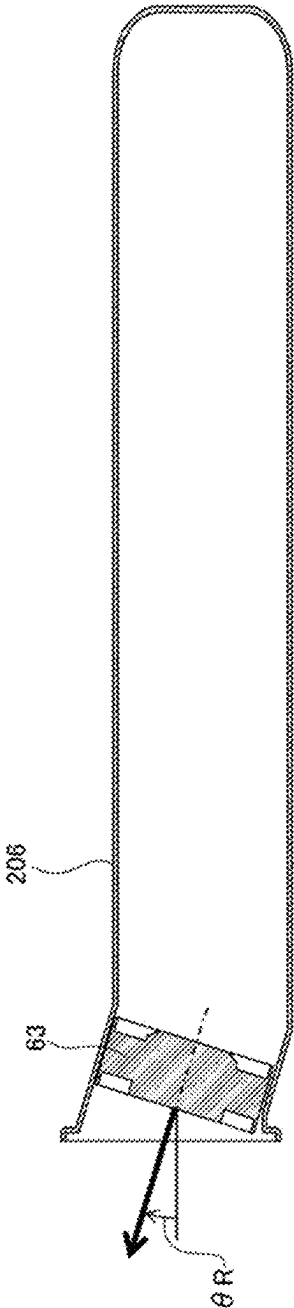


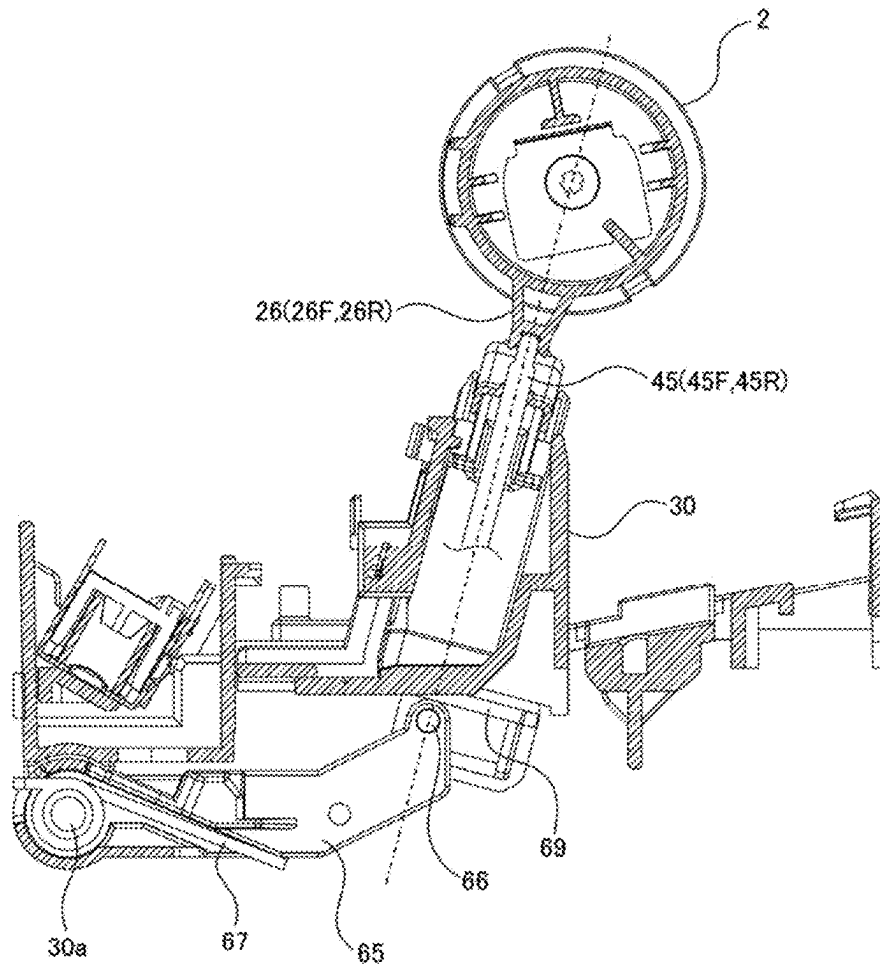
FIG 43

FIG 44

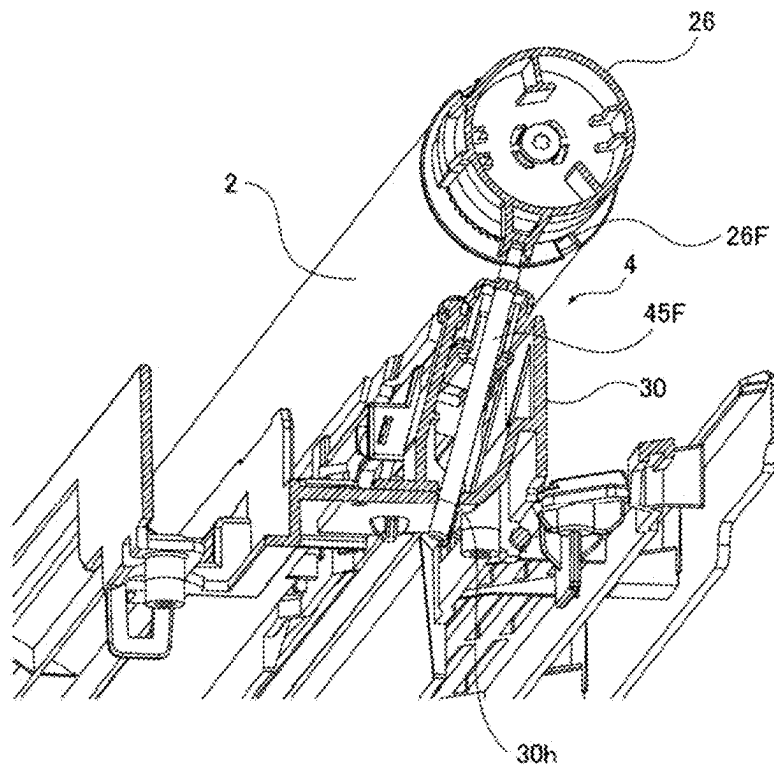


FIG 45

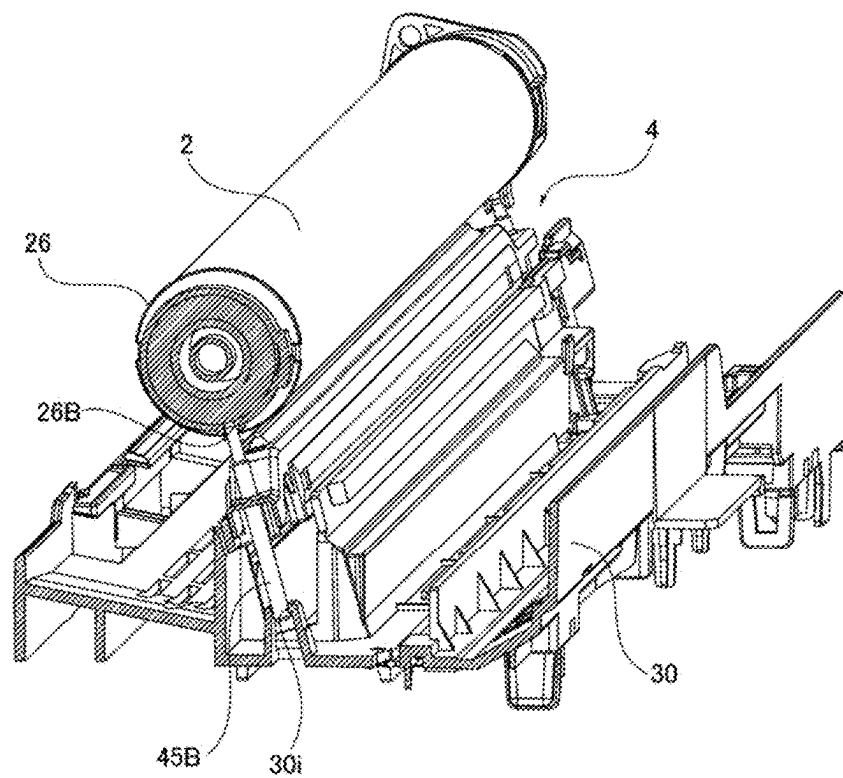


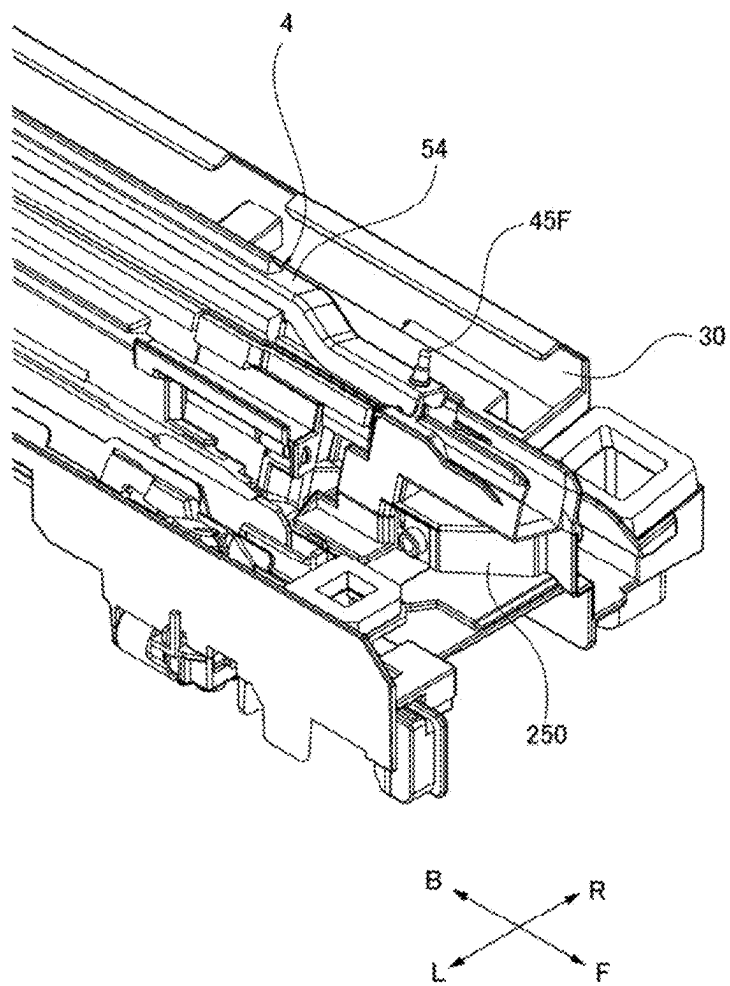
FIG 46

FIG 47

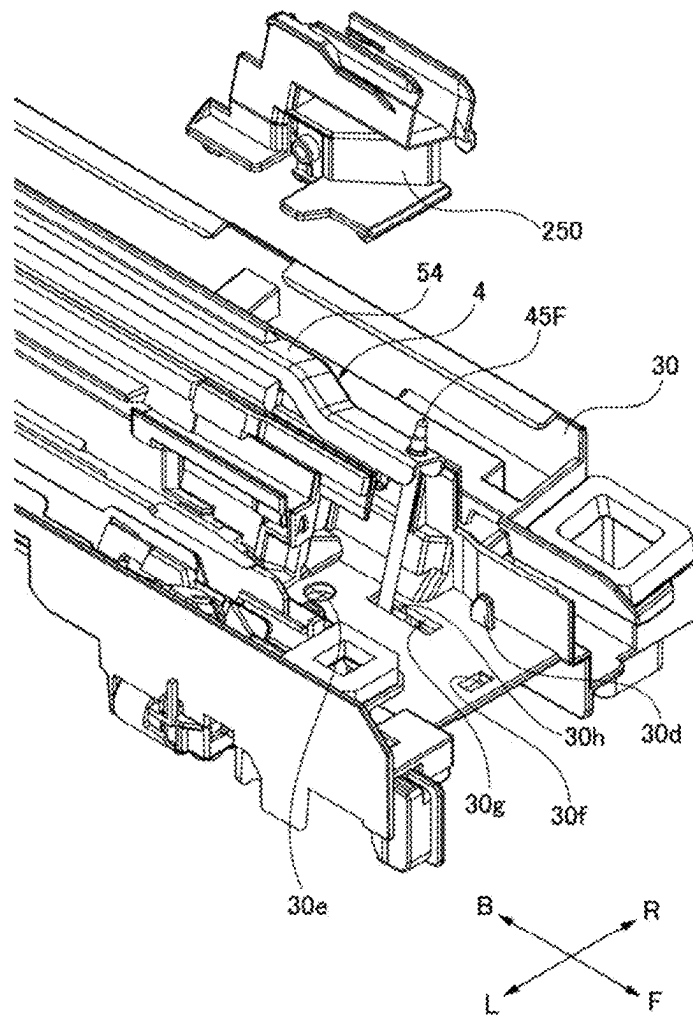


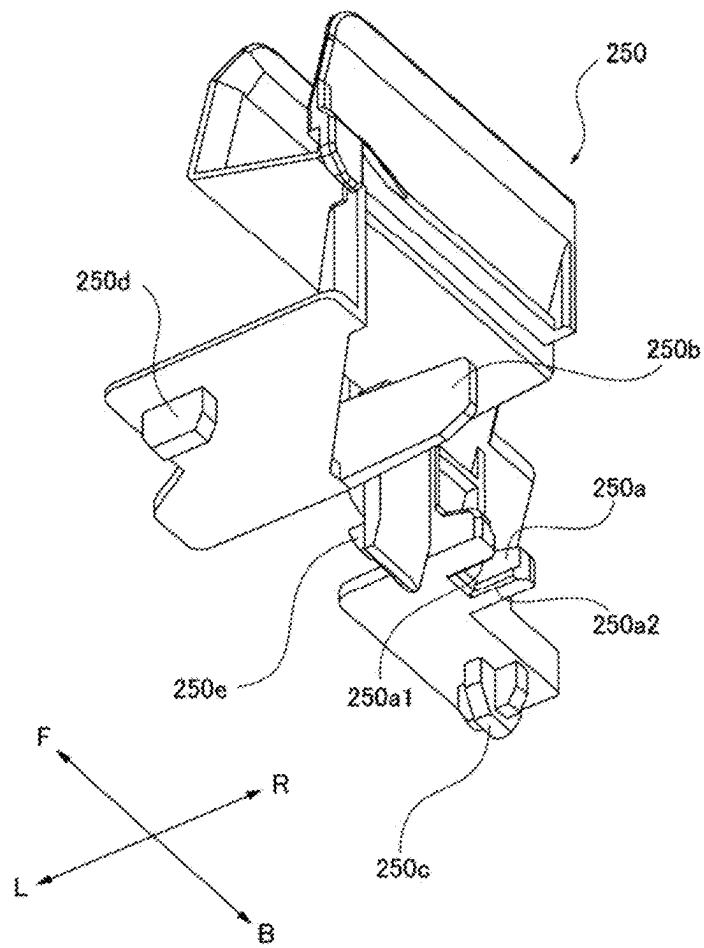
FIG 48

FIG 49

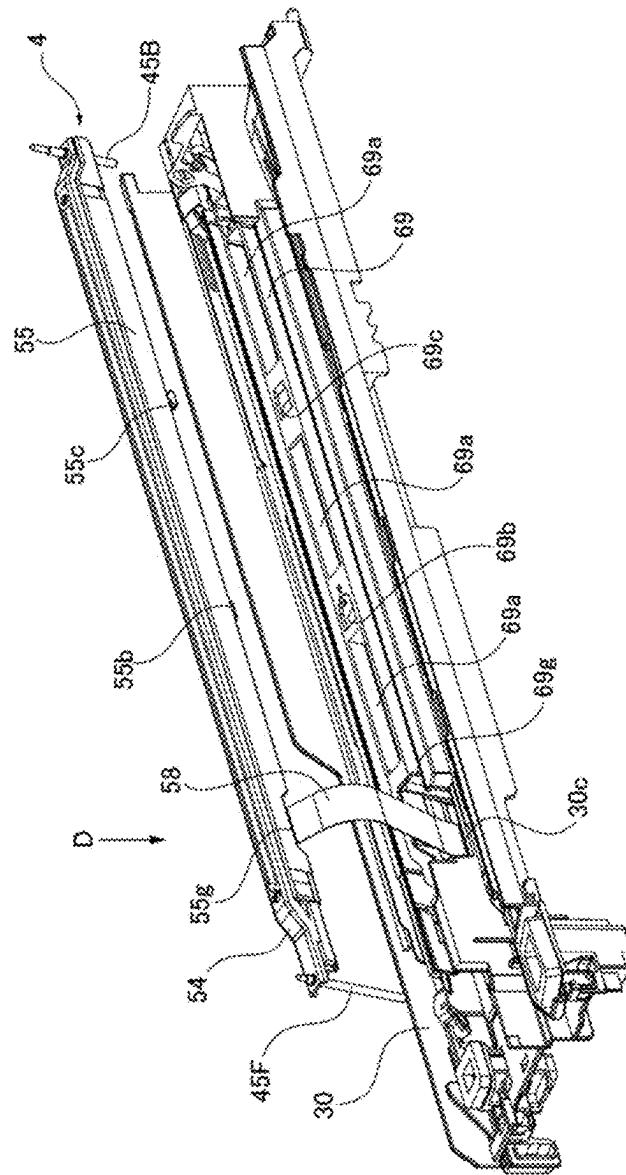
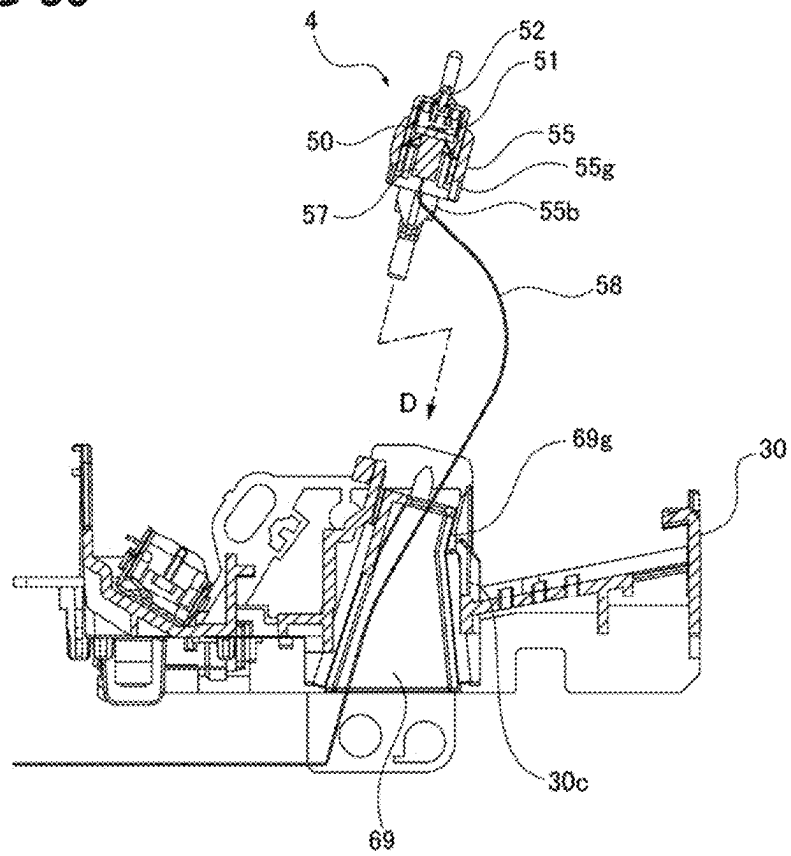


FIG 50



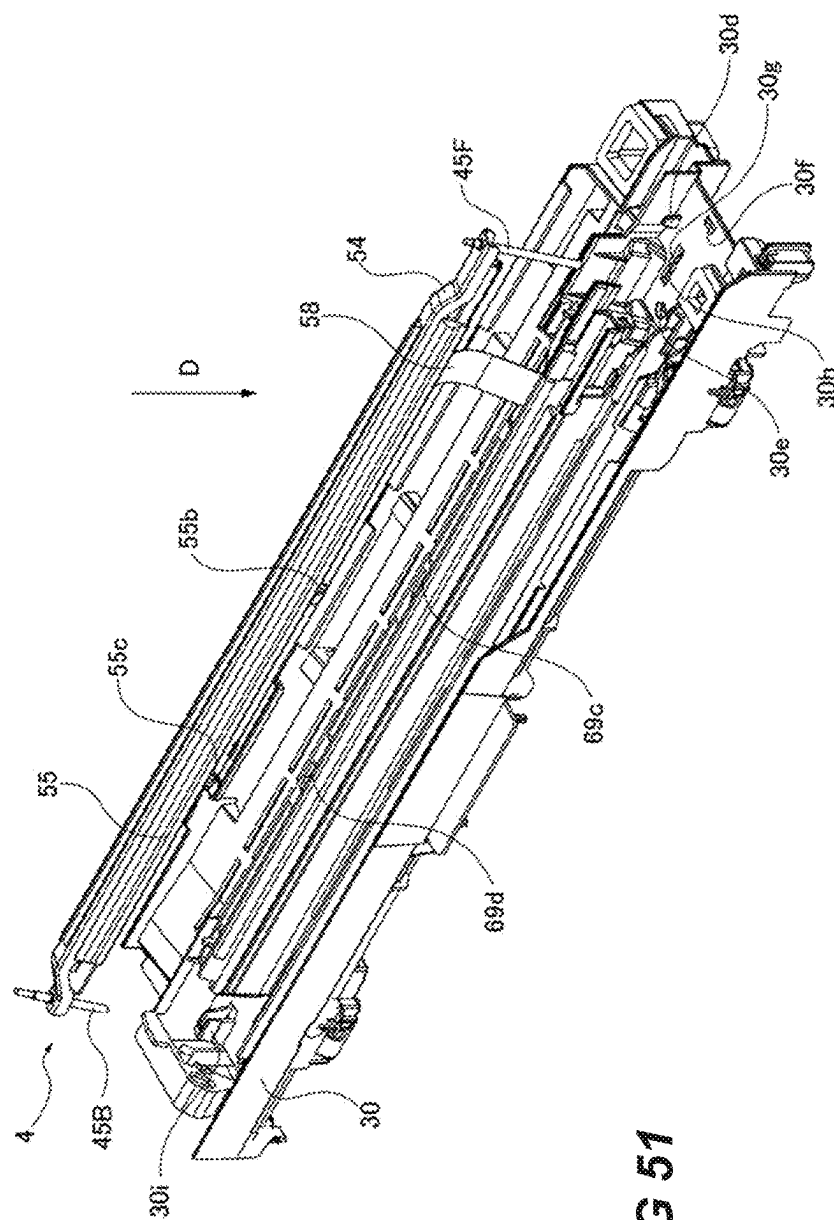


FIG 51

FIG 52

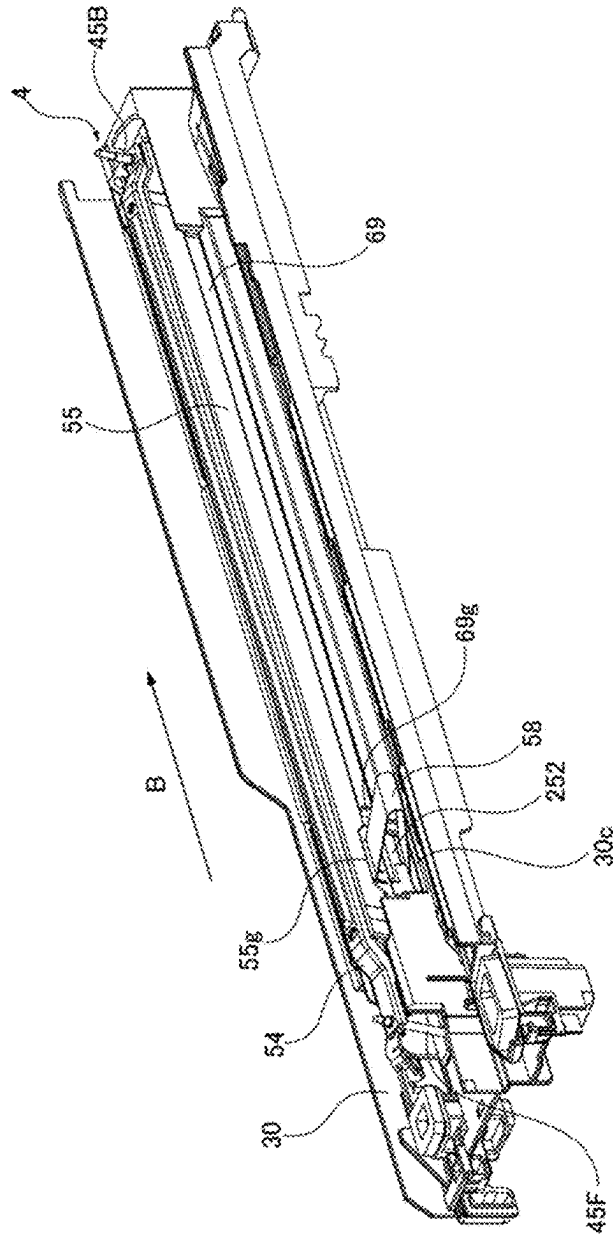
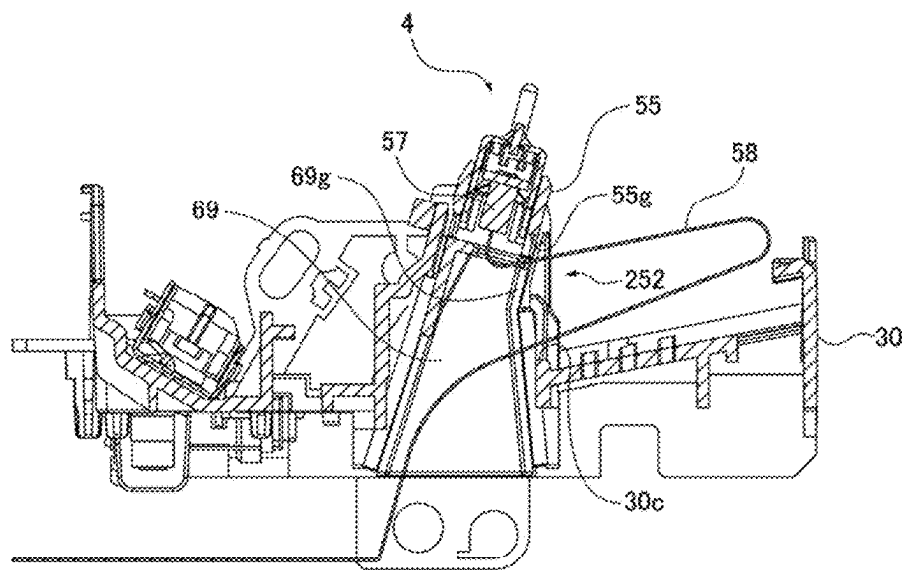
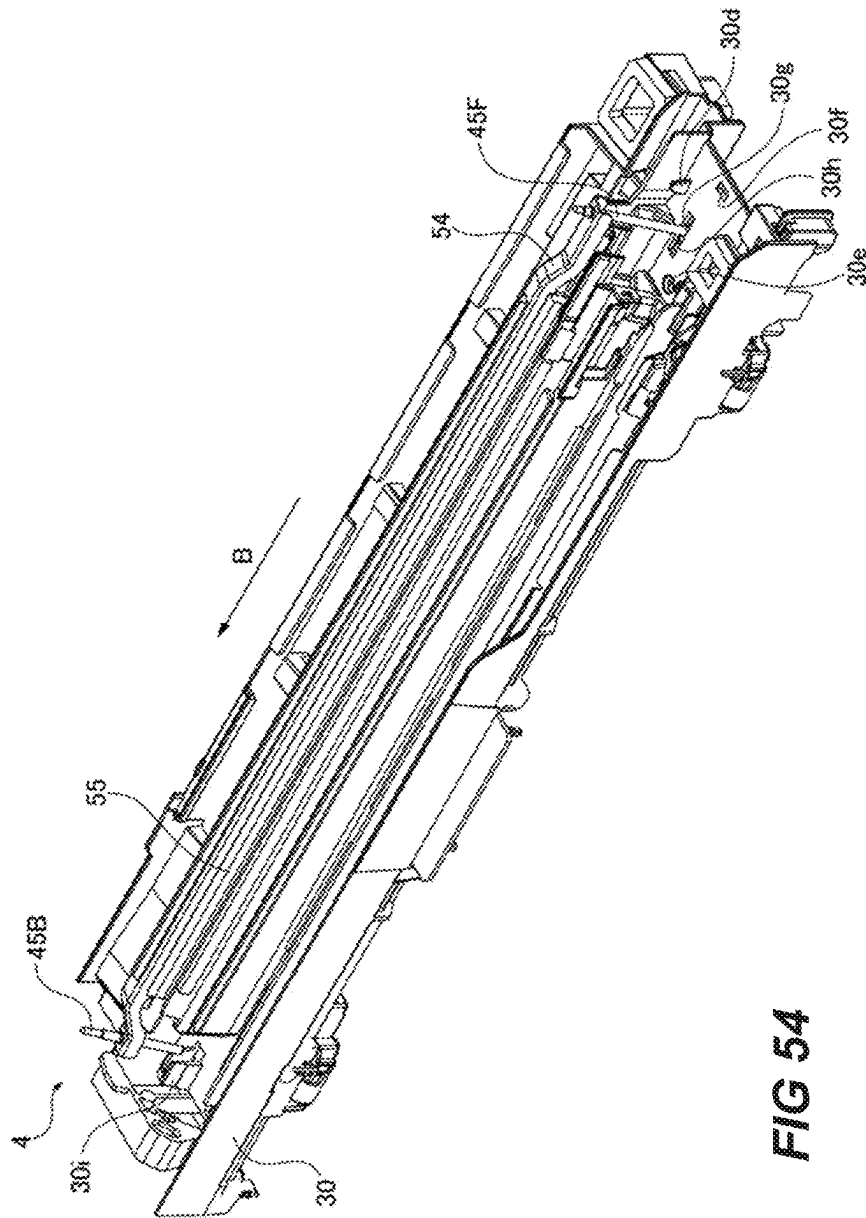


FIG 53



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FIG 55

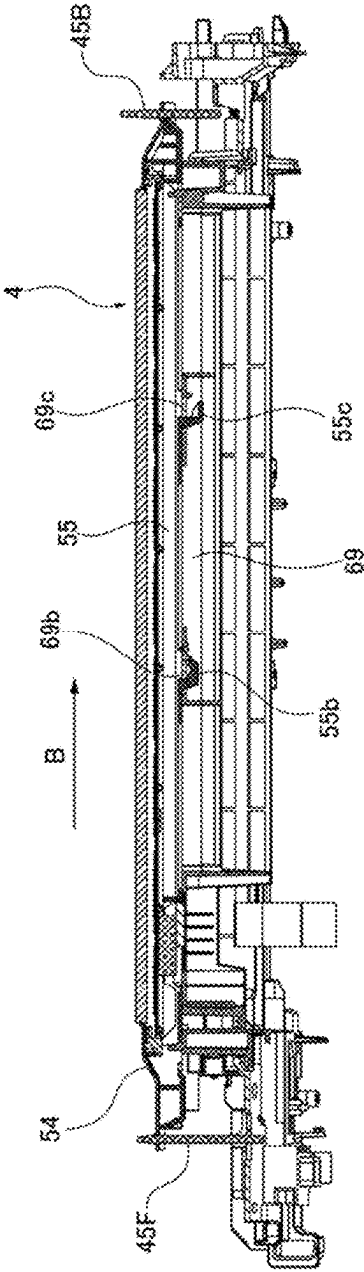


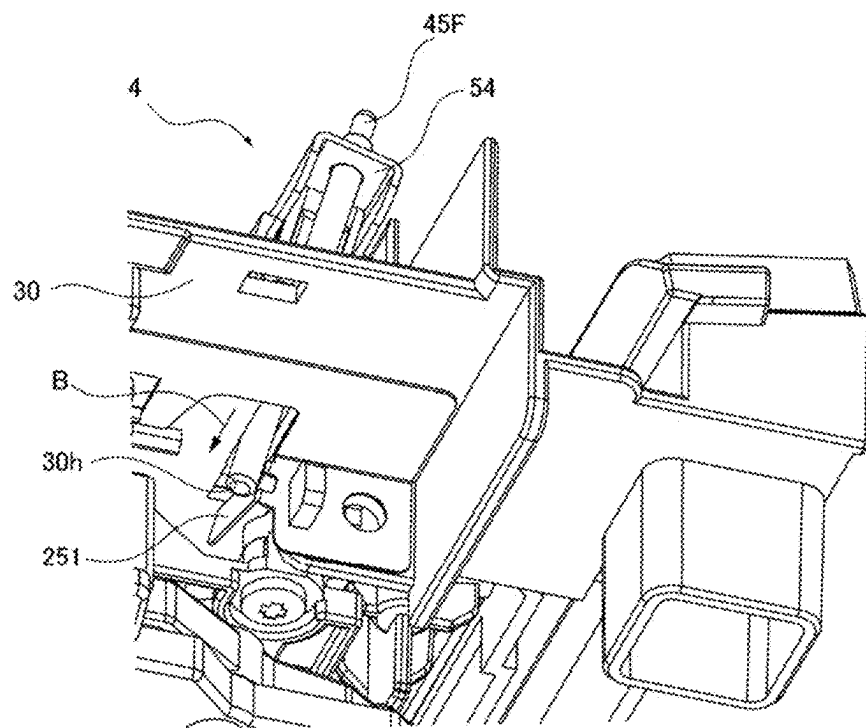
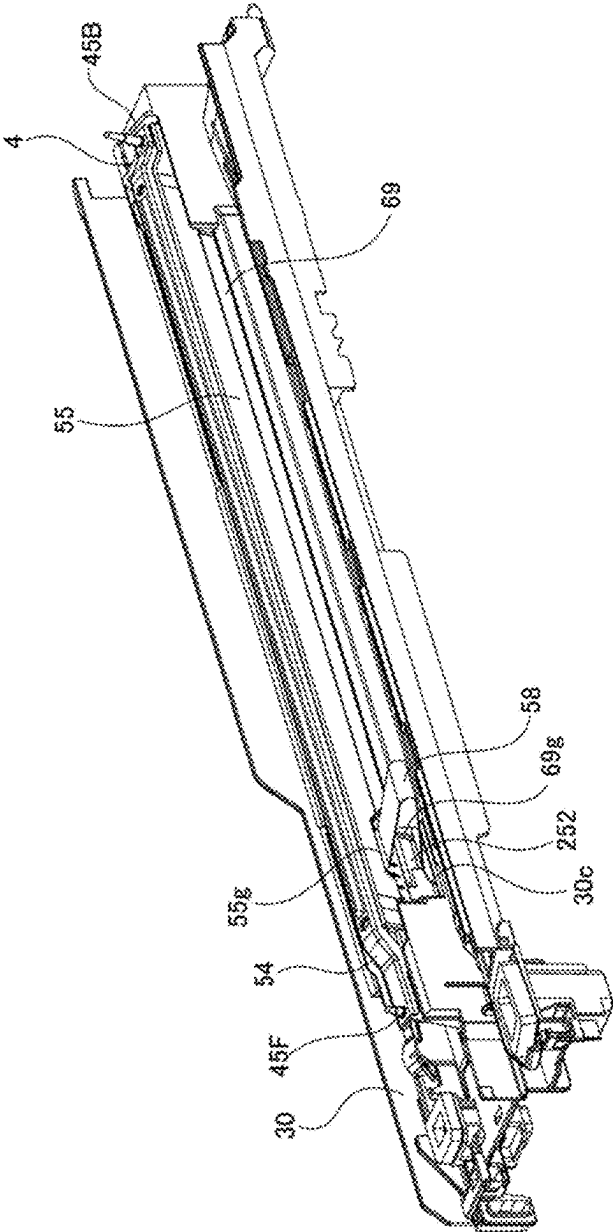
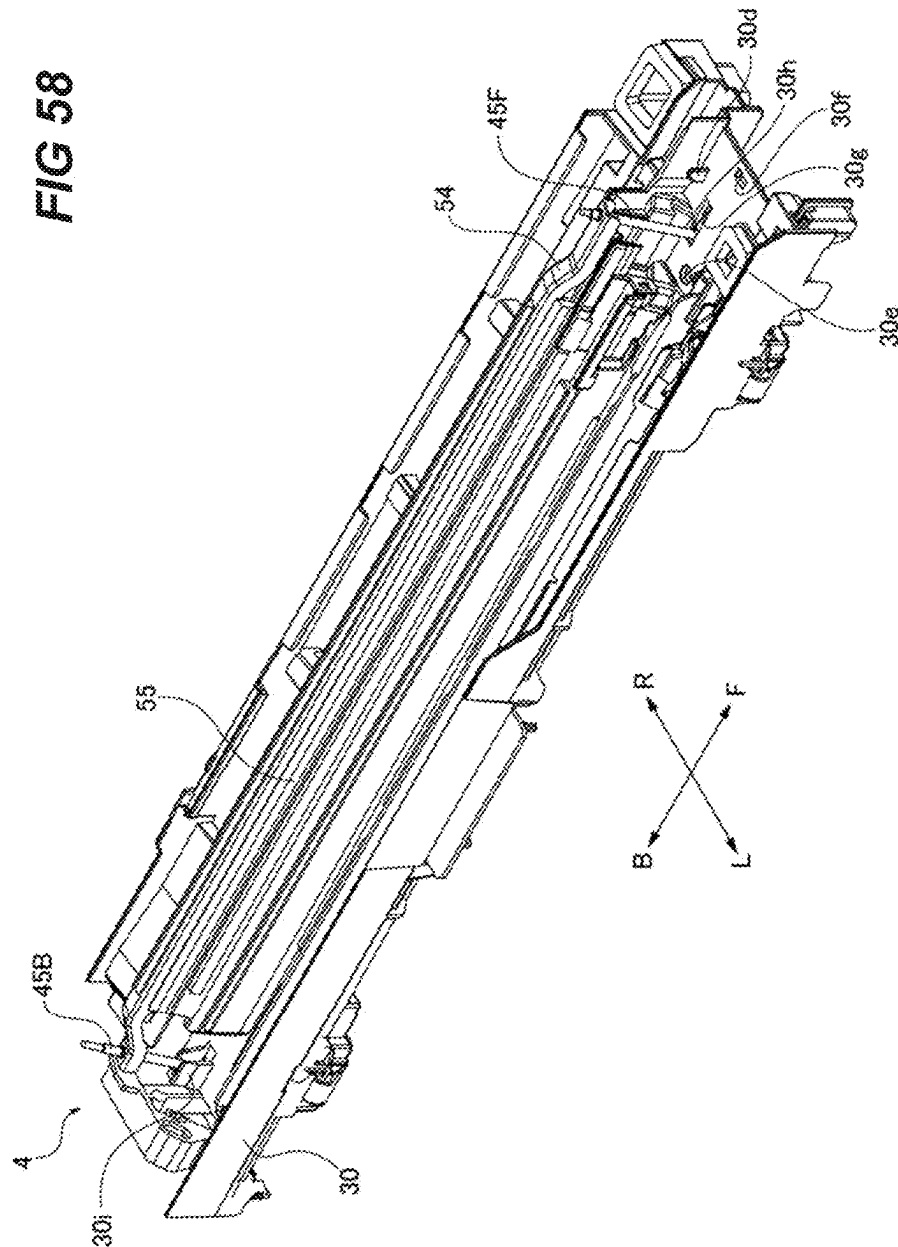
FIG 56

FIG 57



85 G/F



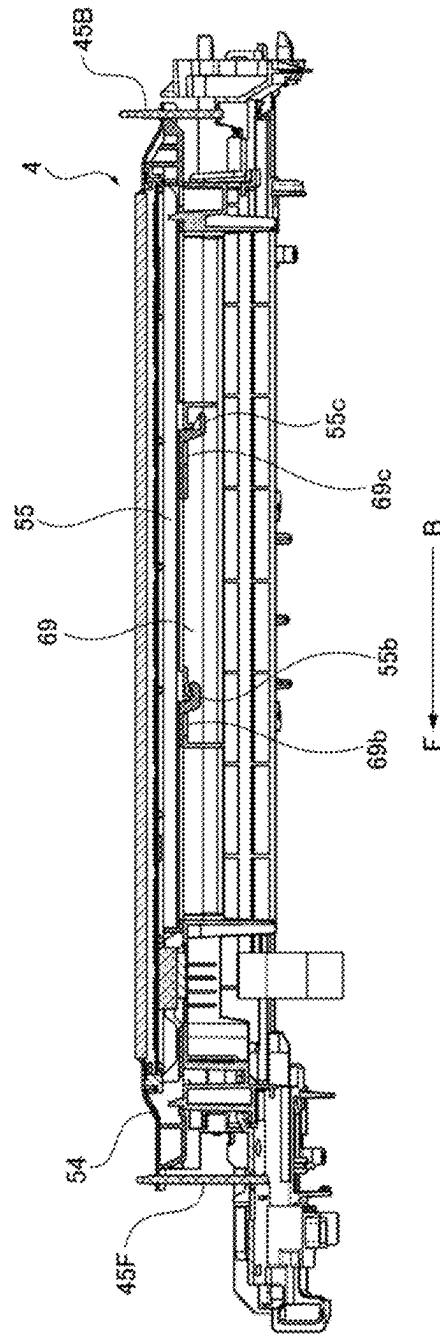


FIG 59

FIG 60

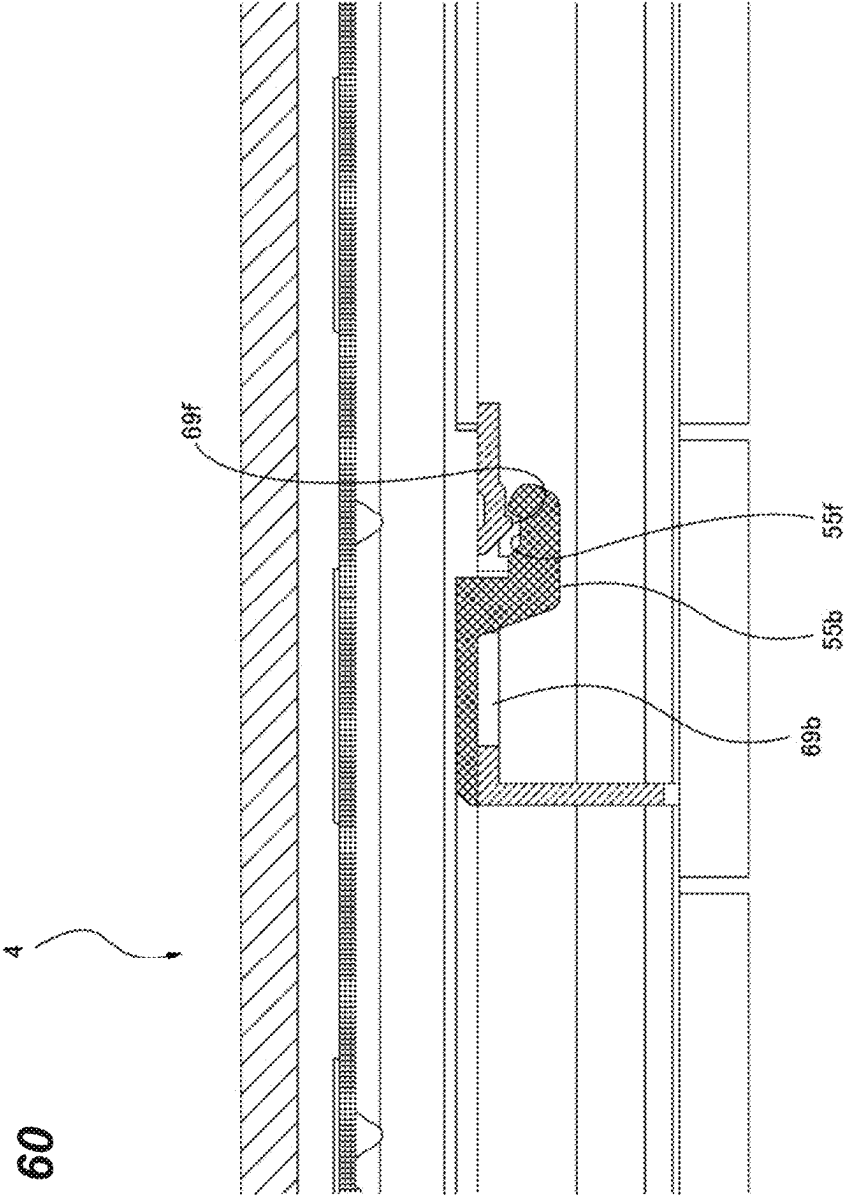


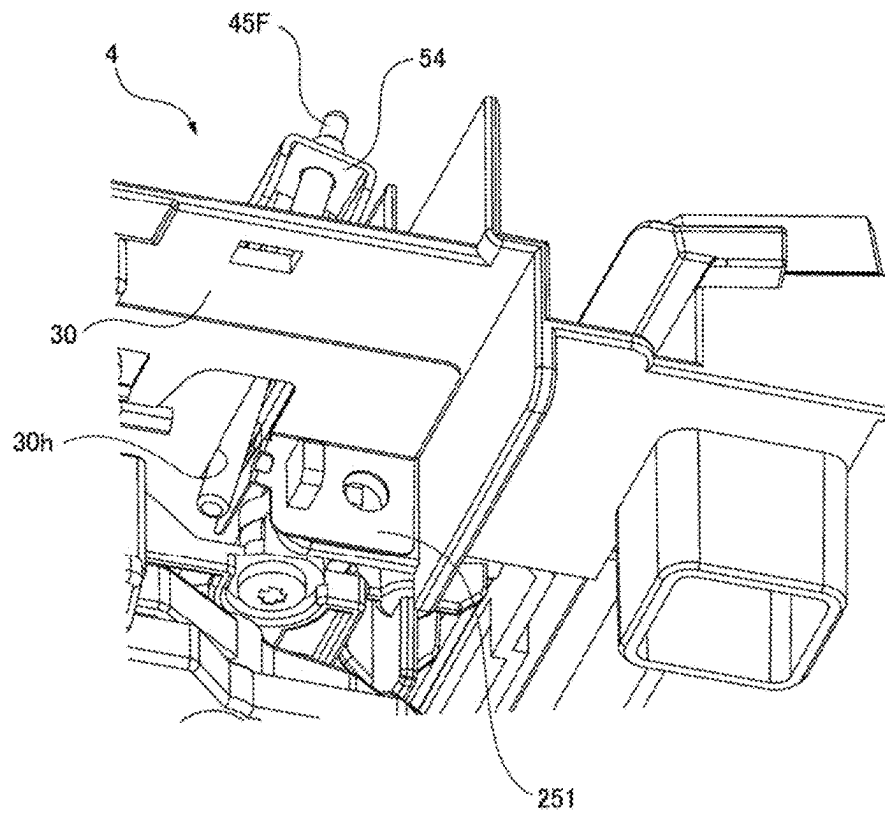
FIG 61

FIG 62

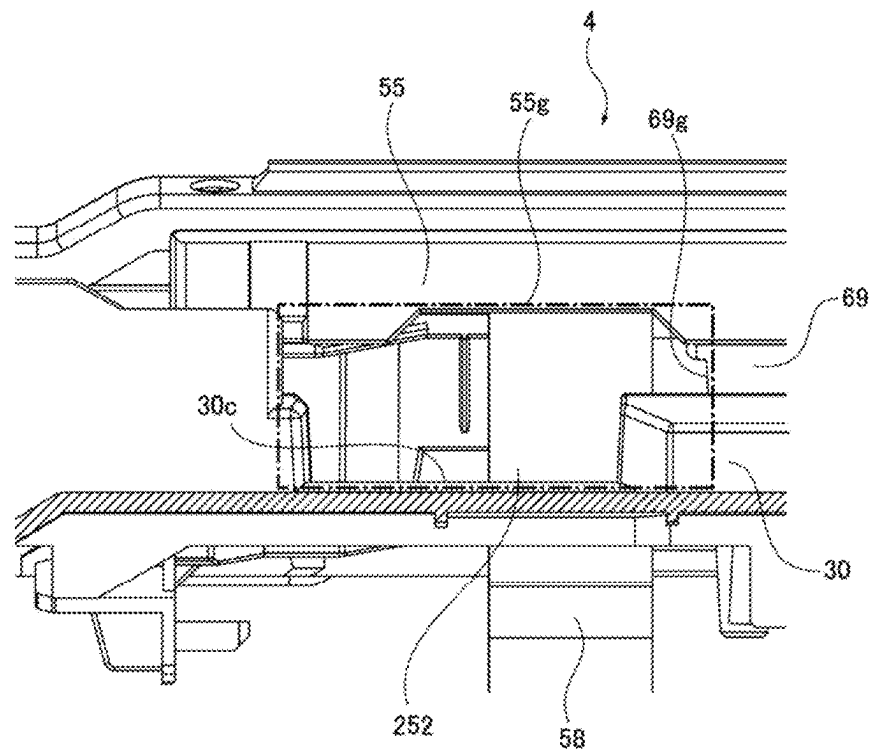


FIG 63

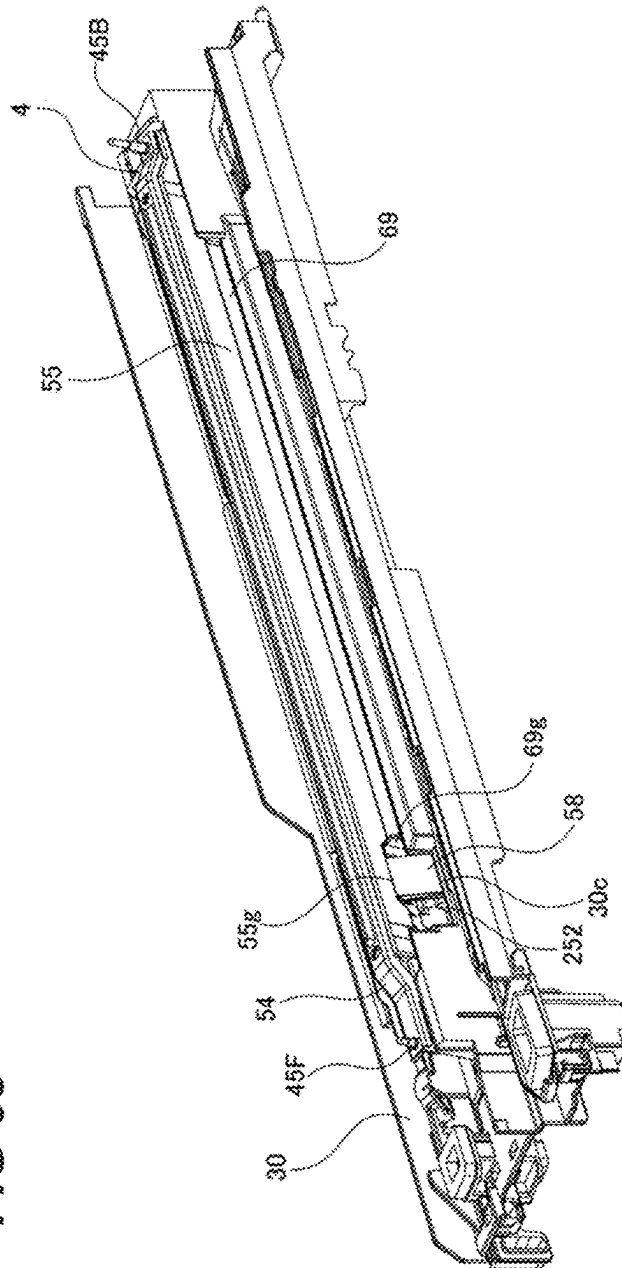


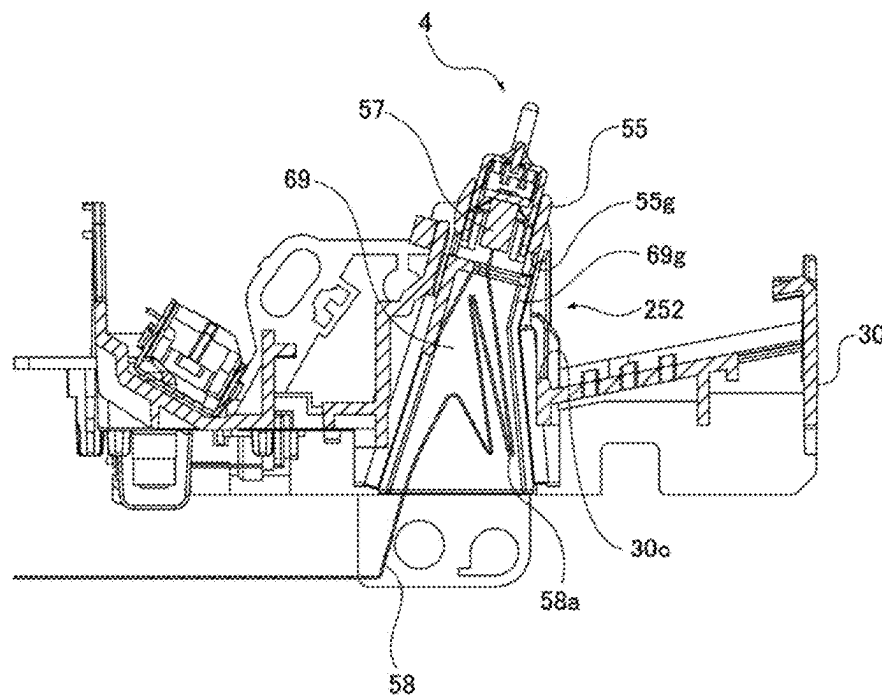
FIG 64

FIG 65

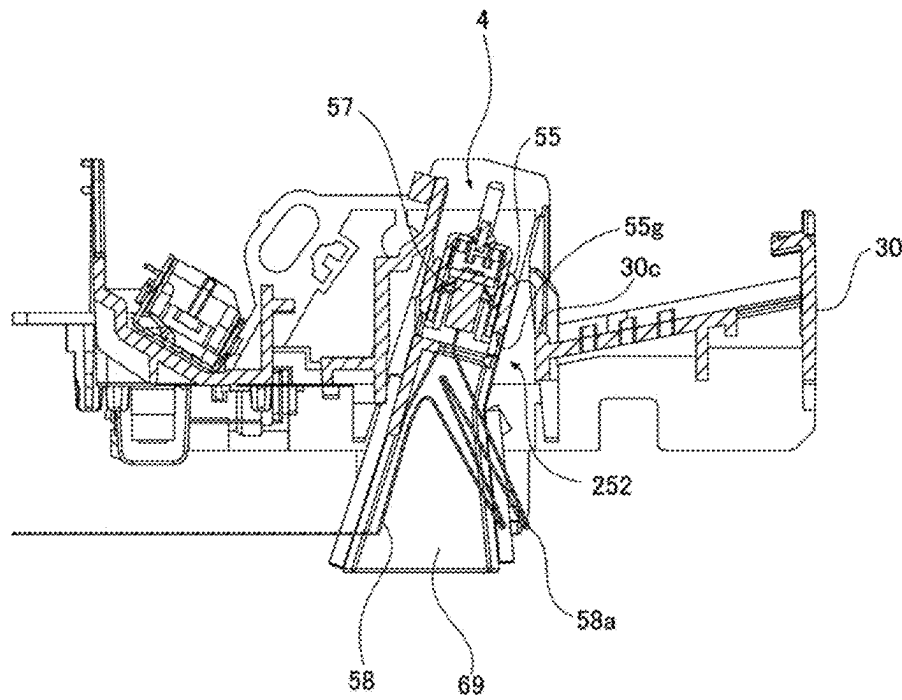


FIG 66

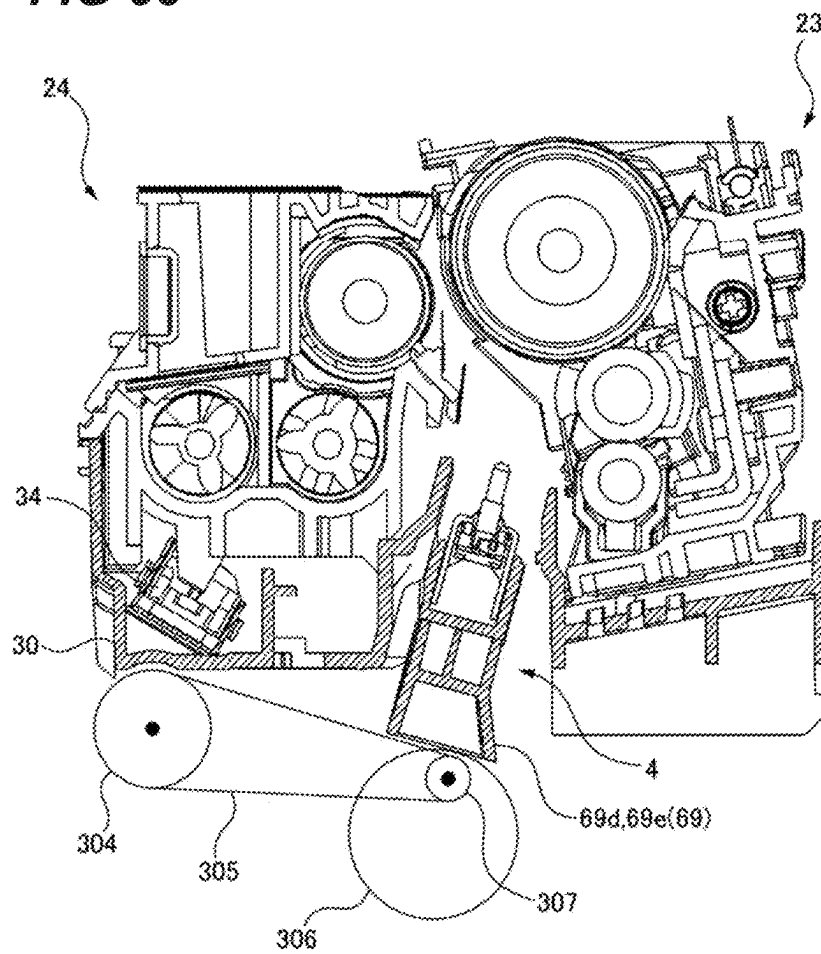


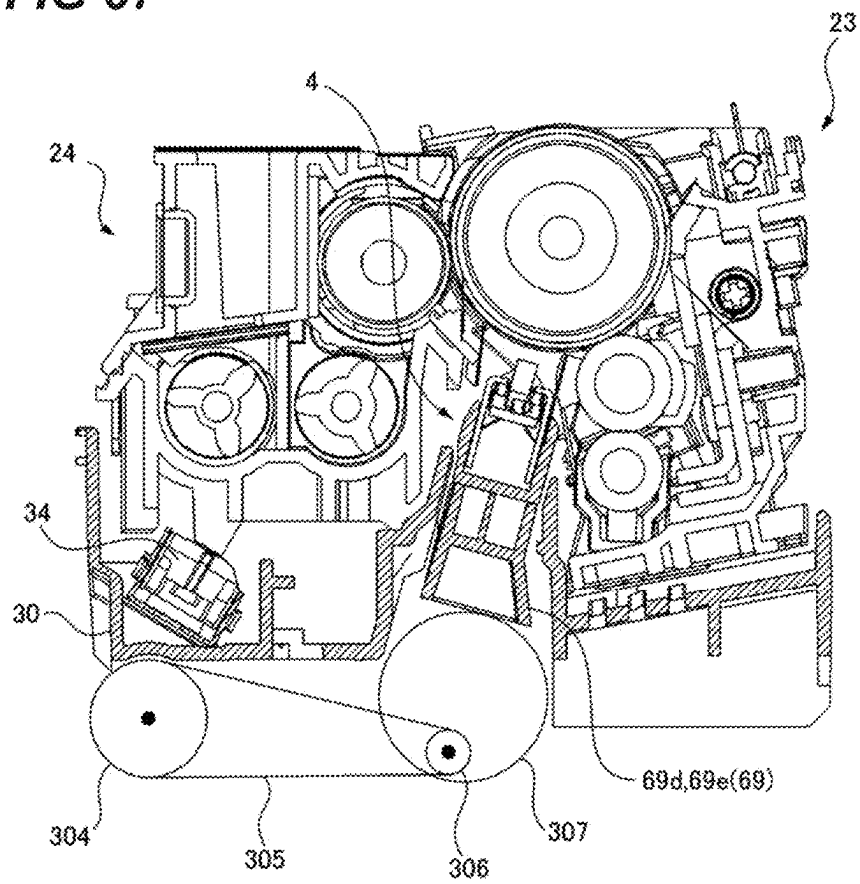
FIG 67

FIG 68

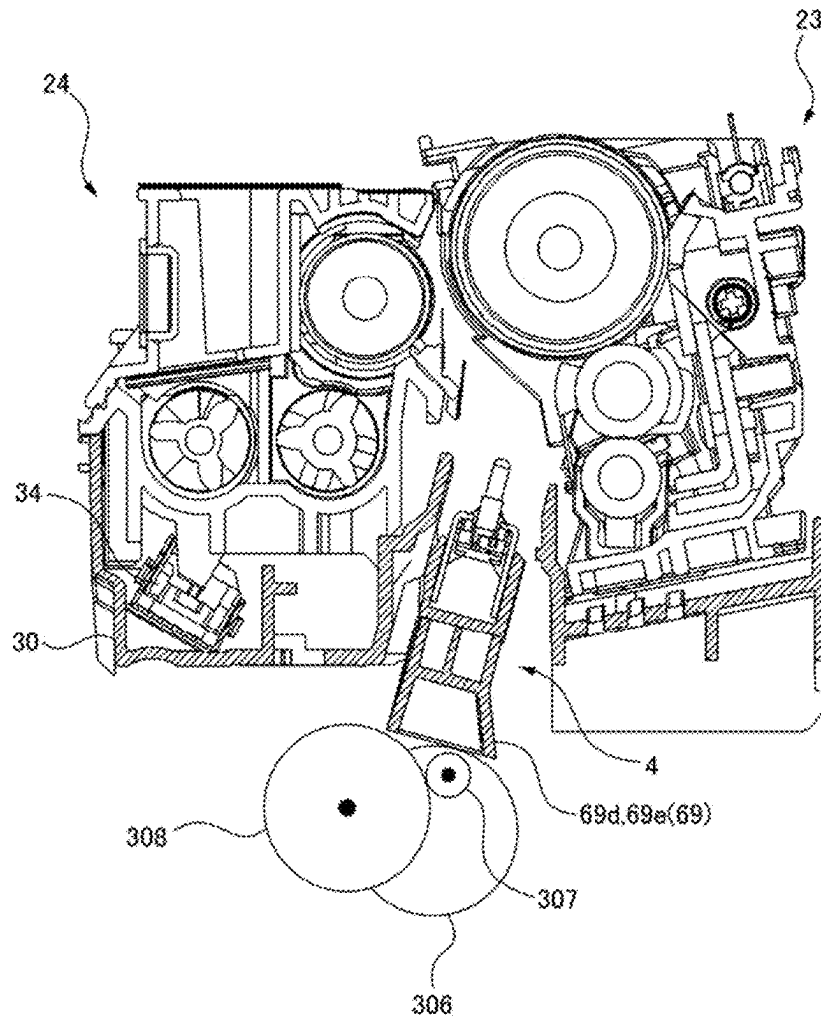
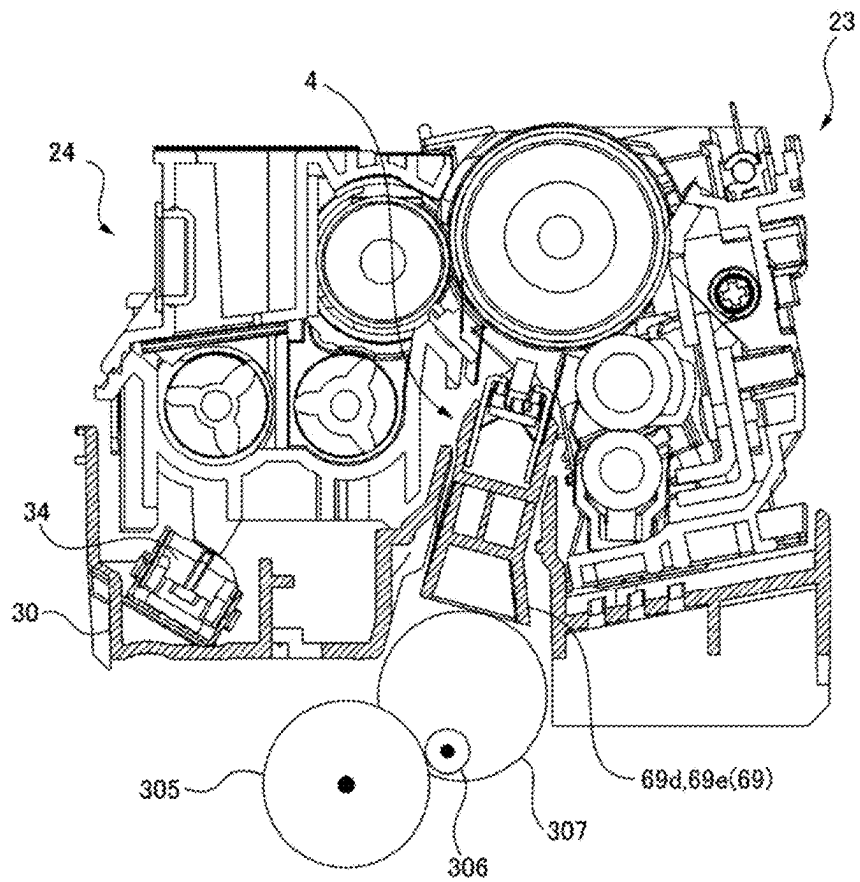


FIG 69



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IMAGE FORMING APPARATUS INCLUDING DUCT STRUCTURE THAT SUPPORTS LIGHT EMITTING ELEMENTS FOR EXPOSING A PHOTOCONDUCTOR

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to an image forming apparatus including an exposure head that exposes a photoconductor.

Description of the Related Art

Conventionally, as techniques related to an exposure head used in an electrophotographic system image forming apparatus, techniques described in Japanese Patent Application Laid-Open No. 2007-140349 are known.

An exposure head described in Japanese Patent Application Laid-Open No. 2007-140349 includes a substrate in which a plurality of light emitting elements is aligned along the axial direction of a photoconductor, and a lens that condenses light emitted from the plurality of light emitting elements, and the light emitted from the plurality of light emitting elements is condensed through the lens to expose a photosensitive drum.

Therefore, the exposure head described in Japanese Patent Application Laid-Open No. 2007-140349 is disposed close to the photoconductor together with a developing device that develops a latent image formed on the photoconductor with toner. Therefore, the exposure head is disposed close to the developing device.

The exposure head includes a light emitting diode (LED) as a light emitting element. Alternatively, organic electro luminescence (EL) may be included as light emitting elements. The organic EL may be referred to as an organic light emitting diode (OLED).

It is known that the exposure head dissipates heat according to the duration of the light emission time and the magnitude of the light emission amount of such a light emitting element. As described above, since the exposure head is disposed close to the developing device that causes the toner to adhere to the photoconductor, the cooling unit is often required in order to reduce the influence of heat from the exposure head on the toner. Japanese Patent Application Laid-Open No. 2007-140349 discloses a configuration in which an airflow is formed in a space between an exposure head and a developing device as a cooling unit of the exposure head.

However, in the configuration described in Japanese Patent Application Laid-Open No. 2007-140349, an airflow is also formed around the developing device. Therefore, there is a problem that the airflow for cooling the exposure head scatters the toner of the developing device disposed close to the exposure head inside the image forming apparatus.

SUMMARY OF THE INVENTION

An image forming apparatus according to the present invention includes: a rotatable photoconductor; an exposure portion including a substrate on which a plurality of chips each including a plurality of light emitting portions that emit light for exposing the photoconductor is mounted, and a support portion that supports the substrate; a fan that generates an airflow for cooling the exposure portion; and a duct

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that communicates with the fan and communicates with the support portion to guide the airflow generated by the fan to the substrate.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an image forming apparatus;

FIG. 2 is a schematic cross-sectional view of the image forming apparatus in FIG. 1;

FIG. 3 is a schematic cross-sectional view of the image forming apparatus in FIG. 1;

FIG. 4 is a perspective view of an image forming apparatus;

FIG. 5 is a partially enlarged perspective view of the image forming apparatus;

FIG. 6 is a partially enlarged perspective view of the image forming apparatus;

FIG. 7 is a partially enlarged perspective view of the image forming apparatus;

FIG. 8 is a perspective view of a cartridge tray;

FIG. 9 is a perspective view of the cartridge tray;

FIG. 10 is a cross-sectional view of an exposure head;

FIG. 11 is a perspective view of the exposure head;

FIG. 12 is a perspective view of the exposure head;

FIGS. 13A, 13B, and 13C are views showing a substrate in an exposure head, and FIGS. 13D and 13E are views showing a lens array;

FIG. 14 is a perspective view of the substrate as viewed from a surface (substrate front surface) on which an LED is mounted;

FIG. 15 is a perspective view of the substrate as viewed from a surface (substrate back surface) on which a connector is mounted;

FIG. 16 is a perspective view of the exposure head with the substrate assembled to a housing as viewed from below;

FIG. 17 is an enlarged view of a connector side of the exposure head in FIG. 16;

FIG. 18 is a perspective view illustrating the exposure head in a state where the lens array is assembled to the housing in FIG. 16;

FIG. 19 is a perspective view of the connector side of the exposure head in which the housing and a housing support member are integrally configured as viewed from below;

FIG. 20 is a perspective view of the cartridge tray;

FIG. 21 is a bottom view of the cartridge tray and an elevating duct;

FIG. 22 is a cross-sectional view taken along line X-X in FIG. 20;

FIG. 23 is a cross-sectional view taken along line X-X in FIG. 20;

FIG. 24 is a cross-sectional view taken along line Y-Y in FIG. 20;

FIG. 25 is a cross-sectional view taken along line Y-Y in FIG. 20;

FIG. 26 is a cross-sectional view of an exposure cooling airflow in a direction perpendicular to an optical axis;

FIG. 27 is a perspective view of the exposure head, the elevating duct, and a rotating arm;

FIG. 28 is a perspective view of the exposure head, the elevating duct, and the rotating arm;

FIG. 29 is a cross-sectional view taken along line E-E in FIG. 37;

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FIG. 30 is a cross-sectional view taken along line E-E in FIG. 37;

FIG. 31 is a side view of a development stay;

FIG. 32 is a side view of the development stay;

FIG. 33 is a cross-sectional view taken along line X-X in FIG. 20;

FIG. 34 is a cross-sectional view taken along line A-A in FIG. 2;

FIG. 35 is a perspective view of a duct unit;

FIG. 36 is a perspective view of the duct unit as viewed from below;

FIG. 37 is a cross-sectional view of an intake side of the exposure cooling airflow;

FIG. 38 is an enlarged cross-sectional view of the intake side of the exposure cooling airflow;

FIG. 39 is a cross-sectional view of an exhaust side of the exposure cooling airflow;

FIG. 40 is a cross-sectional view immediately before the duct unit is assembled;

FIGS. 41A and 41B are enlarged cross-sectional views of a positioning shape of the duct unit, and FIGS. 41C and 41D are enlarged cross-sectional views immediately before the duct unit is assembled;

FIG. 42 is a cross-sectional view taken along line F-F in FIG. 39;

FIG. 43 is a cross-sectional view of a photosensitive drum, the exposure head, and the rotating arm;

FIG. 44 is a cross-sectional perspective view illustrating positioning on the front side of the photosensitive drum and the exposure head;

FIG. 45 is a cross-sectional perspective view illustrating positioning on the rear side of the photosensitive drum and the exposure head;

FIG. 46 is a perspective view illustrating a state after a positioning member is attached;

FIG. 47 is a perspective view illustrating a state before the positioning member is attached;

FIG. 48 is a perspective view illustrating a shape of the positioning member;

FIG. 49 is a right perspective view of a state where the exposure head is removed from the elevating duct;

FIG. 50 is a front cross-sectional view of a state where the exposure head is removed from the elevating duct;

FIG. 51 is a left perspective view of a state where the exposure head is removed from the elevating duct;

FIG. 52 is a right perspective view of a state where the exposure head is placed in the elevating duct;

FIG. 53 is a front cross-sectional view of a state where the exposure head is placed in the elevating duct;

FIG. 54 is a left perspective view of a state where the exposure head is placed in the elevating duct;

FIG. 55 is a cross-sectional view of a state where the exposure head is placed in the elevating duct;

FIG. 56 is a perspective view of the vicinity of a conductive member in a state where the exposure head is placed in the elevating duct;

FIG. 57 is a right perspective view of a state where the exposure head is attached to the elevating duct;

FIG. 58 is a left perspective view of a state where the exposure head is attached to the elevating duct;

FIG. 59 is a cross-sectional view of a state where the exposure head is attached to the elevating duct;

FIG. 60 is an enlarged cross-sectional view of an engagement portion illustrated in FIG. 59;

FIG. 61 is a perspective view of the vicinity of the conductive member in a state where the exposure head is attached to the elevating duct;

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FIG. 62 is a right cross-sectional view illustrating a harness opening portion;

FIG. 63 is a right perspective view in a state where extra length processing of an FFC is performed;

FIG. 64 is a front cross-sectional view in a state where the extra length processing of the FFC is performed;

FIG. 65 is a front cross-sectional view illustrating a state of the FFC in a retracted position;

FIG. 66 is a cross-sectional view of the photosensitive drum, the exposure head, and an eccentric cam;

FIG. 67 is a cross-sectional view of the photosensitive drum, the exposure head, and the eccentric cam;

FIG. 68 is a cross-sectional view of the photosensitive drum, the exposure head, and the eccentric cam; and

FIG. 69 is a cross-sectional view of the photosensitive drum, the exposure head, and the eccentric cam.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, embodiments of the present invention will be described with reference to the drawings. The dimensions, materials, shapes, relative arrangements, and the like of the components described below are not intended to limit the scope of the present invention only thereto, unless otherwise specified.

(Image Forming Apparatus)

First, a schematic configuration of an image forming apparatus 100 will be described with reference to FIGS. 1, 2, and 3. FIG. 1 is a perspective view of the image forming apparatus 100. FIGS. 2 and 3 are schematic cross-sectional views of the image forming apparatus in FIG. 1. The image forming apparatus 100 illustrated in FIGS. 1 to 3 is a copying machine including a reading device, but the embodiment may be another image forming apparatus such as a printer not including a reading device. Further, the embodiment is not limited to the color image forming apparatus including a plurality of photosensitive drums 2 as illustrated in FIGS. 2 and 3, and may be a color image forming apparatus including one photosensitive drum 2 or an image forming apparatus forming a monochrome image.

The image forming apparatus 100 illustrated in FIGS. 2 and 3 includes four image forming portions 1Y, 1M, 1C, and 1K (hereinafter, also collectively and simply referred to as “image forming portion 1”) that form toner images of respective colors of yellow, magenta, cyan, and black.

The image forming portions 1Y, 1M, 1C, and 1K include photosensitive drums 2Y, 2M, 2C, and 2K (hereinafter, also collectively and simply referred to as a “photosensitive drum 2”), which are examples of photoconductors, respectively. The photosensitive drum 2 may be a photosensitive belt.

The image forming portions 1Y, 1M, 1C, and 1K include charging rollers 3Y, 3M, 3C, and 3K (hereinafter, also collectively and simply referred to as a “charging roller 3”) as charging portions that respectively charge the photosensitive drums 2Y, 2M, 2C, and 2K.

In addition, the image forming portions 1Y, 1M, 1C, and 1K include light emitting diode (LED, hereinafter referred to as LED) exposure heads 4Y, 4M, 4C, and 4K (hereinafter, also collectively and simply referred to as an “exposure head 4”) as exposure units that expose the photosensitive drums 2Y, 2M, 2C, and 2K.

Further, the image forming portions 1Y, 1M, 1C, and 1K include developing units 24Y, 24M, 24C, and 24K (hereinafter, also collectively and simply referred to as a “developing unit 24”) as developing units that develop the electrostatic latent image on the photosensitive drum 2 with toner and develop the toner image of each color on the

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photosensitive drum 2. Note that Y, M, C, and K attached to the reference numerals indicate the colors of the toner.

The image forming apparatus 100 illustrated in FIGS. 2 and 3 is an image forming apparatus adopting a so-called “lower surface exposure system” in which the photosensitive drum 2 is exposed from below, that is, the exposure head 4 is disposed below the photosensitive drum 2. Hereinafter, a description will be given on the premise of an image forming apparatus adopting a lower surface exposure system. Note that, although not illustrated, as an embodiment, an image forming apparatus adopting an “upper surface exposure system” in which the photosensitive drum is exposed from above may be used.

The image forming apparatus 100 includes an intermediate transfer belt 9 to which a toner image formed on the photosensitive drum 2 is transferred, and a primary transfer roller 6 (Y, M, C, and K) that sequentially transfers the toner image formed on the photosensitive drum 2 to the intermediate transfer belt 9. The intermediate transfer belt 9 is disposed above the image forming portion 1. In addition to the intermediate transfer method using the intermediate transfer belt 9, a direct transfer method of directly transferring from the photosensitive drum 2 to a sheet may be used.

In addition, the image forming apparatus 100 includes a secondary transfer roller 16 as a transfer portion that transfers the toner image on the intermediate transfer belt 9 onto a recording sheet P conveyed from a feeding portion 11, and a fixing device 19 as a fixing unit that fixes the secondarily transferred image onto the recording sheet P.

Toner bottles 22Y, 22M, 22C, and 22K (hereinafter, also collectively and simply referred to as a “toner bottle 22”) that accommodate replenishment toners of the respective colors are detachably replaceable units with respect to the image forming apparatus 100. The toner bottle 22 is disposed above the intermediate transfer belt 9. In the toner bottle 22, an appropriate amount of toner is appropriately supplied from the corresponding toner bottle to each developing unit included in the four image forming portions by a toner supply mechanism (not illustrated).

In addition, the image forming apparatus 100 includes the feeding portion 11 that feeds the recording sheet P. The feeding portion 11 includes sheet cassettes 12a and 12b, feeding rollers 13a and 13b, and a registration roller 15. The sheet cassettes 12a and 12b are disposed below the image forming portion 1. The recording sheets P accommodated in the sheet cassettes 12a and 12b are fed one by one by the feeding rollers 13a and 13b, and conveyed to a secondary transfer portion T2 at a predetermined timing by the registration roller 15.

(Image Forming Process)

Next, an image forming process of the image forming apparatus 100 will be briefly described. The charging roller 3Y charges the surface of the photosensitive drum 2Y. The exposure head 4Y exposes the surface of the photosensitive drum 2Y charged by the charging roller 3Y. As a result, an electrostatic latent image is formed on the photosensitive drum 2Y. Next, the developing unit 24Y develops the electrostatic latent image formed on the photosensitive drum 2Y with yellow toner. The yellow toner image developed on the surface of the photosensitive drum 2Y is transferred onto the intermediate transfer belt 9 by the primary transfer roller 6Y. Magenta, cyan, and black toner images are also formed by a similar image forming process, and are transferred so as to be superimposed on the intermediate transfer belt 9.

The toner image of each color transferred onto the intermediate transfer belt 9 is conveyed to the secondary transfer portion T2 by the intermediate transfer belt 9. The toner

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images conveyed to the secondary transfer portion T2 are collectively transferred to the recording sheet P conveyed from the feeding portion 11 by the secondary transfer roller 16. The recording sheet P to which the toner image has been transferred is conveyed to the fixing device 19. The fixing device 19 fixes the toner image on the recording sheet P by heat and pressure. The recording sheet P subjected to the fixing process by the fixing device 19 is discharged to a discharge tray 21 disposed above the toner bottle 22 by a discharge roller 20.

(Drum Unit and Developing Unit)

A replaceable drum unit 23 and the developing unit 24 in the image forming apparatus 100 of the present embodiment will be described by way of example.

The photosensitive drum 2 and the charging roller 3 described above are integrally unitized (drum unit and drum cartridge) together with a cleaning device (not illustrated). An example of the configuration will be described with reference to FIGS. 4, 5, 6, and 7. FIGS. 4 and 5 are perspective views illustrating schematic structures around the drum unit 23 (Y, M, C, and K) and around the developing unit 24 (Y, M, C, and K) included in the image forming apparatus 100. FIG. 6 is a view illustrating how the drum unit 23 is inserted into and removed from the image forming apparatus 100 from the outside of the apparatus main body. FIG. 7 is a view illustrating how the developing unit 24 is inserted into and removed from the image forming apparatus 100 from the outside of the apparatus main body.

Drum units 23Y, 23M, 23C, and 23K (hereinafter, also collectively and simply referred to as a “drum unit 23”) including the photosensitive drum 2 are attached to the image forming apparatus 100. The drum unit 23 is a cartridge to be replaced by an operator such as a user or a maintenance person. The drum unit 23 rotatably supports the photosensitive drum 2. Specifically, the photosensitive drum 2 is rotatably supported by the frame body of the drum unit 23. The drum unit 23 may not include the charging roller 3 or the cleaning device.

In addition, developing units 24Y, 24M, 24C, and 24K (hereinafter, also collectively and simply referred to as a “developing unit 24”) separate from the drum unit 23, which is a photoconductor unit, are attached to the image forming apparatus 100. The developing unit 24 includes developing sleeves 5Y, 5M, 5C, and 5K (hereinafter, also collectively and simply referred to as a “developing sleeve 5”) as developer carriers that carry the developer, and screws 7Y, 7M, 7C, and 7K (hereinafter, also collectively and simply referred to as a “screw 7”) that supply the developer to the developing sleeve 5 and agitate the developer. The developing unit 24 is a cartridge in which the developing sleeve 5 and the screw 7 are integrated, and is detached from the apparatus main body of the image forming apparatus 100 and replaced by an operator as illustrated in FIGS. 5 and 7.

Here, the toner is circulated and conveyed at high speed by the screw 7 inside the developing unit 24. The rotation speed of the screw 7 is relatively very high with respect to the rotation speed of the developing sleeve 5 and the photosensitive drum 2, and coating on the developing sleeve 5 can be uniformly performed without unevenness.

In addition, the image forming apparatus 100 includes a cartridge tray 30 (30Y, 30M, 30C, and 30K) for each image forming portion (see FIGS. 8 and 9). The drum unit 23 and the developing unit 24 are supported by the cartridge tray 30 of each image forming portion, guided in the axial direction of the photosensitive drum, and inserted into and removed from the apparatus main body of the image forming apparatus 100.

In addition, the image forming apparatus **100** includes a front plate **100F** formed of a sheet metal and a back plate **100B** similarly formed of a sheet metal (see FIG. **34**). The front plate **100F** is a side wall provided on the front side of the image forming apparatus **100**. The front plate **100F** forms a part of a housing of the apparatus main body on the front side of the apparatus main body of the image forming apparatus **100**. The back plate **100B** is a side wall provided on the back side of the image forming apparatus **100**. The back plate **100B** forms a part of the housing of the apparatus main body on the back side of the apparatus main body of the image forming apparatus **100**. The front plate **100F** and the back plate **100B** are arranged to face each other on one side and the other side in the axial direction of the photosensitive drum, and a sheet metal (not illustrated) as a beam is bridged therebetween. Each of the front plate **100F**, the back plate **100B**, and the beam (not illustrated) constitutes a part of a frame body (housing) of the image forming apparatus. Here, regarding the image forming apparatus of the present embodiment or the constituent members thereof, the forward side or the front side is a side on which the drum unit **23** and the developing unit **24** are taken in and out (inserted and removed) with respect to the apparatus main body of the image forming apparatus **100**.

One ends of the cartridge trays **30** are attached to the front plate **100F** (see FIG. **34**) and the other ends are attached to the back plate **100B** (see FIG. **34**) in the axial direction of the photosensitive drum. The cartridge tray **30** will be described later.

Since the drum unit **23** and the developing unit **24** deteriorate due to repetition of the image forming process, they take the form of a unit (cartridge) that can be maintained by replacement or attachment or detachment.

FIG. **3** illustrates the arrangement of the drum unit **23**, the developing unit **24**, and the exposure head **4** when replacement or attachment or detachment is performed. In the image forming apparatus illustrated in FIG. **3**, unlike the image forming apparatus illustrated in FIG. **2**, it can be seen that the developing unit **24** and the exposure head **4** are retracted and separated from the photosensitive drum **2**.

This is because, when the state in which the developing unit **24** and the exposure head **4** are disposed close to the photosensitive drum **2** as illustrated in FIG. **2** is maintained, each unit may be damaged due to dynamic interference at the time of attachment and detachment of the unit, or the unit may not be taken out.

Therefore, at the time of attaching and detaching the unit, the developing unit **24** and the exposure head **4** are retracted from the photosensitive drum **2** and separated from each other as illustrated in FIG. **3** by a retraction mechanism by a development stay **31**, a rotating arm **65**, an elevating duct **69**, and the like, which will be described later.

The drum unit **23** and the developing unit **24** are inserted and removed from the front side of the image forming apparatus **100**, and are mounted at predetermined positions (mounting positions) of the apparatus main body of the image forming apparatus **100**.

The image forming apparatus **100** includes inner doors **102Y**, **102M**, **102C**, and **102K** (hereinafter, also collectively and simply referred to as an "inner door **102**") that cover front sides of both the drum unit **23** and the developing unit **24** mounted at the mounting positions. As illustrated in FIGS. **8** and **9**, one end of the inner door **102** is fixed to the front side of the cartridge tray **30** by a hinge, and is rotatable with respect to the cartridge tray **30** by the hinge.

The inner door **102** is a member necessary for protecting each unit and making it difficult for the photosensitive drum

2 to be exposed to light in a process other than the image forming process, and is disposed at a position facing the front in the attachment and detachment direction of each color unit.

Furthermore, a front cover **101** forming an exterior of the apparatus is provided on the front side of the image forming apparatus **100**. One end of the front cover **101** is fixed to the front side of the apparatus main body of the image forming apparatus **100** by a hinge, and is rotatable with respect to the apparatus main body of the image forming apparatus **100** by the hinge. The front cover **101** is provided on the front side of the inner door **102** in the axial direction of the photosensitive drum. In the closed state illustrated in FIG. **1**, the front cover **101** covers the entire plurality of inner doors **102** arranged in the left-right direction to form an exterior on the front side of the apparatus.

Therefore, replacement work of the drum unit **23** and the developing unit **24** is performed by an operator in the following procedure. The operator opens the front cover **101** as illustrated in FIG. **4**, then opens the inner door **102** as illustrated in FIG. **5**, and takes out the drum unit **23** (FIG. **6**) or the developing unit **24** (FIG. **7**) in the apparatus main body. Then, the replacement work is completed by inserting a new drum unit **23** or developing unit **24**, closing the inner door **102**, and further closing the front cover **101**.

The retraction mechanism of the developing unit **24** and the exposure head **4** retracts the developing unit **24** and the exposure head **4** from the photosensitive drum **2** in conjunction with the operation of opening the inner door **102**. The retraction mechanism (the development stay **31**, the rotating arm **65**, and the elevating duct) will be described later.

Here, in the following description, the front plate side is defined as a front side (front side or forward side), and the back plate side is defined as a back side (back side or backward side) with respect to the apparatus main body. When the photosensitive drum **2K** on which the electrostatic latent image related to the black toner image is formed is used as a reference, a side on which the photosensitive drum **2Y** on which the electrostatic latent image related to the yellow toner image is formed is disposed is defined as a left side. When the photosensitive drum **2Y** on which the electrostatic latent image related to the yellow toner image is formed is used as a reference, a side on which the photosensitive drum **2K** on which the electrostatic latent image related to the black toner image is formed is disposed is defined as a right side. Furthermore, a direction perpendicular to the front-back direction and the left-right direction defined here and upward in the vertical direction is defined as an upward direction, and a direction perpendicular to the front-back direction and the left-right direction defined here and downward in the vertical direction is defined as a downward direction. The defined forward direction F, backward direction B, rightward direction R, leftward direction L, upward direction U, and downward direction D are illustrated in FIG. **1**.

In addition, the axial direction of the photosensitive drum **2** described in the following description is a direction that coincides with the front-back direction (near-far direction) illustrated in FIG. **1**. The longitudinal direction of the exposure head **4** also coincides with the front-back direction illustrated in FIG. **1**. That is, the axial direction of the photosensitive drum **2** and the longitudinal direction of the exposure head **4** coincide with each other. In addition, one end side in the axial direction of the photosensitive drum **2** means a front side defined herein, and the other end side means a back side defined herein. One end side and the other end side in the front-back direction also correspond to the

front side and the back side defined here. One end side in the left-right direction means the left side defined here, and the other end side means the right side defined here. (Exposure Head)

Next, the exposure head 4 will be described with reference to FIGS. 10 to 19. FIG. 10 is a schematic cross-sectional view of the exposure head 4 included in the image forming apparatus of the present embodiment. FIG. 11 is a perspective view of the exposure head 4 as viewed from above. FIG. 12 is a perspective view of the exposure head 4 as viewed from below.

The exposure head 4 has an elongated shape (longitudinal shape) extending in the axial direction of the photosensitive drum 2. The exposure head 4 includes a substrate 50, a light emitting element mounted on the substrate 50, a lens array 52, and a holding member that holds the substrate 50 and the lens array 52. The holding member includes a housing 54 to be described later and a housing support member 55 that supports the housing 54. Here, the exposure head 4 includes an LED (Light Emitting Diode) 51 as a light emitting element that emits light.

(Substrate and Lens Array)

Here, the substrate 50 and the lens array 52 of the exposure head 4 will be described with reference to FIGS. 13A to 13E, 14, and 15. First, the substrate 50 will be described. FIG. 13A is a schematic perspective view of the substrate 50. FIG. 13B illustrates an alignment of the plurality of LEDs 51 provided on the substrate 50, and FIG. 13C illustrates an enlarged view of FIG. 13B. FIG. 14 is a perspective view of the substrate as viewed from the surface (substrate front surface) on which the LED is mounted. FIG. 15 is a perspective view of the substrate as viewed from the surface (substrate back surface) on which the FFC connector is mounted. One side of the arrow in the drawing indicates the front side of the image forming apparatus 100, and the other side indicates the back side of the image forming apparatus 100.

An LED chip 53 is mounted on the substrate 50. As illustrated in FIGS. 13A, 14, and 15, the LED chip 53 is provided on one surface of the substrate 50, and a long FFC connector 57 is provided on the other surface. Here, one surface of the substrate 50 is a surface (upper surface, front surface) on a side where the LED chip 53 is provided. The other surface of the substrate is a surface (lower surface, back surface) opposite to the side where the LED chip 53 is provided.

The FFC connector 57 is attached to the other surface (lower surface, back surface) of the substrate 50 such that the longitudinal direction thereof is along the longitudinal direction of the substrate 50. The long FFC connector 57 is provided on the front side of the image forming apparatus 100 (one side in the longitudinal direction of the substrate 50). Wiring for supplying a signal to each LED chip 53 is provided on the substrate 50. One end of a flexible flat cable 58 (See FIG. 26, hereinafter FFC) as an example of a cable is connected to the FFC connector 57.

Note that a control circuit portion of the image forming apparatus 100 is provided with a substrate (not illustrated) including a controller and a connector. The other end of the FFC 58 is connected to the connector. That is, the FFC 58 electrically connects the substrate (control circuit portion) of the apparatus main body and the substrate 50 of the exposure head 4. A control signal (drive signal) is input to the substrate 50 of the exposure head 4 from the control circuit portion of the apparatus main body of the image forming apparatus 100 via the FFC 58 and the FFC connector 57. The control signal is transferred to each LED chip 53. The LED

chip 53 is driven (light emission and turn-off operation) by the control signal input to the substrate 50.

The LED chip 53 mounted on the substrate 50 will be described in more detail. As illustrated in FIGS. 13B and 13C, LED chips 53-1 to 53-29 (29 pieces) in which the plurality of LEDs 51 (an example of the light emitting element) are arranged are aligned on one surface of the substrate 50. In each of the LED chips 53-1 to 53-29, 516 LEDs 51 are aligned in the longitudinal direction. In the longitudinal direction of the LED chip 53, the center-to-center distance k2 between the adjacent LEDs 51 corresponds to the recording resolution of the image forming apparatus 100. Since the recording resolution of the image forming apparatus 100 of the present embodiment is 1200 dpi, the LEDs 51 are aligned such that the center-to-center distance k2 of the adjacent LEDs 51 is 21.16 μ m in the longitudinal direction of the LED chips 53-1 to 53-29. Therefore, the exposure range of the exposure head 4 of the present embodiment is about 314 mm. The length of the photosensitive layer in the axial direction of the photosensitive drum 2 is 314 mm or more. Since the length of the long side of the A4-size recording sheet and the length of the short side of the A3-size recording sheet are 297 mm, the exposure head 4 of the present embodiment has an exposure range in which an image can be formed on the A4-size recording sheet and the A3-size recording sheet.

The LED chips 53-1 to 53-29 are aligned in a staggered manner in the axial direction of the photosensitive drum 2. Specifically, the LED chips 53-1 to 53-29 are alternately arranged in two rows along the axial direction of the photosensitive drum 2. That is, as illustrated in FIG. 13B, the odd-numbered LED chips 53-1, 53-3, . . . 53-29 counted from the left side are mounted in a row in the longitudinal direction of the substrate 50. Further, the even-numbered LED chips 53-2, 53-4, . . . 53-28 counted from the left side are mounted in a row in the longitudinal direction of the substrate 50. The LED chip 53 is arranged in this manner. As a result, as illustrated in FIG. 13C, in the longitudinal direction of the LED chip 53, the center-to-center distance k1 of the LEDs 51 arranged at one end of one LED chip 53 and the other end of the other LED chip 53 in the different adjacent LED chips 53 can be made equal to the center-to-center distance k2 of the adjacent LEDs 51 on one LED chip 53.

In the present embodiment, the light emitting element is a semiconductor LED which is a light emitting diode, but may be, for example, an organic light emitting diode (OLED). This OLED is also called organic electro-luminescence (organic EL), and is a current-driven light emitting element. The OLEDs are arranged on a line along the main scanning direction (axial direction of the photosensitive drum 2) on a thin film transistor (TFT) substrate, for example, and are electrically connected in parallel by power supply wiring similarly provided along the main scanning direction.

Next, the lens array 52 which is a lens assembly will be described. FIG. 13D is a schematic view of the lens array 52 as viewed from the photosensitive drum 2 side. FIG. 13E is a schematic perspective view of the lens array 52. As illustrated in FIG. 13D, the lens array 52 condenses the light emitted from the light emitting element on the photosensitive drum 2. The lens array 52 is a lens assembly including a plurality of lenses. The plurality of lenses are arranged in two rows along the alignment direction of the plurality of LEDs 51. Each lens is alternately arranged such that one of the lenses in the other row is arranged so as to be in contact with both of the adjacent lenses in the alignment direction of

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the lenses in one row. Each lens is a cylindrical rod lens made of glass, and has a light incident surface **52b** on which the light emitted from the LED **51** is incident and a light exit surface **52a** from which the light incident from the light incident surface is emitted (see FIG. 10). The material of the lens is not limited to glass, and may be plastic. The shape of the lens is not limited to the cylindrical shape, and may be, for example, a polygonal prism such as a hexagonal prism.

A dotted line **Z** illustrated in FIG. 13E indicates the optical axis of the lens. The exposure head **4** is moved in a direction substantially along the optical axis of the lens (hereinafter, also referred to as an optical axis direction) indicated by the dotted line **Z** by a retraction mechanism (the rotating arm **65** and the elevating duct **69** in FIGS. 24 and 25) to be described later. The optical axis of the lens here means a line connecting the center of the light exit surface of the lens and the focal point of the lens. The lens array **52** is a lens assembly having a plurality of lenses, and the "optical axis" described above is an optical axis of any lens among the plurality of lenses. Here, strictly speaking, the plurality of lenses included in the lens array **52** may be slightly inclined to each other. This is due to tolerance during assembly. However, the deviation of the tolerance is not considered when the direction of the optical axis is defined. Therefore, it is considered that the optical axes of the plurality of lenses are in the same direction. The lens array **52** has a function of condensing the light emitted from the LED **51** on the surface of the photosensitive drum **2**.

Amounting position of the lens array **52** with respect to the housing **54** is adjusted at the time of assembling the exposure head **4** such that a distance between the light emitting surface of the LED **51** and the light incident surface of the lens is substantially equal to a distance between the light exit surface of the lens and the surface of the photosensitive drum **2**.

(Housing)

As illustrated in FIG. 10, the housing **54** holds the lens array **52** and the substrate **50**. In the present embodiment, the housing **54** is a metal member formed by bending a plate material obtained by plating a galvanized steel plate or a cold-rolled steel plate.

The housing **54** is made of metal as described above. For example, the housing **54** is formed by pressing a sheet metal such as an iron thin plate into a U shape. Hereinafter, the shape of the housing **54** will be described.

As illustrated in FIG. 10, the housing **54** has a flat portion (opposing surface) **54U** in which a first opening **54a** into which the lens array **52** is inserted is formed. The flat portion **54U** faces the photosensitive drum **2** in the optical axis direction of the lens of the lens array **52**. Note that the flat portion **54U** is not limited to a flat surface, and may be a slightly curved surface. The housing **54** has an extending portion **54R** extending in a direction away from the photosensitive drum **2** from one side in the lateral direction of the flat portion **54U**. The housing **54** has an extending portion **54L** extending in a direction away from the photosensitive drum **2** from the other side in the lateral direction of the flat portion **54U**.

The extending portion **54R** and the extending portion **54L** form a substrate support portion for supporting the substrate **50** inserted from a second opening **54b** in the housing **54**. The flat portion **54U** and the substrate support portion (the extending portions **54R** and **54L**) are integrated to form the housing **54** that holds the lens array **52** and the substrate **50**, and a cross section thereof is formed in a substantially U-shape. Since the housing **54** is formed in a substantially U shape, the second opening **54b** is formed on the side

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opposite to the flat portion **54U**. The second opening **54b** is formed between the substrate support portions (the extending portions **54L** and **54R**) extending from the flat portion **54U** to the side away from the photosensitive drum.

The substrate **50** is inserted from the second opening **54b**, that is, from the lower side of the U-shaped housing **54**, and is adhered to the inside of each substrate support portion (the inside of the extending portion **54L** and the inside of the extending portion **54R**) with an adhesive. Since the position of the substrate **50** in the focusing direction is determined by a jig (not illustrated), the exposure head **4** does not include a positioning unit in the focusing direction of the substrate **50**.

In addition, the lens array **52** is also adhered to the flat portion **54U** with an adhesive in a state of being inserted into the first opening **54a** formed in the flat portion **54U**. Note that the lens array **52** is fixed to the flat portion **54U** (housing **54**) after the position and inclination in the focusing direction are adjusted by a jig such that the distance in the focusing direction between all the LED chips **53** and the lens array **52** mounted on the substrate **50** becomes a predetermined value. The lens array **52** is fixed to the flat portion **54U** at a plurality of positions in the longitudinal direction by an adhesive. That is, the exposure head **4** of the present embodiment has a plurality of adhering locations for adhering and fixing the lens array **52** inserted into the first opening **54a** to the flat portion **54U** in the longitudinal direction of the flat portion **54U**.

After the substrate **50** and the lens array **52** are positionally adjusted and fixed to the housing **54**, a gap between the substrate **50** inserted into the second opening **54b** and the housing **54** (the extending portions **54L** and **54R**) is sealed by a sealant **59** in the longitudinal direction as illustrated in FIGS. 16 and 17. FIG. 16 is a perspective view of a state in which the substrate **50** on which the LEDs **51** are mounted is assembled to the housing **54** as viewed from the lower side of the exposure head **4**. FIG. 17 is an enlarged view of the front side of the exposure head illustrated in FIG. 16. This prevents the LED **51** from being contaminated by toner and dust from the outside. Here, the sealant **59** merely seals the gap (boundary portion) between the substrate **50** and the housing **54**, and most of the FFC connector **57** and the substrate **50** are exposed.

Similarly, the sealant **59** is applied to a gap between the lens array **52** inserted into the first opening **54a** and the housing **54** (flat portion **54U**), and the gap is sealed by the sealant **59** in the longitudinal direction as illustrated in FIG. 18. FIG. 18 is a perspective view of the exposure head **4** in a state where the lens array **52** is assembled to the housing **54**. More specifically, as illustrated in FIG. 10, the sealant **59** seals a gap between the side wall of the lens array **52** and the edge of the first opening **54a** along the longitudinal direction of the housing **54**. As a result, it is possible to reduce the possibility that dust such as toner flows in from the gap between the side wall of the lens array **52** and the first opening **54a** and the light emitted from the LED **51** is blocked by the dust. Note that, as a matter of course, the gap sealed by the sealant **59** seals not only the gap between the side wall on one side of the lens array **52** and the edge of the first opening **54a**, but also the gap between the side wall on the other side of the lens array **52** and the edge of the first opening **54a**. The side wall on the other side of the lens array **52** refers to a side wall opposite to the side wall on one side of the lens array **52**. Also here, a gap (boundary portion) between the housing **54** and the lens array **52** is sealed with the sealant **59**. This prevents the LED **51** from being contaminated by toner and dust from the outside.

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As described above, the substrate **50** and the lens array **52** are held by the housing **54**, so that the LED **51** and the incident surface **52b** of the lens face each other. As a result, the light emitted from the LED **51** is incident on the incident surface **52b** of the lens, and is emitted from the exit surface **52a** of the lens toward the photosensitive drum **2**. Here, in the present embodiment, the light emitted from the three LEDs **51** (the plurality of LEDs **51**) can pass through the same one lens. In addition, even light emitted from one LED **51** can pass through a plurality of lenses because the light travels radially. That is, the light emitted from the plurality of LEDs **51** passes through the lens array **52** (some of the plurality of lenses included in the lens array **52**) to expose the photosensitive drum **2**.

(Housing Support Member)

As illustrated in FIGS. **11** and **12**, the housing support member **55** supports the housing **54** holding the substrate **50** and the lens array **52** in the longitudinal direction, and is provided integrally with the housing **54**. The housing support member **55** is a member having a longitudinal shape extending in the axial direction of the photosensitive drum **2**. The housing support member **55** is formed in a U shape as illustrated in FIG. **10**. The housing support member **55** includes a left side wall **55L** which is a first side wall, a right side wall **55R** which is a second side wall facing the left side wall **55L**, and a bottom surface portion **55D** facing the flat portion **54U** of the housing **54** between the left side wall **55L** and the right side wall **55R**. The bottom surface portion **55D** of the housing support member **55** is provided with a plurality of openings **55a** in the longitudinal direction which is the axial direction of the photosensitive drum **2**.

The opening **55a** of the housing support member **55** is provided at a position facing a surface (back surface of the substrate **50**) of the substrate **50** opposite to the mounting surface (front surface of the substrate **50**) on which the LED **51** is mounted. The opening **55a** is provided between the left side wall **55L** and the right side wall **55R** in the lateral direction orthogonal to the longitudinal direction.

As illustrated in FIG. **22**, the left side wall **55L** which is the first side wall is provided between the housing **54** that holds the substrate **50** and the developing unit **24** which is a developing unit on one side in the lateral direction orthogonal to the axial direction of the photosensitive drum **2**. As illustrated in FIG. **11**, the left side wall **55L** is provided across the axial direction of the photosensitive drum **2** so as to separate the housing **54** and the developing unit **24**.

As illustrated in FIG. **22**, the right side wall **55R** which is the second side wall is provided between the housing **54** and the drum unit **23** which is an adjacent portion adjacent to the housing **54** on the other side in the lateral direction. Similarly to the left side wall, the right side wall **55R** is provided across the axial direction of the photosensitive drum **2** so as to separate the housing **54** and the drum unit **23**.

By providing the housing support member **55** integrally with the housing **54**, the airflow sent from a duct unit **60** described later is blown to the back surface of the substrate **50** through the opening **55a** between the left side wall **55L** and the right side wall **55R** of the housing support member **55**. Moreover, the airflow blown onto the back surface of the substrate **50** is blown in a direction orthogonal to the back surface of the substrate **50**.

As described above, the airflow blown from the opening **55a** of the housing support member **55** to the back surface of the substrate **50** is separated from the developing unit **24** adjacent to the exposure head **4** by the left side wall **55L** and separated from the drum unit **23** by the right side wall **55R**. Therefore, the airflow for cooling the exposure head **4**

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introduced into the back surface of the substrate **50** does not leak to the side of the developing unit **24** adjacent to the exposure head **4**, and the toner of the developing unit **24** can be prevented from scattering inside the image forming apparatus.

(Engagement Claw)

As illustrated in FIGS. **11** and **12**, the exposure head **4** includes an engagement claw **55b** and an engagement claw **55c** which are first engagement portions. The engagement claw **55b** and the engagement claw **55c** are provided on the housing support member **55** of the exposure head **4**, and engage with the elevating duct **69** by snap-fitting.

In the housing support member **55** of the exposure head **4**, a bottom surface portion **55D** between the opening **55a** at one end (front side) in the longitudinal direction and the opening **55a** adjacent thereto is defined as a first bottom surface portion **55D1**. Similarly, the bottom surface portion **55D** between the opening **55a** at the other end (back side) in the longitudinal direction and the opening **55a** adjacent thereto is defined as a second bottom surface portion **55D2**. The engagement claw **55b** that engages with the elevating duct **69** is provided on the lower surface of the first bottom surface portion **55D1**. The engagement claw **55c** that engages with the elevating duct **69** is provided on the lower surface of the second bottom surface portion **55D2**. The first bottom surface portion **55D1** and the second bottom surface portion **55D2**, that is, the bottom surface portion **55D**, are facing portions facing an upper surface portion **69U** of the elevating duct **69**.

The engagement claws **55b** and **55c** as the first engagement portions are formed on the bottom surface portion **55D** so as to protrude toward the elevating duct **69** in the moving direction of the elevating duct **69** described later, and are further formed to extend in the axial direction of the photosensitive drum **2** orthogonal to the protruding direction.

Specifically, the engagement claws **55b** and **55c** of the exposure head **4** are formed so as to protrude toward the elevating duct **69**, and have a substantially L shape formed so as to extend in the axial direction of the photosensitive drum **2** orthogonal to the protruding direction. As will be described later, by sliding the exposure head **4**, the substantially L-shaped claw ends of the engagement claws **55b** and **55c** are engaged with the edges of the engagement holes **69b** and **69c** by snap-fitting and integrated with the elevating duct **69**.

(Shielding Wall)

The housing support member **55** includes a shielding wall **76**. The shielding wall **76** will be described with reference to FIG. **19**. FIG. **19** is a perspective view of the exposure head **4** in which the housing **54** and the housing support member **55** are integrally formed as viewed from below.

The housing support member **55** includes the shielding wall **76** for partitioning a connector region where the FFC connector **57** is disposed on the back surface of the substrate **50** and a duct region where the opening **55a** on the back side in the front-back direction is present from the connector region for the reason described later.

The connector **57** is disposed at a position deviated to one end side from the center in the longitudinal direction of the substrate **50**. On the other hand, a plurality of openings **55a** is formed further on the other end side than the connector **57** in the longitudinal direction of the substrate **50**. The shielding wall **76** is provided so as to partition the opening **55a** and the connector **57**.

The shielding wall **76** is provided on the bottom surface portion **55D** of the housing support member **55**. The shielding wall **76** is provided on the surface side of the bottom

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surface portion 55D facing the back surface of the substrate 50 so as to protrude in the direction of the back surface of the substrate 50. The shielding wall 76 is provided between the left side wall 55L and the right side wall 55R of the housing support member 55. The shielding wall 76 partitions a duct region on the back surface of the substrate 50 held by the housing 54 and a connector region of the FFC connector 57 mounted on the substrate 50 between the left side wall 55L and the right side wall 55R. Here, the duct region on the back surface of the substrate 50 is a region facing the opening 55a provided in the bottom surface portion 55D of the housing support member 55, and is a region (range La illustrated in FIG. 20) communicated with the opening portion 61 of the duct unit 60 through a closed space formed by the elevating duct 69 and the cartridge tray 30 to be described later. The connector region mounted on the back surface of the substrate 50 is a region on the back surface of the substrate where the FFC connector 57 is mounted, is a region outside the duct region (range La) in the axial direction of the photosensitive drum 2, is a region further on the front side than the duct region in the front-back direction, and is a region indicated by a range Lc in FIG. 20.

As described above, the shielding wall 76 that partitions the range La (duct region) and the range Lc (connector region) illustrated in FIG. 20 is provided in the housing support member 55. As a result, the air blown to the back surface of the substrate 50 held by the housing 54 from the opening 55a of the housing support member 55 is prevented from leaking in the direction of the FFC connector 57 and the cooling capacity for the substrate 50 is prevented from deteriorating.

The shielding wall 76 partitions the duct region and the connector region such that the connector region is outside the duct region and on one end side in the axial direction of the photosensitive drum 2. Further, the shielding wall 76 partitions the duct region and the connector region such that the connector region is on the front side in the front-back direction of the image forming apparatus 100. Therefore, the FFC connector 57 is disposed as far forward as possible in order to make the range La as long as possible. As a result, the air blown to the back surface of the substrate 50 is blocked by the FFC 58 connected to the FFC connector 57, and is prevented from flowing in an unintended direction. In other words, the air blown toward the exposure head 4 from the duct unit 60 to be described later is blown to the back surface of the substrate 50 without being blocked by the FFC 58 connected to the FFC connector 57. The air blown against the back surface of the substrate 50 tends to flow in the space between the left side wall 55L and the right side wall 55R of the housing support member 55 along the longitudinal direction of the substrate 50. At this time, the airflow in the direction toward the connector region is blocked by the shielding wall 76, and the airflow blown against the back surface of the substrate 50 flows through the duct region from one side (front side) to the other side (back side). Therefore, the airflow blown from the duct unit 60 to be described later to the back surface of the substrate 50 can flow in an intended direction, and scattering of toner in the image forming apparatus due to the airflow flowing in an unintended direction can be prevented.

In FIG. 26, a range Li and a range Lo are ranges when the range La that is the duct region is divided into two equal parts of the intake side and the exhaust side. The range Li is a range on the intake side when the range La is divided into two equal parts of the intake side and the exhaust side. The range Lo is a range on the exhaust side when the range La is divided into two equal parts of the intake side and the

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exhaust side. A range Ls is a range in which the cross-sectional area of the airflow in the duct is locally narrowed in the range Li on the intake side in the range La that is the duct region.

As described above, the exposure head 4 is configured as an integrated head unit by the substrate including the LED, the lens array including the plurality of lenses, the housing 54, and the housing support member 55.
(Elevating Duct)

The image forming apparatus 100 includes the elevating duct 69. The elevating duct 69 will be described with reference to FIGS. 22, 23, 27, and 28. FIGS. 22 and 23 are cross-sectional views taken along line X-X in FIG. 20. FIGS. 27 and 28 are perspective views of the exposure head 4, the elevating duct 69, and the rotating arm 65.

The elevating duct 69 is an exposure support member that detachably supports the exposure head 4, and is provided in the apparatus main body of the image forming apparatus 100 together with a cartridge tray 30 to be described later.

The elevating duct 69 is provided between a developing support member 301 that supports the developing unit 24 of the cartridge tray 30 to be described later and a drum support member 302 that supports the drum unit 23. The elevating duct 69 is provided to be movable between an exposure position (see FIGS. 22 and 24) at which the photosensitive drum 2 is exposed and a retracted position (see FIGS. 25 and 23) retracted from the exposure position between the developing support member 301 and the drum support member 302 of the cartridge tray 30. Both end portions of the elevating duct 69 in the longitudinal direction are supported from below by the rotating arm 65 to be described later. The elevating duct 69 is moved in a direction (first direction, moving direction) orthogonal to the axial direction of the photosensitive drum 2 integrally with the exposure head 4 by the rotating arm 65. The elevating duct 69 is moved to the exposure position or the retracted position by the rotation of the rotating arm 65.

The elevating duct 69 has a longitudinal shape extending in the front-back direction (the axial direction of the photosensitive drum) similarly to the exposure head 4 so as to be able to support the entire exposure head 4, and a central portion thereof has a shape having openings in the vertical direction. The elevating duct 69 forms a duct in which one opening 69a communicates with the opening 55a of the exposure head 4 and the other opening portion 64 communicates with the opening portion 61 of the duct unit 60. The elevating duct 69 forms a part of a duct that cools the exposure head 4 while supporting the exposure head 4.

The elevating duct 69 has an upper surface portion 69U (see FIG. 26) facing the bottom surface portion 55D of the housing support member 55. The upper surface portion 69U of the elevating duct 69 is provided with a plurality of openings 69a in the longitudinal direction which is the axial direction of the photosensitive drum 2. The elevating duct 69 includes a duct left wall 69L, a duct right wall 69R facing the duct left wall 69L, a duct front wall 69F, and a duct back wall 69B facing the duct front wall 69F. The duct left wall 69L, the duct right wall 69R, the duct front wall 69F, and the duct back wall 69B are provided integrally with the upper surface portion 69U so as to surround the periphery of the opening 69a provided in the upper surface portion 69U. As a result, the elevating duct 69 has a shape in which the upper surface portion 69U, the duct left wall 69L, the duct right wall 69R, the duct front wall 69F, and the duct back wall 69B are integrally formed and an opening is provided in the vertical direction. The elevating duct 69 forms a duct (closed

space) that allows an airflow from the duct unit **60** described later to flow to the exposure head **4** through between the duct walls and the opening **69a** of the upper surface portion **69U**.

The plurality of openings **69a** is provided at positions facing the plurality of openings **55a** provided in the bottom surface portion **55D** of the housing support member **55**, respectively. In other words, similarly to the opening **55a** of the housing support member **55**, the opening **69a** of the elevating duct **69** is provided at a position facing the back surface of the substrate **50**. In other words, the elevating duct **69** has the opening **69a** communicating with the opening **55a** of the exposure head **4**.

The duct left wall **69L**, which is the first duct wall, is provided on the left side, which is one side in the lateral direction orthogonal to the axial direction of the photosensitive drum **2**, through the opening **69a**. That is, the duct left wall **69L** is provided along the axial direction of the photosensitive drum **2** from the duct front wall **69F** to the duct back wall **69B** at a position facing the developing unit **24** as a developing unit.

The duct left wall **69L** has a first inclined surface **69L1** inclined in a direction away from the duct right wall **69R** from the upstream side to the downstream side in the moving direction from the exposure position to the retracted position. The first inclined surface **69L1** is similarly inclined from the duct front wall **69F** to the duct back wall **69B**. The first inclined surface **69L1** is provided at a position facing the developing support member **301** on the duct left wall **69L**, and forms a predetermined angle $\theta 2$ with respect to the moving direction (direction of arrow G illustrated in FIG. **22**).

The duct right wall **69R**, which is the second duct wall, is provided on the right side, which is the other side in the lateral direction, through the opening **69a**. That is, the duct right wall **69R** is provided along the axial direction of the photosensitive drum **2** from the duct front wall **69F** to the duct back wall **69B** at a position facing the drum unit **23** which is an adjacent portion adjacent to the elevating duct **69**.

The duct right wall **69R** has a second inclined surface **69R1** inclined in a direction away from the duct left wall **69L** from the upstream side to the downstream side in the moving direction from the exposure position to the retracted position. The second inclined surface **69R1** is similarly inclined from the duct front wall **69F** to the duct back wall **69B**. The second inclined surface **69R1** is provided at a position facing the drum support member **302** on the duct right wall **69R**, and forms a predetermined angle $\theta 1$ with respect to the moving direction (direction of arrow G illustrated in FIG. **22**).

The duct front wall **69F**, which is a third duct wall, is provided outside the plurality of openings **69a** provided in the upper surface portion **69U** in the axial direction of the photosensitive drum.

The duct back wall **69B**, which is a fourth duct wall, is provided outside the plurality of openings **69a** provided in the upper surface portion **69U** in the axial direction of the photosensitive drum.

The plurality of openings **69a** provided in the upper surface portion **69U** of the elevating duct **69** are provided between the duct left wall **69L** and the duct right wall **69R** and between the duct front wall **69F** and the duct back wall **69B**.

In other words, the elevating duct **69** is formed in a hollow shape having no opening at a position facing the developing unit **24** and the drum unit **23** and having an opening in the vertical direction by these duct walls.

Accordingly, the elevating duct **69** allows an air flow generated by the duct unit **60** described later to flow to the back surface of the substrate **50** of the exposure head **4** through the opening portion **64** described later and the opening **69a** between the duct left wall **69L** and the duct right wall **69R**. Therefore, the elevating duct **69** can allow the airflow from the duct unit **60** to flow to the back surface of the substrate **50** of the exposure head **4** without leaking the airflow to the side of the adjacent developing unit **24** or drum unit **23**, and the scattering of the toner inside the apparatus can be reduced.

Further, the duct front wall **69F** and the duct back wall **69B** extend downward further than the duct left wall **69L** and the duct right wall **69R** in the moving direction. In other words, the duct front wall **69F** and the duct back wall **69B** are ribs protruding in the direction of the duct unit **60** described later on both outer sides in the longitudinal direction of the opening **69a** communicating with the exposure head **4**.

The elevating duct **69** integrally supports the exposure head **4** and communicates with the duct unit **60** to be described later, thereby forming a duct (a part of a second cooling duct) that allows an airflow from the duct unit **60** to flow to the exposure head **4** through the opening **69a**. The duct walls **69F** and **69B** as the ribs form a part of the duct at the exposure position.

The duct front wall **69F** and the duct back wall **69B** form an opening portion **64** (see FIG. **21**) communicating with an opening portion **61** (see FIG. **35**) of the duct unit **60** described later together with the cartridge tray **30** described later.

Further, the elevating duct **69** includes a first engagement portion **69d** and a second engagement portion **69e** which are engaged with the rotating arm **65** at both end portions in the longitudinal direction. The first engagement portion **69d** is provided outside the duct front wall **69F** on one end side in the longitudinal direction. Further, the first engagement portion **69d** is provided outside a region provided with the FFC connector **57** located outside the duct front wall **69F** on one end side in the longitudinal direction. The second engagement portion **69e** is provided outside the duct back wall **69B** on the other end side in the longitudinal direction.

Therefore, the region (range Lm in FIG. **20**) where the first engagement portion **69d** is provided and the region (range Ln in FIG. **20**) where the second engagement portion **69e** is provided are provided outside the duct region (range La in FIG. **20**) where the opening **69a** is surrounded by the duct wall. In other words, the duct region (range La in FIG. **20**) where the opening **69a** is surrounded by the duct wall is provided between the region (range Lm in FIG. **20**) where the first engagement portion **69d** is provided and the region (range Ln in FIG. **20**) where the second engagement portion **69e** is provided.

A region (range Lc in FIG. **20**) where the FFC connector **57** is provided is provided outside the duct region (range La in FIG. **20**) where the opening **69a** is surrounded by the duct wall to form a duct, and between the duct region and the region (range Lm in FIG. **20**) where the first engagement portion **69d** is provided.

In addition, the range La forming the duct includes most of the substrate **50** on which the LED **51** is mounted, and the exposure head **4** can be sufficiently cooled by blowing the airflow to the range La. Note that the range Lc is a mounting portion of the FFC connector **57** of a signal line that transmits a drive signal to the substrate **50** on which the LED **51** is mounted. The range Lc is not provided with an opening

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for forming a duct, but is configured to enable necessary and sufficient cooling in the range La as described above.

As a result, the air taken in from the outside of the apparatus by the duct unit 60 to be described later is blown against the back surface of the substrate 50 from the opening 55a of the exposure head 4 through the elevating duct 69 (see FIG. 37). The airflow blown from the opening 55a of the exposure head 4 to the back surface of the substrate 50 is exhausted to the outside of the apparatus by the duct unit 60 through the elevating duct 69 (see FIG. 39). (Engagement Hole)

As illustrated in FIG. 26, the elevating duct 69 has an engagement hole 69b and an engagement hole 69c which are second engagement portions. The engagement hole 69b and the engagement hole 69c which are the second engagement portions are engaged with the engagement claw 55b and the engagement claw 55c which are the first engagement portions of the exposure head 4 by snap-fitting.

In the elevating duct 69, the upper surface portion 69U between the opening 69a at one end (front side) in the longitudinal direction and the opening 69a adjacent thereto is defined as a first upper surface portion 69U1. Similarly, the upper surface portion 69U between the opening 69a at the other end (back side) in the longitudinal direction and the opening 69a adjacent thereto is defined as a second upper surface portion 69U2. The first upper surface portion 69U1 is provided with the engagement hole 69b to be engaged with the engagement claw 55b of the exposure head 4. The second upper surface portion 69U2 is provided with the engagement hole 69c to be engaged with the engagement claw 55c of the exposure head 4. The first upper surface portion 69U1 and the second upper surface portion 69U2, that is, the upper surface portion 69U are support portions that face the bottom surface portion 55D of the exposure head 4 (housing support member 55) and detachably support the exposure head 4.

The engagement holes 69b and 69c as the second engagement portions are formed in the upper surface portion 69U facing the exposure head 4 at positions corresponding to the engagement claws 55b and 55c.

Thus, the exposure head 4 is configured to be detachable from the image forming apparatus 100. The exposure head 4 is moved (in the downward direction D illustrated in FIG. 49) so that the engagement claws 55b and 55c of the exposure head 4 are dropped inside the engagement holes 69b and 69c of the elevating duct 69, respectively, and the bottom surface portion 55D of the exposure head 4 abuts against the upper surface portion 69U of the elevating duct 69. That is, the exposure head 4 is moved in the direction orthogonal to the axial direction with respect to the elevating duct 69 to engage the engagement claws 55b and 55c with the engagement holes 69b and 69c in the protruding direction.

Thereafter, by moving the exposure head 4 along the upper surface portion 69U of the elevating duct 69 (backward direction B illustrated in FIG. 52), the engagement claws 55b and 55c of the exposure head 4 are respectively engaged with the engagement holes 69b and 69c of the elevating duct 69 by snap-fitting.

That is, the exposure head 4 is moved in the axial direction with respect to the elevating duct 69 to engage the engagement claws 55b and 55c with the engagement holes 69b and 69c in the extending direction orthogonal to the protruding direction. In this manner, the exposure head 4 is connected to the elevating duct 69 in the image forming apparatus 100, and the exposure head 4 is integrated with the elevating duct 69. The procedure of detaching the exposure

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head 4 from the elevating duct 69 is reverse to the procedure described above. The replacement and attachment or detachment configuration of the exposure head will be described later in detail.

(Cartridge Tray)

The image forming apparatus 100 includes the cartridge tray 30. The cartridge tray 30 will be described with reference to FIGS. 8, 9, 20, 22, and 21. FIGS. 8, 9, and 20 are perspective views of the cartridge tray 30. FIG. 21 is a view of the cartridge tray 30 as viewed from the lower surface.

The cartridge tray 30 is a support member that supports the drum unit 23 and the developing unit 24 described above, and guides and supports the attaching and detaching operation along the axial direction of the photosensitive drum 2. The cartridge tray 30 axially supports a rotation shaft 102a of the inner door 102 such that the inner door 102 is rotatable within a predetermined range.

The cartridge tray 30 is provided for each image forming portion. Each cartridge tray 30 includes the developing support member 301 that guides and supports the attaching and detaching operation of the developing unit 24 along the axial direction of the photosensitive drum 2, and the drum support member 302 that guides and supports the attaching and detaching operation of the drum unit 23 along the axial direction of the photosensitive drum 2. In the cartridge tray 30, the developing support member 301 and the drum support member 302 are integrally formed. The cartridge tray 30 is not limited to a configuration to be provided for each image forming portion.

The elevating duct 69 is movably disposed between the developing support member 301 and the drum support member 302 of the cartridge tray 30. In the elevating duct 69, the first engagement portion 69d and the second engagement portion 69e at both ends in the longitudinal direction are supported by the rotating arm 65 from below between the developing support member 301 and the drum support member 302. Although described later, the rotating arm 65 is rotatably provided on the developing support member 301 of the cartridge tray 30. The exposure head 4 is detachably attached to the elevating duct 69 movably arranged on the cartridge tray 30. In other words, the cartridge tray 30 is a support member that supports the exposure head 4, and guides and supports the attaching and detaching operation along the axial direction of the photosensitive drum 2.

(Developing Support Member)

The developing support member 301 is a developing support member that guides and supports the attaching and detaching operation of the developing unit 24 along the axial direction of the photosensitive drum, and is a member having a longitudinal shape extending in the axial direction of the photosensitive drum 2. The developing support member 301 includes a first developing guide portion 301a, a second developing guide portion 301b facing the first developing guide portion 301a, and a developing bottom surface portion 301c provided between the first developing guide portion 301a and the second developing guide portion 301b. In the developing support member 301, the first developing guide portion 301a, the second developing guide portion 301b, and the developing bottom surface portion 301c are integrally formed.

The developing bottom surface portion 301c faces the bottom surface portion 24D of the frame body of the developing unit 24 with a space therebetween, and is provided over the longitudinal direction which is the axial direction of the photosensitive drum 2. The first developing guide portion 301a is provided on one end side in the lateral direction orthogonal to the longitudinal direction of the

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developing bottom surface portion **301c**, and is provided between the elevating duct **69** and the developing unit **24** so as to separate the elevating duct **69** and the developing unit **24**. The second developing guide portion **301b** is provided on the other end side in the lateral direction of the developing bottom surface portion **301c**, and is provided so as to face the first developing guide portion **301a**. The first developing guide portion **301a** and the second developing guide portion **301b** abut on the frame body of the developing unit **24**, and guide, in the longitudinal direction, the developing unit **24** inserted and removed in the longitudinal direction.

The first developing guide portion **301a** includes a facing portion **301d** facing the first inclined surface **69L1** of the elevating duct **69**. The facing portion **301d** has a first tray inclined surface which is inclined similarly to the first inclined surface **69L1** of the elevating duct **69**.

The first developing guide portion **301a** includes a partition wall portion **301e** on the downstream side of the facing portion **301d** in the moving direction from the retracted position to the exposure position of the elevating duct **69**. The partition wall portion **301e** is provided between the exposure head **4** and the developing unit **24** so as to separate the exposure head **4** located at the exposure position illustrated in FIG. 22 from the developing unit **24**. The partition wall portion **301e** is an end portion (upper end portion) on the developing sleeve side in the first developing guide portion **301a**.

As illustrated in FIGS. 2, 8, and 22, the developing support member **301** (cartridge tray **30**) is disposed immediately below the developing unit **24**. The upper surface of the developing support member **301** (cartridge tray **30**) and the bottom surface of the developing unit **24** form a duct that is a closed space, and function as an intermediate path of a development cooling airflow to be described later in addition to a guide of the attaching and detaching operation of the developing unit.

As will be described later, in the duct (closed space) formed by the upper surface of the cartridge tray **30** (developing support member **301**) and the upper surface of the developing unit **24**, the opening on one end side (front side) in the longitudinal direction communicates with an opening **41a** of a front-side duct **41** that takes in air from the outside of the apparatus through the opening **102c** of the inner door **102**. Further, the duct has an opening on the other end side (back side) in the longitudinal direction that communicates with an opening **42a** of a back-side duct **42** that exhausts air to the outside of the apparatus. The duct between the developing unit **24** and the developing support member **301** forms one closed space communicating with the front-side duct **41** and the back-side duct **42** (see FIG. 34).

(Drum Support Member)

The drum support member **302** is a photoconductor support member that guides and supports the attaching and detaching operation of the drum unit **23** along the axial direction of the photosensitive drum, and is a member having a longitudinal shape extending in the axial direction of the photosensitive drum **2**. The drum support member **302** includes a first drum guide portion **302a**, a second drum guide portion **302b** facing the first drum guide portion **302a**, and a drum bottom surface portion **302c** provided between the first drum guide portion **302a** and the second drum guide portion **302b**. In the drum support member **302**, the first drum guide portion **302a**, the second drum guide portion **302b**, and the drum bottom surface portion **302c** are integrally formed.

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The drum bottom surface portion **302c** faces the bottom surface portion **23D** of the frame body of the drum unit **23** and is provided over the longitudinal direction which is the axial direction of the photosensitive drum **2**. The first drum guide portion **302a** is provided on one end side in the lateral direction orthogonal to the longitudinal direction of the drum bottom surface portion **302c**, and is provided between the elevating duct **69** and the drum unit **23** so as to separate the elevating duct **69** and the drum unit **23**. The second drum guide portion **302b** is provided on the other end side in the lateral direction of the drum bottom surface portion **302c**, and is provided so as to face the first drum guide portion **302a**. Each of the first drum guide portion **302a** and the second drum guide portion **302b** abuts on the frame body of the drum unit **23** and guides, in the longitudinal direction, the drum unit **23** inserted and removed in the longitudinal direction.

The first drum guide portion **302a** includes a facing portion **302d** facing the second inclined surface **69R1** of the elevating duct **69**. The facing portion **302d** has a second tray inclined surface which is inclined similarly to the second inclined surface **69R1** of the elevating duct **69**.

(Relationship Between Cartridge Tray and Elevating Duct)

Here, a relationship between the cartridge tray **30** and the elevating duct **69** will be described.

The elevating duct **69** is movably disposed between the first developing guide portion **301a** and the first drum guide portion **302a** in the cartridge tray **30**, and is moved between the exposure position illustrated in FIG. 24 and the retracted position illustrated in FIG. 25 by rotation of the rotating arm **65** described later. That is, the first developing guide portion **301a** and the first drum guide portion **302a** of the cartridge tray **30** function as guide members that guide the elevating duct **69** in the moving direction thereof.

When the exposure head **4** is detachably mounted, the opening **55a** of the exposure head **4** and the opening **69a** of the elevating duct **69** communicate with each other and the elevating duct **69** is integrated with the exposure head **4**. The elevating duct **69** is separated from the developing unit **24** by the first developing guide portion **301a** of the cartridge tray **30**, and is separated from the drum unit **23** by the first drum guide portion **302a** of the cartridge tray **30**. Further, a gap between the elevating duct **69** and the first developing guide portion **301a** is sealed by a seal **72** which is a sealing member described later at the exposure position illustrated in FIG. 22. Similarly, a gap between the elevating duct **69** and the first drum guide portion **302a** is sealed by a seal **71** which is a sealing member described later at the exposure position illustrated in FIG. 22.

In this manner, the cartridge tray **30** and the elevating duct **69** disposed between the first developing guide portion **301a** and the first drum guide portion **302a** of the cartridge tray **30** form a duct that is a closed space communicating with the opening **55a** of the exposure head **4**.

(Relationship Among Cartridge Tray, Elevating Duct, and Duct Unit)

Further, the cartridge tray **30** and the elevating duct **69** form the opening portion **64** communicating with the opening portion **61** of the duct unit **60** on a side facing the duct unit **60** to be described later.

The opening portion **64** formed by the cartridge tray **30** and the elevating duct **69** will be described with reference to FIGS. 22 and 21.

As illustrated in FIG. 21, the opening portion **64** formed by the cartridge tray **30** and the elevating duct **69** is formed by the developing support member **301**, the drum support member **302**, and the elevating duct **69** between the devel-

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oping support member 301 and the drum support member 302. More specifically, the opening portion 64 is formed by the developing guide portion 301a of the developing support member 301, the drum guide portion 302a of the drum support member 302, and the duct front wall 69F and the duct back wall 69B of the elevating duct 69 between the developing guide portion 301a and the drum guide portion 302a.

In this manner, the cartridge tray 30 and the elevating duct 69 form the opening portion 64 communicating with the opening portion 61 of the duct unit 60 on the side facing the duct unit 60 to be described later. Then, by attaching the duct unit 60 to be described later to the image forming apparatus 100, the opening portion 61 of the duct unit 60 is pressed from below against the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69. As a result, the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 communicates with the opening portion 61 of the duct unit.

Note that a space between the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 and the opening portion 61 of the duct unit is sealed by a sealing member 207 to be described later.

(Relationship Between Rib of Elevating Duct and Duct Unit)

Further, the duct front wall 69F and the duct back wall 69B which are ribs of the elevating duct 69 will be described in detail.

As will be described later, among the sealing members 207 provided in the opening portion 61 of the duct unit 60, the sealing member 207 provided in the longitudinal direction is sandwiched between the duct unit 60 and the first developing guide portion 301a and the first drum guide portion 302a of the cartridge tray 30 facing the duct unit to seal the space therebetween, as illustrated in FIG. 37. The first developing guide portion 301a and the first drum guide portion 302a of the cartridge tray 30 are disposed on both side surfaces in the left-right direction of the elevating duct 69 (see FIG. 22).

Therefore, the sealing of the left and right ends of the opening of the elevating duct 69, that is, the boundary with the duct unit 60 in the longitudinal direction in the range La illustrated in FIG. 20 is performed by sealing the space between the duct unit 60 and the cartridge tray 30 with the sealing member 207. Therefore, the sealed state is always maintained without being affected by the moving operation of the elevating duct 69.

On the other hand, at the front end and the back end of the opening of the elevating duct 69, that is, at the boundary between the range La and the range Lc and the boundary between the range La and the range Lm illustrated in FIG. 20, another sealing configuration is required. This is because the boundary between the front end and the back end of the opening of the elevating duct 69 and the duct unit 60 cannot be sealed as between the cartridge tray 30 and the duct unit 60.

In the present embodiment, front and back ends between with the opening portion 61 (see FIG. 35) of the duct unit 60 are sealed by the duct front wall 69F and the duct back wall 69B illustrated in FIG. 28.

FIGS. 29 and 30 illustrate cross sections of the exposure head 4, the elevating duct 69, and the duct unit 60. The position of the cross section illustrated in FIGS. 29 and 30 is a position in the direction of arrow E-E illustrated in FIG. 37, and the cartridge tray 30 is not illustrated. FIG. 29 is a cross-sectional view illustrating the arrangement of the elevating duct 69 moved to the exposure position. FIG. 30

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is a cross-sectional view illustrating the arrangement of the elevating duct 69 moved to the retracted position.

The duct front wall 69F and the duct back wall 69B of the elevating duct 69 are disposed outside the duct region (range La in FIG. 20) where the opening 69a of the elevating duct 69 is provided in the axial direction of the photosensitive drum 2. The duct front wall 69F and the duct back wall 69B have such lengths as to protrude toward the duct unit 60 to be described later and to overlap a side surface of the duct unit 60 in a moving direction of the elevating duct 69 (see FIGS. 29 and 30). The duct front wall 69F and the duct back wall 69B have such lengths that the duct front wall 69F and the duct back wall 69B overlap the side surfaces of the sealing member 207 provided in the opening portion 61 of the duct unit 60 when the exposure head 4 is at the exposure position. Both at the exposure position shown in FIG. 29 and at the retracted position shown in FIG. 30, the side surfaces of the duct front wall 69F and the duct back wall 69B on the opening 69a side come into contact with the side surface on the outer side in the longitudinal direction of the sealing member 207 provided in the opening portion 61 of the duct unit 60 described later at the exposure position illustrated in FIG. 22.

In the exposure position shown in FIG. 29, in the elevating duct 69, the side surface of the duct front wall 69F and the side surface of the duct back wall 69B form the front and back end walls of the range La. The sealing between the duct front wall 69F and the duct back wall 69B and the duct unit 60 is performed by bringing side surfaces of the sealing member 207 on the upper surface of the duct unit 60 into contact with side surfaces of the duct front wall 69F and the duct back wall 69B.

In the retracted position illustrated in FIG. 30, the elevating duct 69 is retracted such that the side surface of the duct front wall 69F and the side surface of the duct back wall 69B overlap the outside of the side surface of the duct unit 60, and does not interfere with the duct unit 60.

As illustrated in FIG. 28, the duct left wall 69L and the duct right wall 69R included in the elevating duct 69 are shorter in length in the moving direction (UD axial direction) than the duct front wall 69F and the duct back wall 69B. The duct left wall 69L and the duct right wall 69R of the elevating duct 69 have such lengths that they do not protrude below the cartridge tray 30 when the exposure head 4 is at the retracted position (see FIG. 23).

In other words, the duct left wall 69L and the duct right wall 69R of the elevating duct 69 in the range La illustrated in FIG. 20 are lower in height in the moving direction than the duct front wall 69F and the duct back wall 69B at both end portions outside the range La. FIG. 28 is a perspective view illustrating a state in which the exposure head 4 and the elevating duct 69 are engaged and integrated. As shown in FIG. 28, in the front-back direction, the duct left wall 69L (and the duct right wall 69R) in the range La is configured to be low with the duct front wall 69F and the duct back wall 69B as boundaries.

This is to prevent the lower ends of the duct left wall 69L and the duct right wall 69R in the range La of the elevating duct 69 from protruding from the cartridge tray 30 and from entering the inside of the duct unit 60 even by the retracting operation of the elevating duct 69 from the photosensitive drum 2.

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Even if the duct left wall 69L and the duct right wall 69R in the range La of the elevating duct 69 are configured to enter the inside of the duct unit 60 without being lowered, sealing is possible. In this case, the third opening portion 201 and the fourth opening portion 202 on the upper surface of the duct unit 60 need to have a size that allows the retracted elevating duct 69 to enter, and further, a space for housing the retracted elevating duct 69 inside the duct unit 60 is required. This is a restriction on the shape of the duct unit 60. In addition, there are restrictions on the assembly order of the duct units 60. Therefore, it is desirable that the duct left wall 69L and the duct right wall 69R in the range La of the elevating duct 69 be kept lower than the duct front and back walls and not protrude downward from the cartridge tray 30 at the retracted position of the exposure head 4.

With such a configuration, when the exposure head 4 is moved to the retracted position, the lower ends of the duct left wall 69L and the duct right wall 69R do not enter the inside of the duct unit 60 and do not hinder the movement.

With such a configuration, the elevating duct 69 which is moved integrally with the exposure head 4 forms a duct which guides air from the duct unit 60 to the exposure head 4, and has a sealing property with respect to the duct unit 60, so that toner scattering into the image forming apparatus can be reduced.

The configuration in which the duct front wall 69F and the duct back wall 69B of the elevating duct 69 overlap the side surface on the outer side of the opening portion 61 of the duct unit 60 at the retracted position illustrated in FIG. 30 is exemplified. However, the present invention is not limited thereto. For example, among the duct left wall 69L, the duct right wall 69R, the duct front wall 69F, and the duct back wall 69B, only the duct front wall 69F and the duct back wall 69B may enter the inside of the opening portions 201 and 202 of the duct unit 60.
(Development Stay)

The cartridge tray 30 includes a development stay 31 slidable along the axial direction of the photosensitive drum 2. The development stay 31 will be described with reference to FIGS. 31 and 32. FIGS. 31 and 32 are side views of the development stay 31 as viewed from the right direction.

The development stay 31, which is a sliding member, is provided movably with respect to the developing support member 301 of the cartridge tray 30. The development stay 31 is provided on the developing bottom surface portion 301c of the developing support member 301 so as to be movable in the axial direction of the photosensitive drum 2. The development stay 31 has an elongated shape (longitudinal shape) extending in the axial direction of the photosensitive drum 2, and includes a development pressurizing frame 32, a development pressurizing frame 33, a development stay link 34, an arm retracting member 68F, and an arm retracting member 68B.

The development stay link 34 is fixed to the front end portion of the development stay 31 and is engaged with the inner door 102 axially supported by the cartridge tray 30. The development pressurizing frame 32 is fixed to one side (front side) in the longitudinal direction of the development stay 31, and the development pressurizing frame 33 is fixed to the other side (back side) in the longitudinal direction of the development stay 31. The development pressurizing frame 32 and the development pressurizing frame 33 are provided at positions facing the developing unit 24.

In a state where the inner door 102 is opened, a link engagement portion 102b engaged with the development stay link 34 is located closer to the back end (the lower end in the opened state) of the inner door 102 than the rotation

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shaft 102a. Therefore, according to the rotation of the inner door 102, the link engagement portion 102b of the inner door 102 moves in the rotation direction along a locus of a circle whose radius is the distance between the rotation shaft 102a and the link engagement portion 102b. That is, as illustrated in FIG. 32, by opening the inner door 102, the link engagement portion 102b also rotates and moves to the apparatus back side.

As a result, the development stay link 34 engaged with the link engagement portion 102b of the inner door 102 is slid in the direction of the arrow B which is the back side of the apparatus, and the two development pressurizing frames 32 and 33 integrally configured through the development stay 31 are also slid in the direction of the arrow B. This means that, as illustrated in FIG. 32, the two development pressurizing frames 32 and 33 are out of the position where the developing unit 24 is held. When the development pressurizing frames 32 and 33 deviate from the holding position, the developing unit 24 moves in a direction of an arrow D, which is a direction in which the development pressurizing frames are retracted from the photosensitive drum 2 by their own weight.

From the above description, it can be understood that the developing unit 24 retracts from the photosensitive drum 2 in conjunction with the operation of opening the inner door 102. When the inner door 102 is closed, the developing unit 24 is moved in the direction of the photosensitive drum 2 and pressed through a procedure reverse to the opening operation.

In this manner, the development stay 31 is slid and moved in the front-back direction by the development stay link 34 engaged with the link engagement portion 102b in conjunction with the opening and closing operation of the inner door 102. The development stay 31 moves the developing unit 24 to a developing position (see FIG. 22) where the developing sleeve 5 is close to the photosensitive drum 2 at the time of development and a separated position (see FIG. 23) where the developing sleeve 5 is separated from the photosensitive drum 2 at the time of non-development by sliding movement in the front-back direction.

As illustrated in FIG. 31, the development stay 31 is slid in the forward direction F in conjunction with the operation of closing the inner door 102. At this time, the developing unit 24 is moved upward (arrow U) along the inclined surface of the development pressurizing frame 32 and the development pressurizing frame 33 of the development stay 31. As a result, the developing sleeve 5 of the developing unit 24 is moved in a direction approaching the photosensitive drum 2 of the drum unit 23.

Also, as illustrated in FIG. 32, the development stay 31 is slid in the backward direction B in conjunction with the operation of opening the inner door 102. At this time, the developing unit 24 is moved downward (arrow D) along the inclined surface of the development pressurizing frame 32 and the development pressurizing frame 33 of the development stay 31. As a result, the developing sleeve 5 of the developing unit 24 is moved in a direction away from the photosensitive drum 2 of the drum unit 23, and the developing sleeve 5 is separated from the photosensitive drum 2 as compared with the time of development.

Furthermore, the development stay 31 includes the arm retracting member 68F for rotating the rotating arm 65, which is a rotational member to be described later, and the arm retracting member 68B. The arm retracting member 68F and the arm retracting member 68B are integrally formed with the development stay 31. The arm retracting member 68F is fixed to one side (apparatus front side) in the

longitudinal direction of the development stay 31, and is provided on a surface on the opposite side of the development pressurizing frame 32. The arm retracting member 68B is fixed to the other side (apparatus back side) in the longitudinal direction of the development stay 31, and is provided on a surface on the opposite side of the development pressurizing frame 33. The arm retracting member 68F and the arm retracting member 68B are moved in the same direction as the development stay 31 is slid in the front-back direction in conjunction with the operation of opening and closing the inner door 102, and rotate the rotating arm 65.

The development stay 31 releases the engagement with the rotating arm 65 by moving to one side in the axial direction. As a result, the rotating arm 65 is rotated in one direction to move the exposure head 4 to the exposure position integrally with the elevating duct 69. On the other hand, the development stay 31 is engaged with the rotating arm 65 by moving to the other side in the axial direction. As a result, the rotating arm 65 is rotated in the other direction to move the exposure head 4 to the retracted position integrally with the elevating duct 69.

(Rotating Arm)

As illustrated in FIGS. 24 and 25, the rotating arm 65, which is a rotational member, is rotatably provided on the developing support member 301 of the cartridge tray 30. One end portion of the rotating arm 65 in the left-right direction orthogonal to the axial direction of the photosensitive drum 2 is supported to be rotatable about the axis along the axial direction of the photosensitive drum 2. The axis which is the rotation center of the rotating arm 65 may not be parallel to the axial direction of the photosensitive drum 2, and may have an angle in the range of 0 to $\pm 200^\circ$ with respect to the axial direction of the photosensitive drum 2. In the rotating arm 65, an engagement boss 66 which is the other end portion in the left-right direction supports the engagement portions 69d and 69e which are both end portions of a region outside the opening 69a of the elevating duct 69 in the axial direction.

Specifically, one end portion of the rotating arm 65 in the left-right direction rotates about a rotation shaft 30a integrally provided on the cartridge tray 30. The rotation shaft 30a is integrally provided on the back surface of the developing bottom surface portion 301c of the developing support member 301 on the side opposite to the surface facing the developing unit 24 with the axis parallel to the moving direction of the development stay 31 as the center. In other words, the moving direction of the development stay 31 may be the axial direction that is the rotation center of the photosensitive drum 2 or the longitudinal direction of the drum unit 23, the developing unit 24, and the exposure head 4. This is a condition under which the rotating arm 65 can be arranged most compactly with respect to the width dimension of the cartridge tray 30 illustrated in FIG. 25 in consideration of the operation of the arm retracting members 68F and 68B to be described later. As a result, the influence on the arrangement and operation of the components around the rotating arm 65 can also be minimized.

The rotating arm 65 includes the engagement boss 66 at a distal end that is the other end portion in the left-right direction. The engagement boss 66 is provided at an end portion of the rotating arm 65 on a side opposite to the rotation shaft 30a. The rotating arm 65 supports both end portions of a region outside the opening 55a in the longitudinal direction of the exposure head 4 from below. That is, the rotating arm 65 supports from below both end portions in the longitudinal direction of the elevating duct 69 that supports the exposure head 4 in the longitudinal direction, in

a region (range Lm in FIG. 20) outside the duct region (range La in FIG. 20) of the exposure head 4. Specifically, the rotating arm 65 supports the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e at both ends in the longitudinal direction of the elevating duct 69 from the lower side by the engagement boss 66 provided at the distal end.

The rotating arm 65 presses the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e at both ends in the longitudinal direction of the elevating duct 69 upward by the force of an arm pressure spring 67 which is a biasing member. Here, a torsion coil spring is used as the arm pressure spring 67. In FIG. 24, the exposure head 4 is disposed close to the photosensitive drum 2, but this is maintained by the rotating arm 65 pressing the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e at both ends of the elevating duct 69 upward. This pressing is ensured by a predetermined spring pressure by the arm pressure spring 67.

In this manner, the rotating arm 65 does not directly press the exposure head 4 but presses the elevating duct 69 that supports the exposure head 4.

The rotating arm 65 constitutes a moving mechanism (retraction mechanism) that moves up and down the elevating duct 69 together with the arm pressure spring 67 and the rotation shaft 102a provided in the cartridge tray 30. That is, the moving mechanism for moving (elevating) the elevating duct 69 includes the rotation shaft 102a, the rotating arm 65 which is a rotational member rotating about the rotation shaft 102a, and the arm pressure spring 67 which is a biasing member that applies a force to the rotating arm 65.

The rotating arm 65 rotates about the rotation shaft 30a in response to the sliding movement of the development stay 31 to move the exposure head 4 to the exposure position (see FIG. 24) or the retracted position (see FIG. 25). That is, in conjunction with the operation of opening and closing the inner door 102, the rotating arm 65 rotates in one direction to move the exposure head 4 to the exposure position where the photosensitive drum 2 is exposed, and rotates in the other direction to move the exposure head 4 to the retracted position retracted from the exposure position.

Specifically, as illustrated in FIG. 32, the development stay 31 is slid in the backward direction B in conjunction with the operation of opening the inner door 102. At this time, the arm retracting member 68F and the arm retracting member 68B moved in the backward direction B are engaged with the rotating arm 65. As a result, the rotating arm 65 is rotated, and the rotating arm 65 is pushed downward against the force of the arm pressure spring 67 (see FIG. 25). The rotation of the rotating arm 65 causes the engagement boss 66 provided at the distal end of the rotating arm 65 to push down engagement ribs 69d1 and 69e1 provided at the lower ends of the engagement portions 69d and 69e of the elevating duct 69, thereby retracting the exposure head 4 integrated with the elevating duct 69 from the photosensitive drum 2. That is, the exposure head 4 is moved from the exposure position to the retracted position.

The retraction of the rotating arm 65 is performed by inserting the wedge-shaped arm retracting member 68F and the arm retracting member 68B between the upper surface of the rotating arm 65 and the lower surface of the cartridge tray 30 at a position close to the rotation shaft 30a.

Further, as illustrated in FIG. 31, the development stay 31 is slid and moved in the forward direction F in conjunction with the operation of closing the inner door 102. At this time, the arm retracting member 68F and the arm retracting member 68B are moved in the forward direction F, and the

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engagement between the arm retracting member 68F and the arm retracting member 68B, and the rotating arm 65 is released. As a result, the rotating arm 65 is rotated, and the rotating arm 65 is pushed upward by the force of the arm pressure spring 67 (see FIG. 24). The rotation of the rotating arm 65 causes the engagement boss 66 provided at the distal end of the rotating arm 65 to push up the bottom surfaces of the engagement portions 69d and 69e of the elevating duct 69, thereby bringing the exposure head 4 integrated with the elevating duct 69 close to the photosensitive drum 2. That is, the exposure head 4 is moved from the retracted position to the exposure position.

As described above, one end portion of the rotating arm 65 in the direction orthogonal to the axial direction of the photosensitive drum 2 is supported by the rotation shaft 102a so as to be rotatable about the axis parallel to the axial direction. In the rotating arm 65, the engagement boss 66 which is the other end portion supports the engagement portions 69d and 69e which are both end portions of a region outside the opening 69a of the elevating duct 69 in the axial direction. As described above, the moving mechanism of the exposure head 4 is established even within the range Lm illustrated in FIG. 20.

In addition, the rotating arm 65 constituting a moving mechanism for moving the exposure head 4 to the exposure position and the retracted position is provided outside the duct region (range La illustrated in FIG. 20) of the exposure head. Therefore, the arm does not become an obstacle when air is sent from the lower side of the exposure head 4 to the duct region, and air can be directly blown onto the back surface of the substrate 50 of the exposure head 4. As a result, the substrate 50 including the light emitting element of the exposure head 4 can be more effectively cooled.

Further, the exposure head 4 is separated from the adjacent developing unit 24 and drum unit 23 by the housing support member 55 of the exposure head 4. Therefore, the air for cooling the exposure head 4 introduced into the back surface of the substrate 50 does not leak to the side of the developing unit 24 adjacent to the exposure head 4, and the scattering of the toner inside the apparatus can be reduced.

Note that the operation of the rotating arm 65 that moves the exposure head 4 to the exposure position or the retracted position is powered by the sliding movement of the development stay 31 that retracts the developing unit 24, but may be powered via another member linked with the inner door 102.

Furthermore, in the present embodiment, the exposure head 4 is moved to the exposure position or the retracted position using the rotating arm 65, but the exposure head 4 may be moved using another mechanism. FIGS. 66 and 67 are examples of a mechanism for moving the exposure head 4 using an eccentric cam. In FIG. 66, the exposure head 4 is in the retracted position, an eccentric cam 306 is provided below the elevating duct 69, and is in contact with the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e of the elevating duct 69. The eccentric cam 306 includes a first pulley 307 and is connected to a second pulley 304 via a transmission belt 305. When the second pulley 304 rotates, the driving force is transmitted to the first pulley 307 via the transmission belt 305, and the eccentric cam 306 rotates. The rotation of the eccentric cam 306 presses the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e of the elevating duct 69, and the exposure head 4 moves to the exposure position. FIG. 67 is a view illustrating a state in which the exposure head 4 has moved to the exposure position.

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The mechanism for rotating the eccentric cam 306 is not limited to the transmission belt 305, and may be direct transmission by a gear or the like. As shown in FIGS. 68 and 69, a gear 308 that abuts on the first pulley may be provided to transmit the rotation of the gear 308 to the first pulley 307. (Sealing Property of Exposure Head)

The sealing property of the exposure head 4 will be described with reference to FIG. 33. Similarly to FIG. 22, FIG. 33 is cross-sectional views taken along line X-X in FIG. 20. FIG. 33 is a cross-sectional view of the exposure head 4, the elevating duct 69, and the cartridge tray 30 as viewed from the front, in which the drum unit 23 and the developing unit 24 are not illustrated.

As can be understood from FIG. 33, the exposure head 4 forms a cooling duct of the exposure head in a state of being mounted on the elevating duct 69.

As described above, the housing support member 55 of the exposure head 4 is provided with the engagement claws 55b1 and 55b2 for engaging with the elevating duct 69. Meanwhile, in the elevating duct 69, the engagement holes 69b and 69c for engaging with the engagement claws 55b1 and 55b2 are provided in the upper surface portion 69U facing the exposure head 4. Based on this configuration, the exposure head 4 can be engaged with and integrated with the elevating duct 69 according to a procedure of replacement and attachment or detachment of the exposure head 4 described later.

There is a slight gap between the engagement claws 55b1 and 55b2 of the exposure head 4 and the elevating duct 69.

This gap exists as a gap in the vertical direction in FIG. 59, that is, the arrow D direction illustrated in FIG. 49, and is a necessary gap from the viewpoint of assemblability and component accuracy.

First, from the viewpoint of assemblability, when there is no gap, the housing support member 55 becomes resistance of the sliding operation of the exposure head 4 in the direction of the arrow B illustrated in FIG. 52, and the exposure head 4 may not reach the original position illustrated in FIG. 57 due to occurrence of catching during the sliding operation.

Next, from the viewpoint of component accuracy, the housing support member 55 and the elevating duct 69 as a whole have an elongated shape in the moving direction (front-back direction), and it is inevitable that a predetermined amount of warpage occurs in the process of component manufacture.

For example, in a case where the housing support member 55 has warpage in which the central portion in the longitudinal direction is convex by 0.5 mm in the direction of the elevating duct 69 with respect to both end portions thereof, the engagement claws 55b1 and 55b2 do not overlap the edges of the engagement holes 69b and 69c in the first place unless the gap is 0.5 mm or more.

From the above description, it can be understood that a predetermined amount of gap is necessary in order to absorb the warpage of the component and not to generate resistance of the mounting operation of the exposure head 4 on the elevating duct 69.

However, from the viewpoint of a duct for sending air for cooling the exposure head 4, this gap is not desirable, and there is a possibility that toner may be scattered inside the apparatus due to blowing or suction from the gap. Therefore, in order to reduce the risk described above, it is necessary to minimize the gap amount.

As a countermeasure, it is conceivable to provide a seal such as a foamed seal material in the gap. Specifically, a seal is affixed to the exposure head or the elevating duct forming

the gap. However, there is a possibility that curling or peeling off of the seal occurs due to the sliding operation of the exposure head indicated by the arrow B in FIG. 52. In addition, in consideration of the possibility that the seal is disposed in the vicinity of the photosensitive drum 2 and the developing sleeve 5 and fragments of the seal adhere to the photosensitive drum 2 and the developing sleeve 5, it is desirable that the countermeasure is performed by means different from affixing of the seal.

Therefore, in the present embodiment, in order to reduce this gap, the configuration is as illustrated in FIG. 28. FIG. 28 is a perspective view illustrating the elevating duct 69 engaged with the exposure head 4 and the rotating arm 65 engaged with the elevating duct 69.

As described above as the moving mechanism of the exposure head 4, the rotating arm 65 presses the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e at both ends in the longitudinal direction of the elevating duct 69 upward by the force of an arm pressure spring 67 which is a biasing member. Here, when the upper surface portion 69U on which the exposure head 4 is mounted is defined as a first surface, the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e of the elevating duct 69 are second surfaces arranged to face the first surface in the moving direction of the elevating duct 69. The other end portion of the rotating arm 65 presses the bottom surfaces of the first engagement portion 69d and the second engagement portion 69e at both ends of the elevating duct 69 upward by rotating in one direction, thereby moving the exposure head 4 to the exposure position integrally with the elevating duct 69 and maintaining the exposure position of the exposure head 4 with respect to the photosensitive drum 2. As described above, this pressing is ensured by a predetermined spring pressure by the arm pressure spring 67.

In this manner, the rotating arm 65 does not directly press the exposure head 4 but presses the elevating duct 69 that supports the exposure head 4.

Further, the rotating arm 65 applies a force to the elevating duct 69 toward the photosensitive drum 2 with a predetermined spring pressure by the arm pressure spring 67 which is a biasing member. In other words, the rotating arm 65 applies a force to the elevating duct 69 toward the housing support member 55 with a predetermined spring pressure by the arm pressure spring 67.

Here, the spring pressure by the arm pressure spring 67 is set to a sufficiently strong value. Therefore, when the rotating arm 65 applies a force to the elevating duct 69 by the biasing force of the arm pressure spring 67, a gap between the housing support member 55 of the exposure head 4 and the elevating duct 69 can be reduced.

In addition, even if the housing support member 55 of the exposure head 4 or the upper surface portion 69U of the elevating duct 69 engaged with the housing support member 55 is warped by a predetermined amount in the process of component manufacture, both postures can be corrected by the biasing force of the arm pressure spring 67, and the gap between the members caused by warpage can be reduced.

With such a configuration, it is possible to easily perform the attaching and detaching operation of the exposure head 4 with respect to the elevating duct 69, it is possible to reduce a gap in assembly and component accuracy between the exposure head 4 and the elevating duct 69, and it is possible to reduce toner scattering into the image forming apparatus.

(Duct Unit)

The image forming apparatus 100 also includes the duct unit 60 in a detachable manner. The duct unit 60 will be described with reference to FIGS. 35, 37, 39, and 36. FIG. 35 is a perspective view of the duct unit as viewed from above. FIG. 36 is a perspective view of the duct unit as viewed from below. FIG. 37 is a cross-sectional view of the exposure cooling airflow on the intake side, and is a cross-sectional view taken along line Y3-Y3 illustrated in FIG. 34. FIG. 39 is a cross-sectional view of the exposure cooling airflow on the exhaust side, and is a cross-sectional view taken along line Y4-Y4 illustrated in FIG. 34.

The duct unit 60 is an exposure cooling unit that communicates with the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 and cools the exposure head 4 by the airflow through the elevating duct 69.

The duct unit 60 includes an intake fan 62 and an intake duct 205 for sending air from the outside of the image forming apparatus to each exposure head 4. The duct unit 60 includes an exhaust fan 63 and an exhaust duct 206 for exhausting air from each exposure head 4 to the outside of the image forming apparatus. The duct unit 60 is integrally provided with the intake fan 62, the exhaust fan 63, the intake duct 205, and the exhaust duct 206, and is detachably mounted on the apparatus main body of the image forming apparatus 100 immediately below the cartridge tray 30.

The duct unit 60 includes the intake duct 205 and the exhaust duct 206 provided separately from the intake duct 205. In other words, the duct unit 60 includes the intake duct 205 as a first duct and the exhaust duct 206 as a second duct provided separately from the first duct.

The duct unit 60 includes an intake port 203 and an exhaust port 204 on the same surface (left surface) side of the image forming apparatus 100, and the intake fan 62 is disposed in the intake port 203 and the exhaust fan 63 is disposed in the exhaust port 204. In the present embodiment, as illustrated in FIG. 34, the intake fan 62 disposed closer to the front of the image forming apparatus 100 functions as an intake fan that takes in air outside the apparatus, and the exhaust fan 63 disposed closer to the rear functions as an exhaust fan that exhausts air to the outside of the apparatus.

In an exterior cover forming an exterior of the left side surface of the image forming apparatus, a louver (not illustrated) as an opening (first opening, second opening) is formed at a position facing each of the fans 62 and 63. The louver formed in the exterior cover communicates with the intake port 203 and the exhaust port 204 in which the fans 62 and 63 are disposed, respectively. Intake by the intake fan 62 and exhaust by the exhaust fan 63 are performed through the louver formed in the exterior cover forming the exterior of the left side surface of the image forming apparatus.

As illustrated in FIGS. 35 and 37, the duct unit 60 includes a third opening portion 201 (Y, M, C, and K) on the upper surface thereof and closer to the front of the image forming apparatus 100 for each exposure head. The intake port 203 and the opening portion 201 (Y, M, C, and K) of each exposure head are connected by the intake duct 205. The duct unit 60 is configured to discharge air (fresh air) outside the image forming apparatus taken in from the intake port 203 by the intake fan 62 from each opening portion 201.

As illustrated in FIGS. 35 and 39, the duct unit 60 includes a fourth opening portion 202 (Y, M, C, and K) on the upper surface thereof and closer to the rear of the image forming apparatus 100 for each exposure head. The exhaust port 204 and the opening portion 202 of each exposure head are connected by the exhaust duct 206. The duct unit 60 is configured to discharge the air taken in from each opening

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portion 202 by the exhaust fan 63 from the exhaust port 204 to the outside of the image forming apparatus.

In the present embodiment, due to the convenience of component molding, the duct unit 60 is configured by two components of an upper frame body 60a and a lower frame body 60b divided into upper and lower parts as illustrated in FIGS. 35 and 36. Here, the duct unit 60 fixes the outer edge portions of the upper frame body 60a and the lower frame body 60b in the front-back direction by snap-fitting, and also fixes the outer edge portions by snap-fitting at a position straddling the intake duct 205 and the exhaust duct 206. Here, fixing at the position straddling the intake duct 205 and the exhaust duct 206 means fixing by snap-fitting a portion between the intake duct 205 and the exhaust duct 206 of the upper frame body 60a and a portion between the intake duct 205 and the exhaust duct 206 of the lower frame body 60b facing the portion of the upper frame body 60a. By fixing the upper frame body 60a and the lower frame body 60b by snap-fitting in this manner, the leakage of the airflow from the gap between the dividing surfaces of the intake duct 205 and the exhaust duct 206 is reduced.

As shown in FIG. 35, the duct unit 60 includes the opening portion 61 (Y, M, C, and K) on the upper surface thereof. The opening portion 61 of the duct unit 60 includes the third opening portion 201 provided closer to the front side of the apparatus and the fourth opening portion 202 provided closer to the back side of the apparatus. The opening portion 61 (Y, M, C, and K) of the duct unit 60 is provided so as to correspond to each of the exposure heads 4 of the respective colors.

That is, the opening portion 60Y of the duct unit 60 includes an opening portion 201Y provided closer to the front side of the apparatus and an opening portion 202Y provided closer to the back side of the apparatus. The opening portion 60M of the duct unit 60 includes an opening portion 201M provided closer to the front side of the apparatus and an opening portion 202M provided closer to the back side of the apparatus. The opening portion 60C of the duct unit 60 includes an opening portion 201C provided closer to the front side of the apparatus and an opening portion 202C provided closer to the back side of the apparatus. The opening portion 60K of the duct unit 60 includes an opening portion 201K provided closer to the front side of the apparatus and an opening portion 202K provided closer to the back side of the apparatus.

The opening portion 61 of the duct unit 60 is provided at a position facing the opening portion 64 formed by the elevating duct 69 and the cartridge tray 30, and communicates with the opening portion 64 by being pressed to form a closed space.

(Configuration of Sealing Member of Duct Unit)

As described above, the exposure cooling airflow flows from the third opening portion 201 of the duct unit 60 to the first opening portion 73 communicating therewith as illustrated in FIG. 37, and from the fourth opening portion 202 to the second opening portion 74 communicating therewith as illustrated in FIG. 39. That is, the exposure cooling airflow flows from the opening portion 61 of the duct unit 60 including the third opening portion 201 and the fourth opening portion 202 to the cartridge tray 30 and the elevating duct 69 including the first opening portion 73 and the second opening portion 74 communicating therewith. Therefore, in order to prevent toner scattering due to air flow leakage and a decrease in cooling efficiency due to a pressure loss of airflow, it is desirable to seal a gap with each opening portion.

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As illustrated in FIGS. 26 and 38, it is desirable that a gap between the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 and the opening portion 61 of the duct unit 60 communicating therewith is sealed.

Therefore, in the present embodiment, as shown in FIG. 35, the sealing member 207 is provided so as to surround the third opening portion 201 and the fourth opening portion 202 of the duct unit 60. As the sealing member 207, an elastic body such as sponge or rubber made of urethane, silicone, or the like is used. FIG. 21 is a schematic view of the cartridge tray 30 and the elevating duct 69 as viewed from below. In FIG. 21, the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 is indicated by hatching.

As illustrated in FIG. 21, the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 is formed by the developing support member 301, the drum support member 302, and the elevating duct 69 between the developing support member 301 and the drum support member 302. More specifically, the opening portion 64 is formed by the developing guide portion 301a of the developing support member 301, the drum guide portion 302a of the drum support member 302, and the duct front wall 69F and the duct back wall 69B of the elevating duct 69 between the developing guide portion 301a and the drum guide portion 302a.

The sealing member 207 is provided on an upper surface of the duct unit 60 so as to surround four sides of each opening portion 61 including the opening portion (exhaust port) 201 of the intake duct 205 and the opening portion (intake port) 202 of the exhaust duct 206.

Among them, the sealing member 207 provided on the upper surface of one side in the longitudinal direction of the opening portion 61 of the duct unit 60 is sandwiched between the duct unit and the developing guide portion 301a of the developing support member 301 by pressing of the duct unit 60. The sealing member 207 provided on the upper surface of one side in the longitudinal direction of the opening portion 61 of the duct unit 60 is provided from the opening portion 201 on one side (front side) to the opening portion 202 on the other side (back side) in the front-back direction.

The sealing member 207 provided on the upper surface of the other side in the longitudinal direction of the opening portion 61 of the duct unit 60 is sandwiched between the duct unit and the drum guide portion 302a of the drum support member 302 by pressing of the duct unit 60. The sealing member 207 provided on the upper surface of the other side in the longitudinal direction of the opening portion 61 of the duct unit 60 is also provided from the opening portion 201 on one side (front side) to the opening portion 202 on the other side (back side) in the front-back direction.

The sealing member 207 provided on the upper surface of one side in the lateral direction of the opening portion 61 of the duct unit 60 has a side surface in contact with the side surface of the duct front wall 69F of the elevating duct 69 by pressing of the duct unit 60. The side surface of the duct front wall 69F of the elevating duct 69 with which the sealing member 207 is in contact is a surface facing the side surface of the duct back wall 69B in the front-back direction.

The sealing member 207 provided on the upper surface of the other side in the lateral direction of the opening portion 61 of the duct unit 60 has a side surface in contact with the duct back wall 69B of the elevating duct 69 by pressing of the duct unit 60. The side surface of the duct back wall 69B of the elevating duct 69 with which the sealing member 207

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is in contact is a surface facing the side surface of the duct front wall 69F in the front-back direction.

The reason why the side surfaces of the duct front wall 69F and the duct back wall 69B are in contact with the side surface of the sealing member 207 is as follows. That is, this is because the elevating duct 69 is moved to the exposure position illustrated in FIG. 22 and the retracted position illustrated in FIG. 23 by the rotation of the rotating arm 65, and the movement is not hindered by the duct unit 60.

The opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 communicates with the opening 55a of the exposure head 4 integrally supported by the elevating duct 69. Then, when the duct unit 60 is mounted on the image forming apparatus 100, the opening portion 61 of the duct unit is pressed against the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69, and the opening portion 64 and the opening portion 61 communicate with each other. Thus, a duct, which is one closed space, is formed from the duct unit 60 to the exposure head 4 via the cartridge tray 30 and the elevating duct 69.

In this manner, the sealing member 207 is pressed between the opening portion 61 of the duct unit 60 and the developing guide portion 301a and the drum guide portion 302a, and the side surface of the sealing member 207 is brought into close contact with the side surface of the duct front wall 69F and the side surface of the duct back wall 69B to seal the gap. That is, the gap between the opening portion 61 of the duct unit 60 and the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 is sealed by the sealing member 207. This prevents toner scattering due to airflow leakage and cooling efficiency reduction due to pressure loss of airflow.

(Assembly and Removal of Duct Unit)

A configuration for assembling and removing the duct unit 60 to and from the image forming apparatus 100 will be described with reference to FIGS. 2, 35, 37, 38, 21, 40, 41A to 41D, and 36. FIG. 37 is a cross-sectional view illustrating a state in which the duct unit 60 is assembled to the image forming apparatus 100. FIG. 40 is a cross-sectional view illustrating a state immediately before the duct unit 60 is assembled to the image forming apparatus 100 or immediately after the duct unit is removed from the image forming apparatus. FIGS. 41A to 41D are partially enlarged views of FIGS. 37 and 40. FIG. 36 is a perspective view of the duct unit 60 as viewed from the right front lower direction.

The duct unit 60 is inserted from one side (here, the left side) in the left-right direction of the image forming apparatus 100 toward the other side, and is moved from the lower side to the upper side by a guide (guide portion 103, support portion 104) in the apparatus at a timing of positioning. At this time, the opening portion 61 of the duct unit 60 is pressed (engaged with) against the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69 to form an air path in which the opening portion 64 and the opening portion 61 communicate with each other. Hereinafter, description will be given with reference to the drawings.

As described above, the duct unit 60 presses and closely contacts the sealing member 207 between the cartridge tray 30 and the elevating duct 69 to seal the gap, thereby preventing toner scattering due to airflow leakage and cooling efficiency reduction due to airflow pressure loss. Therefore, it is desirable that the duct unit 60 be assembled so as to rise from substantially below with respect to components assembled to the image forming apparatus 100, such as the cartridge tray 30.

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On the other hand, as illustrated in FIG. 2, the sheet cassette 12 is disposed below the duct unit 60, and it is difficult to simply assemble the duct unit 60 to the image forming apparatus 100 from below in terms of arrangement of the units. Therefore, in the present embodiment, the duct unit 60 is assembled from the side surface of the image forming apparatus 100, and slightly rises at the same time as the insertion from the side surface immediately before the assembling position.

Here, it is desirable to assemble the duct unit 60 from the left side surface of the image forming apparatus 100. A roller and a guide related to conveyance of the recording sheet P are arranged on the right side in the image forming apparatus 100, and since it is necessary to remove the roller and the guide related to the conveyance for assembling the duct unit 60, assembling from the right side surface is avoided.

As illustrated in FIGS. 37, 41A, and 41B, the supported portion 209 on the lower surface of the duct unit 60 is supported by the support portion 104 formed in the image forming apparatus 100. Note that the upper surface of the duct unit 60 receives a reaction force downward via the sealing member 207, thereby restricting the height direction. In the present embodiment, the support portion 104 is provided on a part of the sheet metal 100C bridged between the front plate 100F and the back plate 100B constituting a part of the frame body (housing) of the image forming apparatus. The supported portions 209 are provided at a total of four positions, i.e., two positions in the front-back direction on each of the right end and the left end of the duct unit 60. As a result, inclination, distortion, and the like of the duct unit 60 are corrected with respect to the image forming apparatus 100, and thus, it is possible to reliably communicate with the opening portions of the respective colors on the image forming apparatus side and to circulate air.

At this time, the opening portion 64 and the opening portion 61 face each other, and a gap between the opening portion 64 and the opening portion 61 is sealed by the sealing member 207.

As illustrated in FIGS. 37, 41A, and 41B, in the duct unit 60, a fastened portion 210 of the duct unit 60 is fixed to a fastening portion 105 of the sheet metal 100C by a fastening member 211 with respect to the image forming apparatus 100. As illustrated in FIG. 35, the fastened portions 210 are provided at two left end portions of the duct unit 60, and are disposed near the intake fan 62 and the exhaust fan 63. The fastening portions 105 are provided at two left end portions of the sheet metal 100C, and are provided at positions facing the fastened portions 210 in the insertion and removal direction of the duct unit 60. As a result, the intake fan 62 and the exhaust fan 63 are firmly fixed to the image forming apparatus 100, and it is possible to reduce generation of a chattering sound due to vibration and occurrence of an image defect such as a streak due to transmission of vibration to the drum 2 or the like.

As illustrated in FIGS. 40, 41C, and 41D, immediately before the duct unit 60 is assembled to the image forming apparatus 100 or immediately after the duct unit is removed, the sealing member 207 is separated from the cartridge tray 30 and the elevating duct 69 in the height direction. Therefore, the sealing member 207 of the duct unit 60 does not come into sliding contact with the image forming apparatus 100, and the duct unit 60 can be easily assembled to and removed from the image forming apparatus 100. At this time, a guided portion 208 of the duct unit 60 is in contact with the guide portion 103 of the image forming apparatus 100. In the present embodiment, the guide portion 103 is formed of a curved surface continuous with the support

portion **104**, and the guided portion **208** is formed of an inclined surface continuous with the supported portion **209** with respect to the assembling direction of the duct unit **60**. As a result, by assembling the duct unit **60** rightward from the left side surface of the image forming apparatus **100**, the duct unit naturally draws a track upper rightward, and can transition to the assembly state illustrated in FIGS. **37**, **41A**, and **41B**.

Then, when inserted into the image forming apparatus **100**, the duct unit **60**, which is an exposure cooling unit, is guided to the guided portion **208** having the inclined portion inclined along the insertion direction and moves upper rightward. As a result, the sealing member **207** is compressed and brought into close contact with the cartridge tray **30** and the elevating duct **69** to seal the gap between the opening portion **64** and the opening portion **61**.

(Intake Port and Exhaust Port of Duct Unit)

Next, the intake fan **62** and the exhaust fan **63** of the duct unit **60** will be described in more detail.

The exposure cooling airflow is formed as a path separate from the entire image of the development cooling airflow illustrated in FIG. **34** described above and the path of the development cooling airflow as can be understood from FIGS. **22** and **37**. Therefore, the toner leaking from the developing unit **24** is not mixed in the exposure cooling airflow, and the risk of scattering the toner into the apparatus can be reduced.

The airflow is compactly formed by arranging the intake port **203** and the exhaust port **204** of the exposure cooling airflow on the same side surface of the main body of the image forming apparatus. Furthermore, by disposing the intake port **203** and the exhaust port **204** of the exposure cooling airflow on surfaces different from the intake port **101a** and the exhaust port of the development cooling airflow, mutual influence with the development cooling airflow can be minimized.

For example, in the present embodiment, the exhaust port of the development cooling airflow is disposed on the back surface of the apparatus main body, whereas the intake port **203** of the exposure cooling airflow is disposed on the left side surface of the apparatus main body. As a result, it can be said that the airflow for exposure cooling hardly takes in the exhaust heat of image formation by development, and vice versa.

In addition, even in a case where the temperature rise condition of the developing unit **24** and the temperature rise condition of the exposure head **4** are different due to the image and the sheet passing mode, the cooling paths and the fans of the exposure cooling unit and the development cooling unit are separate. Therefore, optimal cooling can be performed in each case, and efficient control with a high degree of freedom can be performed as the airflow control.

(Configuration of Fan of Duct Unit)

Here, details of the arrangement of the intake fan **62** and the exhaust fan **63** will be described with reference to FIG. **42**.

FIG. **42** is a cross-sectional view of the duct unit taken along line F-F in FIGS. **37** and **39** as viewed from above the image forming apparatus. In FIG. **42**, the arrangement of the intake fan **62** and the exhaust fan **63**, which is simply illustrated in FIG. **34**, is illustrated in detail.

In FIG. **42**, the intake fan **62** is disposed at an angle θF with respect to a direction perpendicular to a left side surface of the image forming apparatus **100**, that is, a longitudinal direction of the intake duct **205** and the exhaust duct **206**. Similarly, the exhaust fan **63** is disposed at an angle θR with respect to the longitudinal directions of the intake duct **205**

and the exhaust duct **206**. As a result, the intake direction by the intake fan **62** is a direction from one end side in the longitudinal direction of the substrate **50** toward the center side in the longitudinal direction, and the exhaust direction by the exhaust fan **63** is a direction from the center side in the longitudinal direction of the substrate **50** toward the other end side in the longitudinal direction. As indicated by arrows in FIG. **42**, the angles θF and θR are angles at which the intake angle by the intake fan **62** and the exhaust angle by the exhaust fan **63** are opposite to each other with respect to the horizontal direction (upward and downward). For this reason, the intake of the fresh air by the intake fan **62** hardly takes in the heat included in the exhaust of the exhaust fan **63**, which contributes to the suppression of the decrease in the cooling efficiency due to the circulation of the exposure cooling airflow.

In FIG. **42**, the angle θF and the angle θR are set at relative angles so as to be obliquely upward and obliquely downward, but may be set at relative angles so as to be obliquely leftward and obliquely rightward. Further, a relative angle obtained by combining these angles may be used. That is, it is preferable to set the relative angle such that the central axis of the intake air and the central axis of the exhaust air are separated from each other with increasing distance from the image forming apparatus **100** so that the central axis of the intake air and the central axis of the exhaust air do not intersect with each other outside the image forming apparatus **100**.

It can be understood that, by attaching the fan to the duct unit **60** as described above, scattering of toner to the inside of the image forming apparatus can be reduced, and a cooling unit of the LED exposure device with high efficiency and high degree of freedom can be provided.

(Cooling Configuration of Developing Unit)

Next, a cooling configuration of the developing unit **24** will be described with reference to FIG. **34**. FIG. **34** is a cross-sectional view of the image forming apparatus taken along line A-A in FIG. **2**. In FIG. **34**, a flow of air for cooling the developing unit **24** is indicated by an alternate long and short dash line. The flow of air indicated by the alternate long and short dash line in FIG. **34** is also referred to as a development cooling airflow.

The developing unit **24** contains the screw **7** rotating at a high speed and the toner circulating at a high speed as described above, and with this operation, frictional heat is generated in the bearing portion of the screw **7** and the toner, and the frictional heat is stored in the developing unit **24**, and the temperature rises. When the image formation is completed, the heat storage of the developing unit **24** is completed, and the developing unit is gradually cooled. However, while the image formation is continued, the heat storage is performed as long as the heat capacity of the developing unit **24** allows, and the temperature rises. Since the toner has a property of being easily melted by heat, when the temperature of the developing unit **24** rises to a certain temperature or higher, the toner is fused inside the developing unit **24**, and a coating failure of the developing sleeve **5** occurs, so that the toner image is disturbed, leading to an image defect.

Therefore, there is a need for a cooling configuration in which the developing unit **24** is cooled so that the temperature of the developing unit **24** does not rise to a certain temperature or higher.

The image forming apparatus **100** includes a fan **40** and the front-side duct **41** for sending air from the outside of the apparatus to each developing unit **24**. The image forming apparatus **100** includes the back-side duct **42** for exhausting

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air from each developing unit **24** to the outside of the apparatus, and a toner filter **43**. The image forming apparatus **100** includes the duct formed by the developing unit **24** and the cartridge tray **30**.

The duct formed between the developing unit **24** and the cartridge tray **30** in the axial direction of the photosensitive drum is disposed between the front-side duct **41** disposed on the front side of the apparatus and the back-side duct **42** disposed on the back side of the apparatus. In the duct formed between the developing unit **24** and the cartridge tray **30** in the axial direction of the photosensitive drum, one end side in the axial direction which is the front side of the apparatus communicates with the front-side duct **41**, and the other end side in the axial direction which is the back side of the apparatus communicates with the back-side duct **42**, so that one closed space is formed.

The fan **40** is provided on the right side of the front surface of the apparatus main body of the image forming apparatus **100**, and sucks air outside the apparatus from an intake port **101a** provided on the right side surface side of the image forming apparatus **100** of the front cover **101**. The front-side duct **41** is disposed inside the front cover **101** and extends in the left-right direction, which is the direction in which the developing units **24** are arranged. The front-side duct **41** includes the opening **41a** at a position corresponding to each developing unit **24**. The opening **41a** of the front-side duct **41** is provided at a position facing the opening **102c** of the inner door **102** of each cartridge tray **30** in the axial direction of the photosensitive drum, and communicates with each other by closing the front cover **101**. The opening **102c** of each inner door **102** is provided at a position corresponding to the opening on one end side in the longitudinal direction of the closed space formed between the developing unit **24** and the developing support member **301**, and communicates with each other by closing the inner door **102**.

The back-side duct **42** includes the opening **42a** at a position corresponding to each developing unit **24**. The opening **42a** of the back-side duct **42** is provided at a position corresponding to the opening on the other end side in the longitudinal direction of the closed space formed between the developing unit **24** and the developing support member **301** in the axial direction of the photosensitive drum, and communicates with each other.

As described above, the duct that is the closed space formed between the developing unit **24** and the cartridge tray **30** forms a part of the duct that is one closed space communicating with the front-side duct **41** and the back-side duct **42**. The duct formed between the developing unit **24** and the cartridge tray **30** and the front-side duct **41** and the back-side duct **42** communicating with the duct form a first cooling duct that is the one closed space serving as a flow path of the development cooling airflow.

Note that the first cooling duct forms a closed space serving as a flow path of the development cooling airflow for cooling the developing unit **24**. That is, the first cooling duct is a development cooling unit that cools the developing unit that is a developing unit. However, the first cooling duct serving as the development cooling unit may be partially formed by a duct serving as a closed space formed between the developing unit **24** and the cartridge tray **30**, and the other configuration is not limited to the above-described configuration.

Each of the developing units **24** is cooled by the above-described development cooling airflow (alternate long and short dash line illustrated in FIG. **34**) flowing through the one closed space.

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The development cooling airflow indicated by the alternate long and short dash line in FIG. **34** is first generated by the fan **40** disposed on the front right side of the image forming apparatus and the first cooling duct that is the one closed space described above.

When the fan **40** rotates, the air outside the apparatus is sucked from the intake port **101a** of the front cover **101** provided on the right side surface of the image forming apparatus **100**, and is sent to the developing unit **24** to be cooled via the opening **41a** of the front-side duct **41** disposed inside the front cover **101** and the opening **102c** of the inner door **102**.

The air sent to the developing unit **24** is taken in from the opening on the front side in the front-back direction of the duct formed between the developing unit **24** and the cartridge tray **30**, sent along the axial direction of the photosensitive drum, and exhausted from the opening on the back side in the front-back direction.

The air exhausted from the back side in the front-back direction of the duct formed between the developing unit **24** and the cartridge tray **30** passes through the opening **42a** of the back-side duct **42**, collectively passes through the toner filter **43**, and then is exhausted from the back side of the apparatus to the outside of the apparatus.

Here, the toner filter **43** will be described. The toner filter **43** is disposed immediately before the exhaust port on the back side of the apparatus in the back-side duct **42**. Since the development cooling airflow is an airflow passing around the developing unit **24**, it is inevitable that a small amount of toner is taken into the airflow. Therefore, it is desirable to dispose the toner filter **43** immediately before the exhaust port of the development cooling airflow so as not to discharge even a small amount of toner to the outside of the apparatus.

In the cooling by the airflow, in general, the airflow is mainly formed using an inexpensive fan, and the same applies to the airflow other than the development cooling. (Cooling Configuration of Exposure Head)

Next, a cooling configuration of the exposure head **4** will be described with reference to FIG. **34**. In FIG. **34**, a flow of air for cooling the exposure head **4** is indicated by a broken line. The flow of air indicated by the broken line in FIG. **34** is also referred to as an exposure cooling airflow.

Since the exposure head **4** dissipates heat according to the light emission amount of the light emitting diode (LED) **51** and is disposed close to the developing unit **24** using toner weak against heat, a cooling unit is required. In particular, in a case where the image forming process is repeated at a high frequency, that is, in a case where the image forming process is used in apparatus with high productivity, or in a case where an image with high density is continuously output, the light emission time is long, and the light emission amount is also large. Therefore, the amount of heat generated from the LED **51** and the circuit on the substrate **50** on which the LED is mounted also increases.

As a countermeasure against this, for example, the housing **54** of the exposure head **4** is also used as a heat sink, and the exposure head **4** is configured to easily dissipate heat and hardly store heat. However, even in such a case, it is conceivable that the cooling of the exposure head **4** is not in time, heat storage proceeds, and the heat dissipated to the surroundings also increases. As a result, the toner around the developing sleeve **5** included in the developing unit **24** and a part of the circulating toner inside the developing unit **24** are fused, which may affect the toner coating layer on the surface of the developing sleeve **5**, leading to an image defect.

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Even in a case where the configuration for cooling the developing unit 24 is provided, it is easily assumed that the heat storage caused by the light emission of the LED 51 is superior in a portion where the exposure head 4 of the developing unit 24 is close. Therefore, it is desirable to provide a cooling configuration (exposure cooling airflow) of the exposure head 4 for cooling the exposure head 4 and discharging heat to the outside of the apparatus separately from the cooling configuration (development cooling airflow) of the developing unit 24 to reduce the amount of heat dissipated to the periphery of the exposure head 4.

As illustrated in FIG. 8, the developing unit 24 and the developing sleeve 5 of the developing unit 24 are disposed adjacent to the exposure head 4. The surface of the developing sleeve 5 is coated with toner, and due to its structure, the toner adheres to the vicinity of the bearing portions at both ends of the sleeve, and the toner also adheres to the periphery of the developing unit 24. This is because the developing sleeve 5 and the screw 7 are rotated at a high speed, the stirred toner is stirred up, and the toner is peeled off the surfaces of the developing sleeve 5 and the screw 7. In addition, due to an increase in the internal pressure of the developing unit 24 caused by high-speed rotation of the developing sleeve 5 and the screw 7, toner may be ejected from the gap of the developing unit 24 to the outside.

Therefore, the cooling configuration of the exposure head 4 is desirably a configuration in which these toners are not caught and mixed. In other words, in configuring the exposure cooling airflow separately from the development cooling airflow, it is desirable that the toner around the developing unit 24 adjacent to the exposure head 4 is not caught nor mixed.

The image forming apparatus 100 includes the exposure head 4, the elevating duct 69, the cartridge tray 30, and the duct unit 60. The exposure head 4 is mounted on the elevating duct 69 disposed in the image forming apparatus 100, and is integrated with the elevating duct 69. When the exposure head 4 is mounted on the elevating duct 69, the opening 55a of the housing support member 55 of the exposure head 4 communicates with the opening 69a of the elevating duct 69. The elevating duct 69 is disposed between the first developing guide portion 301a and the first drum guide portion 302a of the cartridge tray 30, and forms a duct that communicates the exposure head 4 and the duct unit 60 together with the cartridge tray 30. The duct unit 60 is mounted on the image forming apparatus 100. When the duct unit 60 is mounted on the image forming apparatus 100, the opening portion 61 of the duct unit 60 communicates with the opening portion 64 formed by the elevating duct 69 and the cartridge tray 30.

In this manner, the housing support member 55, the elevating duct 69, the cartridge tray 30, and the duct unit 60 of the exposure head 4 form the second cooling duct which is one continuous closed space. Each exposure head 4 is cooled by an exposure cooling airflow (broken line illustrated in FIG. 34) flowing through the one closed space formed by the housing support member 55, the elevating duct 69 communicating with the housing support member, and the duct unit 60 communicating with the elevating duct and the cartridge tray 30.

The second cooling duct that is one closed space serving as the flow path of the exposure cooling airflow indicated by the broken line in FIG. 34 is configured separately from the first cooling duct that is one closed space serving as the flow path of the development cooling airflow indicated by the alternate long and short dash line in FIG. 34.

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The first cooling duct that circulates the airflow for cooling the developing unit 24 and the second cooling duct that circulates the airflow for cooling the exposure head 4 are separated by the first developing guide portion 301a of the developing support member 301 and the elevating duct 69. In other words, the first developing guide portion 301a of the developing support member 301 and the elevating duct 69 separate the first cooling duct that circulates the airflow for cooling the developing unit 24 and the second cooling duct that circulates the airflow for cooling the exposure head from each other between the exposure head 4 and the developing unit 24.

The second cooling duct forms a closed space serving as a flow path of an exposure cooling airflow for cooling the exposure head 4. That is, the second cooling duct is an exposure cooling unit that cools the exposure head that is an exposure unit. However, the second cooling duct serving as the exposure cooling unit is not limited to the above-described configuration as long as a closed space different from the first cooling duct is formed by the elevating duct 69 serving as the exposure support member, the cartridge tray 30 serving as the support member, and the duct unit 60 serving as the exposure cooling unit.

As described above, the duct unit 60, the cartridge tray 30, the elevating duct 69, and the housing support member 55 form a continuous closed space to constitute an exposure cooling airflow. As illustrated in FIG. 35, the intake fan 62 and the exhaust fan 63 of the duct unit 60 face the outside of the apparatus only through the exterior cover of the apparatus. The flow path of the exposure cooling airflow is completed by a minimum path formed by directly taking in air from the louver of the exterior cover into the duct unit 60 and directly exhausting air from the duct unit 60. Therefore, the intake and exhaust flow hardly affects the atmosphere air inside the apparatus.

Note that there is a finisher as an option on the sheet ejection side of the image forming apparatus 100, and if the finisher is mounted, substantially the entire region of the left side surface of the image forming apparatus 100 facing the intake fan 62 and the exhaust fan 63 is closed by the finisher. In this case, intake and exhaust by the intake fan 62 and the exhaust fan 63 are performed to the inside of the finisher, but the inside of the finisher has many cavities. Therefore, the louver (not illustrated) of the exterior cover is disposed so as to avoid the front of the main structure inside the finisher. As a result, it is possible to reduce a decrease in performance of the exposure cooling airflow to an extent that there is no problem in practical use.

(Sealing Configuration by Sealing Member)

Next, a sealing configuration of the first cooling duct that is one closed space serving as a flow path of the development cooling airflow and the second cooling duct that is provided separately from the first cooling duct and serves as a flow path of the exposure cooling airflow will be described with reference to FIGS. 22 and 23.

Specifically, a sealing configuration in which a gap between the cartridge tray 30 and the elevating duct 69 is sealed with seals 71 and 72 which are sealing members, and a sealing configuration in which a gap between the developing unit 24 and the cartridge tray 30 is sealed by the seal 70 which is a sealing member will be described.

As illustrated in FIGS. 37 and 39, the cartridge tray 30, the elevating duct 69, and the housing support member 55 form the second cooling duct, and form the exposure cooling airflow for the back surface of the substrate 50 on which the LED 51 is mounted through the opening 55a of the housing support member 55.

The entire width of the lower surface of the elevating duct 69 forms an opening, and the maximum air volume is secured with respect to the substrate 50 disposed immediately above, which is advantageous for cooling the exposure head 4.

Here, when the substrate 50 has a shape extended long in the axial direction of the photosensitive drum 2, cooling efficiency is more excellent when the exposure cooling airflow is a flow orthogonal to the longitudinal direction of the substrate 50 than a flow parallel to the longitudinal direction of the substrate. Also in the duct cross section illustrated in FIG. 37, the airflow flowing through the inside of the second cooling duct with respect to the longitudinal direction of the substrate 50 to be cooled is configured to be a flow substantially orthogonal to the substrate 50. This configuration is advantageous for cooling even at the angle of the exposure cooling airflow.

Here, in the second cooling duct, the cartridge tray 30 and the elevating duct 69 form a part of duct of the second cooling duct. A gap between the cartridge tray 30 and the elevating duct 69 that form the part of the duct is closed by seals 71 and 72 so that the airflow does not leak to the outside of the duct.

That is, the image forming apparatus 100 includes the seals 71 and 72 which are sealing members for sealing the gap between the cartridge tray 30 and the elevating duct 69.

In a state where the exposure head 4 is at the exposure position as illustrated in FIG. 22, the gap between the elevating duct 69 that integrally supports the exposure head 4 and the developing support member 301 of the cartridge tray 30 is sealed with the seal 72 that is a sealing member.

Further, in a state where the exposure head 4 is at the exposure position, the gap between the elevating duct 69 that integrally supports the exposure head 4 and the drum support member 302 of the cartridge tray 30 is sealed with the seal 71 which is a sealing member.

The seal 72 is provided on the first inclined surface 69L1 of the duct left wall 69L which is a side wall of the elevating duct 69 on the developing unit 24 side. As illustrated in FIGS. 27 and 28, the seal 72 is provided on the first inclined surface 69L1 of the duct left wall 69L over the range La from the duct front wall 69F to the duct back wall 69B in the longitudinal direction. The seal 72 seals the gap between the first inclined surface 69L1 of the duct left wall 69L of the elevating duct 69 and the facing portion 301d of the first developing guide portion 301a of the developing support member 301 facing the first inclined surface 69L1 at the exposure position of the exposure head 4. Here, the configuration in which the seal 72 is provided on the elevating duct 69 side is exemplified, but the seal may be provided on the developing support member 301 side.

The seal 71 is provided on the second inclined surface 69R1 of the duct right wall 69R which is a side wall of the elevating duct 69 on the drum unit 23 side. The seal 71 is provided on the second inclined surface 69R1 of the duct right wall 69R over the range La from the duct front wall 69F to the duct back wall 69B in the longitudinal direction. The seal 71 seals the gap between the second inclined surface 69R1 of the duct right wall 69R of the elevating duct 69 and the facing portion 302d of the first drum guide portion 302a of the drum support member 302 facing the second inclined surface 69R1 at the exposure position of the exposure head 4. Here, the configuration in which the seal 71 is provided on the elevating duct 69 side is exemplified, but the seal may be provided on the drum support member 302 side.

FIG. 24 illustrates a state in which the exposure head 4 is disposed close to the photosensitive drum 2. That is, the exposure head 4 is disposed at the exposure position with respect to the photosensitive drum 2. As described above, this is maintained by the rotating arm 65 pressing the bottom surfaces of both end portions of the elevating duct 69 upward. This pressing is ensured by a predetermined spring pressure by the arm pressure spring 67 which is a torsion coil spring. That is, the seals 71 and 72 are crushed by pressing of the elevating duct 69 against the developing support member 301 and pressing of the elevating duct 69 against the drum support member 302 in a state where the exposure head 4 illustrated in FIGS. 22 and 24 is at the exposure position, and ensure sealing of the gap.

On the other hand, in a case where the exposure head 4 is retracted from the photosensitive drum 2, as illustrated in FIG. 25, the rotation of the rotating arm 65 in a direction in which the exposure head 4 is retracted from the photosensitive drum 2 is a starting point. As a result, the engagement boss 66 provided at the distal end of the rotating arm 65 pushes down the engagement ribs 69d1 and 69e1 arranged at the lower end of the elevating duct 69, and retracts the exposure head 4 integrated with the elevating duct 69 from the photosensitive drum 2.

As illustrated in FIGS. 25 and 23, in a state where the exposure head 4 is at the retracted position, the seals 71 and 72 are separated from the developing support member 301 and the drum support member 302 with which the seals abut, respectively, and the sealing of the gap is released.

In other words, when the exposure head 4 is moved to the retracted position, the elevating duct 69 is moved in a direction in which the gap between the developing support member 301 and the drum support member 302 is widened as compared with the gap at the exposure position. That is, when the exposure head 4 is moved to the retracted position, the seals 71 and 72 are separated by moving in a direction of moving the exposure head 4 away from the developing support member 301 and the drum support member 302, and the sealing of the gap is released.

As described above, at the exposure position of the exposure head 4, the gap between the cartridge tray 30 and the elevating duct 69 is sealed with the seals 71 and 72 as sealing members over the range La in the longitudinal direction. As a result, the air flowing to the exposure head 4 through between the cartridge tray 30 and the elevating duct 69 does not leak from the gap to the space around the developing unit 24 and the developing sleeve 5. In addition, the air flowing to the exposure head 4 through between the cartridge tray 30 and the elevating duct 69 does not leak from the gap to the space around the photosensitive drum 2 and the charging roller 3. Therefore, the possibility that the toner is mixed into the exposure cooling airflow, which is the flow of air for cooling the exposure head 4, is small, and the toner scattering to the inside of the image forming apparatus can be reduced.

The development cooling airflow is guided in the front-back direction by a duct formed between the developing unit 24 and the cartridge tray. In the first cooling duct, the duct formed between the cartridge tray 30 and the developing unit 24 forms a part of duct of the first cooling duct. The gap between the cartridge tray 30 and the developing unit 24 that form the part of the duct is closed by the seal 70 so that the development cooling airflow does not leak in the direction of the developing sleeve 5.

That is, the image forming apparatus 100 includes the seal 70 which is a sealing member for sealing the gap between the cartridge tray 30 and the developing unit 24.

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At a pressing position of the developing unit **24** illustrated in FIG. **22**, a gap between the developing unit **24** and the developing support member **301** that separates the developing unit **24** and the exposure head **4** is sealed with the seal **70** that is a sealing member.

The seal **70** is provided in the first developing guide portion **301a** of the developing support member **301** that separates the developing unit **24** and the exposure head **4** between the developing unit **24** and the exposure head **4**. The seal **70** is provided between the frame body of the developing unit **24** and the first developing guide portion **301a** of the developing support member **301** at a portion that is narrowed as the developing unit **24** moves toward the developing position. Here, the seal **70** is provided at an end portion (partition wall portion **301e**) of the first developing guide portion **301a** of the developing support member **301** on the developing roller side in a state where the developing unit **24** is at the developing position. The seal **70** is provided on a surface of the partition wall portion **301e** of the developing support member **301** facing the frame body of the developing unit **24**. The seal **70** is provided from one end portion to the other end portion in the longitudinal direction in the first developing guide portion **301a** of the developing support member **301**. The seal **70** seals the gap between the partition wall portion **301e**, which is an end portion of the developing support member **301** on the developing roller side, and the frame body of the developing unit **24** facing the partition wall portion **301e**. Here, the configuration in which the seal **70** is provided on the developing support member **301** side is exemplified, but may be provided on the developing unit **24** side.

As illustrated in FIG. **31**, the development stay **31** is slid in the forward direction **F** in conjunction with the operation of closing the inner door **102**. At this time, the developing unit **24** is moved upward (arrow **U**) along the inclined surface of the development pressurizing frame **32** and the development pressurizing frame **33** of the development stay **31**. As a result, the developing sleeve **5** of the developing unit **24** is moved in a direction approaching the photosensitive drum **2** of the drum unit **23**, and the developing sleeve **5** is pressed against the photosensitive drum **2**.

When the developing unit **24** is moved to the developing position illustrated in FIGS. **22** and **24**, the developing unit is moved in a direction of narrowing the gap between the developing unit and the first developing guide portion **301a** of the developing support member **301** that separates the developing unit from the exposure head **4**. That is, when the developing unit **24** illustrated in FIGS. **22** and **24** moves to the developing position, the seal **70** is crushed by moving in a direction in which the developing unit **24** approaches the developing support member **301**, and seals the gap.

Also, as illustrated in FIG. **32**, the development stay **31** is slid in the backward direction **B** in conjunction with the operation of opening the inner door **102**. At this time, the developing unit **24** is moved downward (arrow **D**) along the inclined surface of the development pressurizing frame **32** and the development pressurizing frame **33** of the development stay **31**. As a result, the developing sleeve **5** of the developing unit **24** is moved to a separated position away from the photosensitive drum **2** of the drum unit **23** further than the case of the development.

When the developing unit **24** is moved to the retracted position illustrated in FIGS. **25** and **23**, the developing unit is moved in a direction in which the gap between the developing unit and the first developing guide portion **301a** of the developing support member that separates the developing unit from the exposure head **4** is wider than the gap

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illustrated in FIG. **22**. That is, at the retracted position of the developing unit **24** illustrated in FIGS. **25** and **23**, the seal **70** is separated by moving in the direction of moving the developing unit **24** away from the developing support member **301**, and the sealing of the gap is released.

As described above, in a state where the developing unit **24** is at the developing position, the seal **70** seals the gap between the end portion (partition wall portion **301e**) of the developing support member **301** on the developing sleeve side and the frame body of the developing unit **24** facing the end portion. As a result, the air flowing between the developing unit **24** and the cartridge tray **30** does not leak from the gap toward the developing sleeve **5**.

Since the development cooling airflow flows around the developing unit **24** as indicated by the alternate long and short dash line in FIG. **34**, a small amount of toner may be contained. Therefore, by shielding the path between the air flows in this manner, the toner is not mixed from the development cooling airflow. In the exposure cooling airflow, the possibility of scattering toner inside the apparatus can be reduced by considering that the duct is constituted by a minimum path that performs intake and exhaust directly with respect to the outside of the apparatus.

Here, the configuration in which the seals **71**, **72**, and **73** are separated by moving the elevating duct **69** and the developing unit **24** in the direction away from the target member has been exemplified, but the present invention is not limited thereto. By moving the elevating duct **69** and the developing unit **24** in a direction away from the target member, the seals **71**, **72**, and **73** can release the sealing of the gap by reducing the pressing even if they are not separated.

In addition, here, a foamed sealing material made of a rubber sponge material is used for the seals **71**, **72**, and **73** which are sealing members. As the dimension of the seal, a configuration is adopted in which the thickness of the seal is increased with respect to the width of the gap described above, the difference is set as a crushing amount of the seal, and bubbles inside are crushed to increase the sealing force.

The sealing member used for sealing is not limited to the foamed sealing material, and may be, for example, a rubber sheet material (sheet material) such as a urethane sheet. In this case, a sheet longer than the width of the gap described above is used, and sealing is performed by making a sheet end follow an abutting counterpart. In order to increase the sealing force, it is only required to increase the abutting force with respect to the counterpart by setting the length and the abutting angle of the sheet unlike the foamed sealing material.

Even if the material of the seal is either a rubber sponge material or a rubber sheet material, it is necessary to attach the seal by a bonding unit such as a double-sided tape. Since the adhering area is equal to or smaller than the width of the seal and is small, when the seal is subjected to a load such as bending or shifting in the lateral direction thereof, the bonding portion peels off, which may lead to curling, peeling off, and falling off of the seal. Therefore, a configuration is adopted in which, in bonding and attaching the seal, the load applied in the lateral direction is minimized.

As a more general sealing material, a bellows-like rubber material or the like is conceivable, but since a predetermined space is required for expansion and contraction of the bellows, it is difficult to say that the assemblability is good, and it is not assumed to be adopted in the present embodiment.

(Angle of Sealing Member)

Here, the seals **71** and **72** which are sealing members will be described in more detail.

The seals **71** and **72**, which are sealing members, reduce the risk of toner mixing into the flow path of the exposure cooling airflow and the flow path of the development cooling airflow, and reduce toner scattering inside the apparatus. Therefore, it is necessary to prevent the sealing member from being curled, peeled, dropped, or the like.

Here, the moving direction of the exposure head **4** from the retracted position illustrated in FIG. **25** to the exposure position illustrated in FIG. **24** is indicated by an arrow **G** in FIG. **23**. The seals **71** and **72** are attached to the elevating duct **69** at predetermined angles $\theta 1$ and $\theta 2$ with respect to the direction of arrow **G**, which is the moving direction of the exposure head **4**.

Specifically, the seal **71** is provided on the second inclined surface **69R1** of the duct right wall **69R** which is a side wall of the elevating duct **69** on the drum unit **23** side. The second inclined surface **69R1** of the duct right wall **69R** is inclined at a predetermined angle $\theta 1$ with respect to the direction of the arrow **G**. Here, the configuration in which the inclined surface inclined at the angle $\theta 1$ is provided in the elevating duct **69** has been exemplified, but the abutting surface of the seal **71** may be inclined at the angle $\theta 1$.

The seal **72** is provided on the first inclined surface **69L1** of the duct left wall **69L** which is a side wall of the elevating duct **69** on the developing unit **24** side. The first inclined surface **69L1** of the duct left wall **69L** is inclined at a predetermined angle $\theta 2$ with respect to the direction of the arrow **G**. Here, the configuration in which the inclined surface inclined at the angle $\theta 2$ is provided in the elevating duct **69** has been exemplified, but the abutting surface of the seal **72** may be inclined at the angle $\theta 2$.

In the case of the rubber sponge material, the sealing force of the gap by the seals **71** and **72** is determined by the crushing amount of the seal as described above. Therefore, as long as a predetermined seal crushing amount is ensured at the exposure position shown in FIG. **24** where the exposure head **4** is close to the photosensitive drum **2**, there is no influence on sealing of the gap between the elevating duct **69** and the cartridge tray **30** regardless of the angle of attachment of the seals **71** and **72**. In addition, in FIG. **22**, the seals **71** and **72** are mainly subjected to a load in a direction of crushing the thickness, and the load does not cause curling, peeling off, or falling off of the seal.

Next, the load on the seals **71** and **72** due to the moving operation of the exposure head **4** will be described with reference to FIG. **23**.

Similarly to FIG. **22**, the position of the cross section illustrated in FIG. **23** is a position in the direction of arrows **X-X** in FIG. **20**, and FIG. **23** is a cross-sectional view of a range **La** illustrated in FIG. **20**. Note that, in FIG. **22**, the exposure head **4** is at an exposure position close to the photosensitive drum **2**, whereas in FIG. **23**, the exposure head **4** has moved to the retracted position retracted from the photosensitive drum **2**. This is a position corresponding to FIG. **25** in which the exposure head **4** is retracted from the exposure position to the retracted position.

In FIG. **23**, the seals **71** and **72** including the seal **70** are separated from the members with which the seals abut, and the sealing of the gap is released. That is, in a state where the exposure head **4** is retracted from the exposure position and the developing unit **24** is retracted from the developing position, sealing of the gap by the seals **71**, **72**, and **73** is released.

This is because sealing of the gap is not necessary in the middle of the moving operation of the exposure head **4**, and is only required to be performed when the exposure head **4** is positioned at the exposure position illustrated in FIG. **22** with respect to the photosensitive drum **2** and the image forming process starts.

In the state (retracted position) illustrated in FIG. **23**, no load is applied to the seals **71** and **72**. Therefore, the seals **71** and **72** are not curled, peeled, or dropped.

The load on the seals **71** and **72** during the moving operation of the exposure head **4** may be considered by moving the seals **71** and **72** in the direction of the arrow **G** from the state illustrated in FIG. **23**.

In the seal **71**, until the elevating duct **69** is in the state (exposure position) shown in FIG. **22**, the seal **71** provided on the second inclined surface **69R1** of the elevating duct **69** does not abut on the facing portion **302d** of the first drum guide portion **302a** facing the seal, and is arranged not to receive a load.

In the seal **72**, the corner portion of the seal **72** comes into contact with the cartridge tray **30** after the start of movement, but only receives an oblique load with respect to the attachment surface of the seal **72**, and it can be said that the risk of causing the seal **72** to be curled or peeled is small. The oblique load on the seal **72** is determined by the attachment angle $\theta 2$ of the seal **72**, and as the angle $\theta 2$ is larger, the component force that induces curling or peeling of the seal **72** becomes smaller. For example, in a case where the angle $\theta 2$ is 90° , that is, in a case where the seal **72** is attached perpendicularly to the moving direction (the direction of the arrow **G**) of the exposure head **4**, the seal does not receive a load during the moving operation of the exposure head **4**. Therefore, similar to the state of the retracted position illustrated in FIG. **34**, the seal **72** is not curled or peeled off.

However, as the angle $\theta 1$ and the angle $\theta 2$ are increased, the area required for the movement of the seals **71** and **72** by the moving operation of the exposure head **4** is increased, so that it is difficult to make the moving mechanism of the exposure head **4** compact. Therefore, it is necessary to set the angles $\theta 1$ and $\theta 2$ within a predetermined range. In the present embodiment, the angles $\theta 1$ and $\theta 2$ are set to fall within an appropriate range of 20° to 90° ($20^\circ \leq \theta 1 \leq 90^\circ$, $20^\circ \leq \theta 2 \leq 90^\circ$).

As described above, the load on the seals **71** and **72** due to the moving operation of the exposure head **4** can be minimized. Therefore, the seals **71** and **72** are not curled, peeled, or dropped off during the moving operation of the exposure head **4**. In addition, rubbing of the seals **71** and **72** does not become resistance to the moving operation itself of the exposure head **4**. Therefore, the moving operation of the exposure head is stabilized.

With such a configuration, it is possible to provide a cooling unit of the exposure head **4** which is compatible with the moving mechanism of the exposure head **4** and has a sealing property for reducing toner scattering into the apparatus.

(Cooling Control of Exposure Head and Cooling Control of Developing Unit)

Next, cooling control of the exposure head **4** and cooling control of the developing unit **24** will be described.

Cooling control of the exposure head **4** is performed by controlling the intake fan **62** and the exhaust fan **63** in the duct unit **60** based on detection signals of temperature detection sensors (not illustrated) disposed on the substrates **50** of the respective colors. As a result, the intake fan **62** and the exhaust fan **63** of the duct unit **60** are not always rotating,

and start rotating when the temperature detected by the temperature detection sensor reaches a predetermined threshold. As described above, by minimizing the operation of the exposure cooling airflow by the intake fan 62 and the exhaust fan 63 and also minimizing the air volume, it is possible to reduce toner scattering into the image forming apparatus 100 from the viewpoint of control.

On the other hand, the cooling control of the developing unit 24 is performed by controlling the fan 40 provided on the apparatus front surface side based on a detection signal of an internal temperature sensor (not illustrated) arranged inside the image forming apparatus 100 separately from the temperature detection sensor. In other words, the cooling control of the developing unit is performed by controlling the fan 40, and is controlled differently from the intake fan 62 and the exhaust fan 63 in the duct unit 60. As a result, control is performed to perform optimum and minimum cooling for the temperature rise state of the developing unit 24.

Temperature rise conditions of the developing unit 24 and the exposure head 4 are different from each other. Therefore, as described above, the cooling control of the exposure head 4 and the cooling control of the developing unit 24 are different from each other. As a result, in each cooling control, the operation of the fan and the air volume of the fan can be minimized, and the toner scattering into the image forming apparatus 100 can be further reduced.

With such a configuration, it is possible to provide a cooling unit of the exposure head 4 capable of reducing the possibility of occurrence of image defects and toner adhesion to the user without scattering toner inside the image forming apparatus 100.

FIG. 26 is a cross-sectional view taken along line E-E in FIG. 37. In FIG. 26, the FFC 58, the drum unit 23, and the developing unit 24, which are not illustrated in FIG. 37, are illustrated.

In the duct unit 60, the intake fan 62 disposed closer to the front of the image forming apparatus 100 functions as an intake fan that takes in air from the outside of the apparatus. Therefore, when the intake fan 62 rotates, air is taken into the intake duct 205 from the outside of the apparatus through the intake port 203. The air taken in from the outside of the apparatus flows from the left side to the right side of the image forming apparatus 100 along the intake duct 205 as indicated by a dotted line (intra-duct airflow 310) in FIG. 37. As shown in FIG. 37, the air flowing from the left side to the right side of the apparatus in the intake duct 205 flows through the third opening portion 201 for each color provided on the upper surface of the intake duct 205 while branching from the left side of the apparatus to the opening portions 201Y, 201M, 201C, and 201K in this order.

The air sent from the third opening portion 201 of the duct unit 60 is sent upward from the first opening portion 73 communicating with the opening portion 201 through the space between the cartridge tray 30 and the elevating duct 69. The air sent upward through the space between the cartridge tray 30 and the elevating duct 69 is blown to the back surface of the substrate 50 of the exposure head 4 through the opening 69a of the elevating duct 69 and the opening 55a of the exposure head 4 communicating with each other in the vertical direction.

Here, the first opening portion 73 is an opening portion which communicates with the third opening portion 201 of the duct unit 60 on the front side of the apparatus in the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69.

That is, the first opening portion 73 is disposed closer to one end side than the center of the substrate 50 in the longitudinal direction, and faces the third opening portion 201 when the duct unit 60 is mounted.

The air blown against the back surface of the substrate 50 of the exposure head 4 near the front of the image forming apparatus 100 tends to flow in the space between the left side wall 55L and the right side wall 55R of the housing support member 55 along the longitudinal direction of the substrate 50. At this time, in the exposure head 4, the flow of air in the direction toward the connector region is blocked by the shielding wall 76. Therefore, the airflow blown against the back surface of the substrate 50 flows in the duct region from one side (front side) to the other side (back side).

The exposure head 4 and the duct unit 60 communicate with each other by the duct (cooling duct 75) formed by the cartridge tray 30 and the elevating duct 69. Therefore, the airflow blown against the back surface of the substrate 50 is guided in the duct (for example, from the front side to the back side in the longitudinal direction of the substrate), and the substrate 50 is cooled in the process.

At the same time as the intake described above, in the duct unit 60, the exhaust fan 63 disposed close to the back side of the image forming apparatus 100 functions as an exhaust fan that exhausts air from the inside of the duct unit 60 to the outside of the apparatus. Therefore, when the exhaust fan 63 rotates, air is taken in from the fourth opening portion 202 (Y, M, C, and K) for each color provided on the upper surface of the exhaust duct 206. The fourth opening portion 202 communicates with the second opening portion 74. Therefore, air in the duct (cooling duct 75) formed by the cartridge tray 30 and the elevating duct 69 is taken in from the fourth opening portion 202 for each color provided on the upper surface of the exhaust duct 206 through the second opening portion 74 communicating vertically.

Here, the second opening portion 74 is an opening portion which communicates with the opening portion 202 which is the fourth opening portion of the duct unit 60 on the back side of the apparatus in the opening portion 64 formed by the cartridge tray 30 and the elevating duct 69.

That is, the second opening portion 74 is disposed closer to the other end side than the center of the substrate 50 in the longitudinal direction, and faces the fourth opening portion 202 when the duct unit 60 is mounted.

Air is taken in from the fourth opening portion 202 of the duct unit 60 through the second opening portion 74. As a result, in the duct formed by the cartridge tray 30 and the elevating duct 69 and the exposure head 4 integrally supported by the elevating duct 69, an exposure cooling airflow which is a flow of air indicated by a dotted line (intra-duct airflow 311) in FIG. 26 is generated, and the substrate 50 on which the LED 51 is mounted is cooled.

In the duct unit 60, the air taken in from the fourth opening portion 202 of the exhaust duct 206 sequentially merges from the right side of the apparatus to the opening portions 202K, 202C, 202M, and 202Y in the exhaust duct 206 as indicated by a dotted line (intra-duct airflow 312) in FIG. 39, and flows from the right side to the left side of the apparatus. The air inside the apparatus taken into the exhaust duct 206 is finally exhausted to the outside of the apparatus through the exhaust port 204.

In the duct unit 60, the cross-sectional area of the intake duct 205 is smaller than the cross-sectional area of the exhaust duct 206. As a result, the volume of air flowing through the exhaust duct 206, which is the exhaust air volume, is larger than the volume of air flowing through the intake duct 205, which is the intake air volume. As a result,

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it is possible to reliably exhaust air from the exhaust port 204 without leaking the exposure cooling airflow to the outside of the cooling duct 75 formed between the cartridge tray 30 and the elevating duct 69. In addition, with the above-described configuration, the heated air can be prevented from raising the temperature of the developing unit 24 and the like, and toner scattering can be reduced.

In the present embodiment, the air volume balance between intake and exhaust is adjusted by the cross-sectional areas of the intake duct 205 and the exhaust duct 206, but may be adjusted by reducing the air volume of the intake fan 62 with respect to the exhaust fan 63.

(Positioning of Exposure Head)

Next, positioning of the exposure head 4 will be described with reference to FIGS. 26, 43, and 44 to 49.

(Positioning Pin of Exposure Head)

First, positioning pins 45F and 45B of the exposure head 4 will be described.

The housing 54 of the exposure head 4 is provided with a positioning pin 45F as a positioning axis and a positioning pin 45B. Each of the positioning pin 45F and the positioning pin 45B is an example of a metal pin. The housing 54 is an electroconductive member having conductivity, and the positioning pin is also a member having conductivity. The positioning pin 45F and the positioning pin 45B are fixed to both end portions in the longitudinal direction of the housing 54. The positioning pin 45F is fixed to the housing 54 on one side (front side) of the lens array 52 in the axial direction of the photosensitive drum 2, and protrudes from both sides of the housing 54 in the optical axis direction of the lens array 52. The positioning pin 45B is fixed to the housing 54 on the other side (back side) of the lens array 52 in the axial direction of the photosensitive drum 2, and protrudes from both sides of the housing 54 in the optical axis direction of the lens array 52.

In order to ensure the distance between the surface of the photosensitive drum 2 and the light exit surface of the lens array 52 of the exposure head 4 with high accuracy, the positioning pins 45F and 45B adjust the position of the positioning surface at the shaft distal end with reference to the housing 54 and are caulked to the housing 54. Note that the fixing of the positioning pins 45F and 45B to the housing 54 is not limited thereto, and for example, the positioning pin 45F and the positioning pin 45B made of metal may be fixed to the housing 54 made of metal by welding. As described above, in the present embodiment, the positioning pin 45F and the positioning pin 45B are integrated with the housing 54.

Then, the positioning pin 45F and the positioning pin 45B of the exposure head 4 abut on the drum bearing 26 of the drum unit 23 in the moving direction of the elevating duct 69, so that a gap is formed between the lens array 52 and the photosensitive drum 2. In this way, a distance (gap) between the exposure head 4 and the photosensitive drum 2 is determined in a direction orthogonal to the axial direction of the photosensitive drum 2, and the position of the exposure head 4 with respect to the photosensitive drum 2 is determined.

The exposure head 4 is fixed not only by the distance to the photosensitive drum 2 but also by the angle by the positioning pins 45F and 45B. In the image forming apparatus 100 illustrated in FIG. 2, the exposure head 4 is disposed toward the center of the photosensitive drum 2. This arrangement is adopted because it is not necessary to consider the influence of regular reflection on the surface of the photosensitive drum 2 in the mechanism of the LED (light emitting element) 51 included in the exposure head 4.

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FIG. 43 is a cross-sectional view illustrating a relationship among three components, namely, the photosensitive drum 2, a part of the positioning pins 45F and 45B (the distal end on the positioning side), and the rotating arm 65. FIG. 43 is a cross-sectional view taken along line Y-Y in FIG. 24, and only peripheral components of the three components are visualized. Note that the cross-sectional position is moved to the center of the positioning pins 45F and 45B.

FIG. 44 is a perspective view of the positioning pins 45F and 45B as viewed from a front surface cut in a direction orthogonal to the axial direction of the photosensitive drum 2 at the center position of the positioning pins. FIG. 45 is a perspective view of the positioning pins 45F and 45B as viewed from a rear surface cut in a direction orthogonal to the axial direction of the photosensitive drum 2 at the center position of the positioning pins.

In FIGS. 43 and 44, the positioning pin 45F (45B) on the front side of the exposure head 4 and the drum bearing 26 on the front side of the photosensitive drum 2 abut on the end surface of the positioning pin 45F (45B), whereby the optical axis direction of the exposure head 4 is positioned.

Recessed engagement portions 26F and 26B are integrally formed in the drum bearings 26 at positions facing the positioning pins 45F and 45B so as to be engageable with the distal end portions of the positioning pins 45F and 45B. By processing the diameter dimension of the distal end of the positioning pin 45 and the width dimension of the concave shape of the drum bearing 26 with high accuracy, positioning in the direction orthogonal to the optical axis direction of the exposure head 4 and the direction orthogonal to the axial direction of the photosensitive drum 2 is performed with high accuracy. In addition, an inclined surface is formed at an inlet of the positioning pins 45F and 45B so as not to ride on the concave edge when the positioning pins 45F and 45B are engaged with the engagement portions 26F and 26B of the drum bearing 26.

As illustrated in FIG. 26, the positioning pins 45F and 45B and the drum bearing 26 are not in contact with each other in the axial direction of the photosensitive drum 2, and are positioned by a positioning member 250 described later.

Here, the drum bearing 26 is a bearing member that axially supports front and back ends (both ends) of the photosensitive drum 2 in the drum unit 23. The photosensitive drum 2 is axially supported by the drum bearing 26 without a gap by increasing the dimensional accuracy at the engagement location of the drum bearing 26. That is, positioning with high accuracy on the drum bearing 26 can be regarded as positioning with high accuracy on the photosensitive drum 2. The photosensitive drum 2 is rotationally driven according to the image forming process. Therefore, the positioning pin 45 of the exposure head 4 is positioned with respect to the drum bearing 26.

FIGS. 43 and 44 illustrate a cross section of the drum bearing 26 on the front side of the image forming apparatus, but the drum bearing 26 on the rear side has a similar shape. As illustrated in FIG. 45, the positioning pin 45B of the exposure head 4 is also positioned with high accuracy with respect to the drum bearing 26 on the rear side of the image forming apparatus. Therefore, the exposure head 4 is positioned with high accuracy at both end portions in the axial direction of the photosensitive drum 2.

As illustrated in FIG. 43, a pressing position at which the engagement boss 66 of the rotating arm 65 presses the elevating duct 69, an abutting position between the positioning pins 45F and 45B and the engagement portions 26F and 26B of the drum bearing 26, and a center position of the

photosensitive drum 2 are arranged on a substantially straight line as indicated by an alternate long and short dash line.

With this arrangement, the exposure head 4 is pressed toward the center of the photosensitive drum 2, so that an unnecessary rotational moment is not applied to the elevating duct 69. This means that there is no component promoting the inclination of the exposure head 4 with respect to the photosensitive drum 2 in the pressing force, which leads to the positioning accuracy of the distance and angle and the stability of the repeated attaching and detaching operation.

Furthermore, as illustrated in FIGS. 44 and 45, the positioning pins 45F and 45B are supplementarily fitted to the auxiliary fitting portions 30h and 30i of the cartridge tray 30 in the direction orthogonal to the optical axis direction of the exposure head 4 and the direction orthogonal to the axial direction of the photosensitive drum 2 on the lower end peripheral surfaces thereof. As a result, even when a slight rotational moment is generated due to the weight, surface property, dimensional error, and the like of the component, stable positioning accuracy at a distance and an angle, and repeated attaching and detaching operations can be achieved.

(Positioning Member of Exposure Head)

Next, the positioning of the exposure head 4 in the axial direction of the photosensitive drum 2 by the positioning member 250 will be described in detail with reference to FIGS. 46 to 48.

FIG. 46 is a perspective view of the positioning member 250 after attachment, and FIG. 47 is a perspective view of the positioning member before attachment. FIG. 48 is a perspective view illustrating a shape of the positioning member 250.

As illustrated in FIG. 46, the positioning member 250 is attached to the front side of the exposure head 4. As illustrated in FIG. 47, the biasing portion 30d, a round hole portion 30e, a square hole portion 30f, and a claw engagement portion 30g are provided on the front side of the cartridge tray 30.

As illustrated in FIG. 48, a regulating portion 250a, which is third engagement portions, a biasing portion 250b, a cross projection portion 250c, an I-shaped projection portion 250d, and a claw portion 250e are provided on the lower surface of the positioning member 250.

The outer diameter of the cross projection portion 250c is substantially equal to the inner diameter of the round hole portion 30e, and the length of the I-shaped projection portion 250d in the left-right direction is substantially equal to the length of the square hole portion 30f in the left-right direction. By fitting the projection portions 250c and 250d and the hole portions 30e and 30f, the positions of the positioning member 250 in the front-back direction and the left-right direction are determined.

The claw portion 250e has a barbed shape, and the barbed shape is caught by the claw engagement portion 30g, whereby the position of the positioning member 250 in the vertical direction is determined.

The regulating portion 250a, which is the third engagement portion, has a first abutting surface 250a1 abutting on one side in the axial direction of the positioning pin 45F and a second abutting surface 250a2 abutting on the other side in the axial direction of the positioning pin 45F. The first abutting surface 250a1 and the second abutting surface 250a2 face each other in the axial direction. The regulating portion 250a has a concave shape opened on the right side in the left-right direction, and a notch width of the concave shape in the front-back direction and an outer diameter of the

positioning pin 45F are substantially equal to each other. Here, the left-right direction is a second direction orthogonal to the moving direction (first direction) of the elevating duct 69 and the axial direction of the photosensitive drum 2. By fitting the regulating portion 250a, which is the third engagement portion, and the positioning pin 45F, the exposure head 4 in the axial direction of the photosensitive drum 2 is positioned with respect to the positioning member 250.

In this manner, the position of the exposure head 4 in the axial direction of the photosensitive drum 2 can be accurately positioned by the positioning member 250 attached after the attachment of the exposure head 4.

Here, in all the components including the positioning member 250, the cartridge tray 30, and the positioning pin 45F, backlash occurs between the components due to manufacturing variations. In a case where the backlash is large, there is a concern that the position of the exposure head 4 varies due to repeated operations of attaching and detaching the exposure head 4.

In view of the above problem, in the present embodiment, backlash is reduced by the biasing portion 250b and the biasing portion 30d. The biasing portion 250b extends rightward in the left-right direction from the positioning member 250, has a thin thickness in the axial direction of the photosensitive drum 2, and has a shape that is easily elastically deformed in the axial direction of the photosensitive drum 2. On the other hand, the biasing portion 30d has a shape protruding from the upper surface of the cartridge tray 30, and is formed in a shape having rigidity so as not to be deformed in the axial direction of the photosensitive drum 2. When the positioning member 250 is attached to the cartridge tray 30, the distal end of the biasing portion 250b is configured to interfere with (come into contact with) the biasing portion 30d. The front surface of the distal end of the biasing portion 250b and the back surface of the biasing portion 30d come into contact with each other, and the biasing portion 250b is elastically deformed in the back surface direction, so that the reaction force is applied to the positioning member 250 in the back surface direction, that is, from one side to the other side in the axial direction of the photosensitive drum 2.

As described above, by adopting the configuration in which the positioning member 250 is biased in the axial direction of the photosensitive drum 2, it is possible to achieve highly accurate positioning of the exposure head 4 which is hardly affected by repeated operations of attaching and detaching the exposure head 4 and to achieve more precise positioning.

(Replacement and Attachment or Detachment Configuration of Exposure Head)

The replacement and attachment or detachment configuration of the exposure head 4 will be described in detail with reference to FIGS. 33 and 49 to 65. Similarly to FIG. 22, FIG. 33 is cross-sectional views taken along line X-X in FIG. 20. FIG. 33 is a cross-sectional view of the exposure head 4, the elevating duct 69, and the cartridge tray 30 as viewed from the front, in which the drum unit 23 and the developing unit 24 are not illustrated.

As described above, the exposure head 4 is configured to be detachable from the image forming apparatus 100. A procedure for mounting the exposure head 4 will be described in detail with reference to FIGS. 49 to 65.

FIG. 49 is a right perspective view illustrating a state immediately before the exposure head 4 is mounted on the elevating duct 69, FIG. 50 is a front cross-sectional view thereof, and FIG. 51 is a left perspective view thereof.

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In the state of FIGS. 49 and 51, the drum unit 23, the developing unit 24, and the positioning member 250 are removed from the image forming apparatus. The replacement and attachment or detachment of the exposure head 4 are performed in a state where the drum unit 23, the developing unit 24, and the positioning member 250 are removed. In addition, in a state where the photosensitive drum 2 is removed, the exposure head 4 can be manually moved from the retracted position to the exposure position. Therefore, replacement and attachment or detachment of the exposure head 4 are performed in a state where the exposure head 4 is manually moved to the exposure position. At this time, as illustrated in FIG. 50, the FFC 58 connected to the apparatus main body side is previously connected to the FFC connector 57 of the exposure head 4.

As described above, the housing support member 55 of the exposure head 4 is provided with the engagement claws 55b and 55c for engaging with the elevating duct 69. On the other hand, in the elevating duct 69, the engagement holes 69b and 69c for engaging with the engagement claws 55b and 55c are provided in the upper surface portion 69U facing the exposure head 4. Based on this configuration, a procedure of engaging the engagement claws 55b and 55c of the exposure head with the engagement holes 69b and 69c of the elevating duct to be integrated is as follows.

First, as illustrated in FIGS. 49 and 51, by moving the exposure head 4 in the direction of arrow D with respect to the elevating duct 69, the engagement claws 55b and 55c of the exposure head 4 fall into the engagement holes 69b and 69c of the elevating duct 69, respectively. That is, the engagement claws 55b and 55c of the exposure head are engaged with the engagement holes 69b and 69c in the protruding direction. At the same time, the lower end of the positioning pin 45F of the exposure head 4 falls into the auxiliary fitting portion 30h of the cartridge tray 30 with a gap. At this time, the extra length of the FFC 58 connected to the substrate of the apparatus main body and the substrate of the exposure head 4 protrudes from the harness opening portion 252 described later. A state at this time is illustrated in FIGS. 52 to 56.

FIG. 52 is a right perspective view of a state where the exposure head 4 is placed in the elevating duct 69, FIG. 53 is a front cross-sectional view thereof, and FIG. 54 is a left perspective view thereof. FIG. 55 is a cross-sectional view of a state where the exposure head 4 is placed in the elevating duct 69. FIG. 56 is a perspective view of the vicinity of a conductive member in a state where the exposure head 4 is placed in the elevating duct 69.

Next, as illustrated in FIG. 52, the exposure head 4 is slid in the direction of arrow B with respect to the elevating duct 69, and the engagement claws 55b and 55c are engaged with the engagement holes 69b and 69c in the extending direction orthogonal to the protruding direction. A state at this time is illustrated in FIGS. 57 to 61.

FIG. 57 is a right perspective view of a state where the exposure head 4 is mounted on the elevating duct 69, and FIG. 58 is a left perspective view thereof. FIG. 59 is a cross-sectional view taken along line W-W in FIG. 33, and is a cross-sectional view in a state where the exposure head 4 is mounted on elevating duct 69. FIG. 60 is an enlarged cross-sectional view of an engagement portion illustrated in FIG. 59. FIG. 61 is a perspective view of the vicinity of a conductive member in a state where the exposure head 4 is mounted on the elevating duct 69. FIG. 62 is a right cross-sectional view illustrating a harness opening portion. FIG. 63 is a right perspective view in a state where extra length processing of the FFC is performed. FIG. 64 is a front

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cross-sectional view of a state where the extra length processing of the FFC is performed. FIG. 65 is a front cross-sectional view illustrating a state of the FFC in the retracted position.

The engagement claws 55b and 55c of the exposure head 4 are formed so as to protrude toward the elevating duct 69, and have a substantially L shape extending in the arrow B direction which is the sliding movement direction of the exposure head 4. Therefore, by the sliding movement in the direction of the arrow B, the claw ends of the substantially L-shaped engagement claws 55b and 55c are engaged with the edges of the engagement holes 69b and 69c. By this engagement, the exposure head 4 is mounted on the elevating duct 69 and is integrated with the elevating duct 69 at the position shown in FIGS. 57 to 59.

In this way, the posture is stabilized in a state where the engagement claws 55b and 55c of the exposure head 4 are made to pass through the engagement holes 69b and 69c of the elevating duct 69, and then the exposure head 4 is slid to complete the mounting. Accordingly, easy mounting of the exposure head 4 can be achieved with an inexpensive configuration.

(Relationship Between Engagement Claw and Engagement Hole)

Here, the relationship between the engagement claws of the exposure head 4 and the engagement holes of the elevating duct 69 will be described in more detail with reference to FIG. 60. Here, the relationship between the engagement claw 55b of the exposure head 4 and the engagement hole 69b of the elevating duct 69 will be described, but the same applies to the relationship between the engagement claw 55c of the exposure head 4 and the engagement hole 69c of the elevating duct 69.

As illustrated in FIG. 60, the engagement claw 55b has elasticity, and a concave portion 55f is provided at the extended distal end of the engagement claw 55b. A convex portion 69f to be engaged with the concave portion 55f is provided at an edge of the engagement hole 69b corresponding to the concave portion 55f. The convex portion 69f provided at the edge of the engagement hole 69b is disposed at a position corresponding to the concave portion 55f provided at the distal end of the engagement claw 55b at the slide operation completion position of the exposure head 4 with respect to the elevating duct 69, that is, the mounting completion position.

Immediately before the completion of the sliding operation of the exposure head 4 in the direction of the arrow K, the distal end of the engagement claw 55b interferes with the convex portion 69f and the engagement claw 55b is elastically deformed, so that the sliding operation force of the exposure head 4 once rises compared to the operation force immediately before the interference. Thereafter, the concave portion 55f of the engagement claw 55b immediately reaches the convex portion 69f, and the concave portion 55f and the convex portion 69f are engaged with each other, so that the sliding operation force of the exposure head 4 decreases. That is, when the exposure head 4 is moved in the extending direction with respect to the elevating duct 69, the engagement claw 55b is elastically deformed until the concave portion 55f and the convex portion 69f are engaged with each other, so that the operation force for moving the exposure head 4 in the extending direction is changed. By the rapid increase or decrease of the sliding operation force of the exposure head 4, it is possible to provide a click feeling indicating that the sliding operation of the exposure head 4 is completed.

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As described above, the sliding operation force of the exposure head 4 is changed by the elastic deformation of the engagement claw 55b until the concave portion 55f of the engagement claw 55b and the convex portion 69f of the engagement hole 69b are engaged, whereby the completion of the mounting of the exposure head 4 can be clearly presented.

As described above, as illustrated in FIGS. 44 and 45, the positioning pins 45F and 45B of the exposure head 4 are supplementarily fitted to the auxiliary fitting portions 30h and 30i of the cartridge tray 30 on the lower end peripheral surface thereof. As illustrated in FIGS. 54 and 58, this fitting is simultaneously performed by the sliding operation of the exposure head 4 in the direction of the arrow B.

Furthermore, in the present embodiment, the earth connection between the housing 54 of the exposure head 4 and the image forming apparatus 100 is also simultaneously performed by the above-described sliding operation of the exposure head 4 in the direction of arrow B. The housing 54 of the exposure head 4 and the positioning pins 45F and 45B are caulked and fixed, and conduction is established. As illustrated in FIGS. 56 and 61, the earth connection with the image forming apparatus 100 is performed by the peripheral surface of the positioning pin 45F of the exposure head 4 coming into contact with the conductive member 251 provided in the cartridge tray 30 on the apparatus side. The conductive member has conductivity. Specifically, the conductive member 251 is made of a metallic thin plate. Therefore, a sufficient contact pressure can be obtained by contact and deformation by the sliding operation of the exposure head 4. The conductive member 251 is electrically connected to the frame body sheet metal of the image forming apparatus 100 via a circuit board on which a harness, a resistance element for noise removal, and a capacitor (not illustrated) are mounted.

In this manner, the positioning pin 45F of the exposure head 4 can be electrically connected to the conductive member 251 by the sliding operation, and the exposure head 4 can be connected to the ground. As a result, the exposure head 4 can be connected to the ground or the like with easy work to reduce generation of radiation noise such as electromagnetic waves. That is, a stable grounding characteristic can be obtained only by a sliding operation for mounting the exposure head 4 on the elevating duct 69.
(Extra Length Processing of FFC)

Here, extra length processing of the FFC 58 will be described.

FIG. 62 is a right cross-sectional view illustrating a harness opening portion. The harness opening portion 252 includes a first opening forming portion 55g formed in the housing support member 55 constituting the exposure head 4, a second opening forming portion 69g formed in the elevating duct 69 to be left on the main body side of the image forming apparatus 100 at the time of replacement, and a second opening forming portion 30c formed in the cartridge tray 30.

In other words, the harness opening portion 252 is formed to be surrounded by the first opening forming portion 55g provided in the vicinity of the connector 57 of the exposure head 4 and the second opening forming portions 69g and 30c provided in the vicinity of the first opening forming portion 55g and provided in the apparatus main body. Here, a configuration in which the second opening forming portion constituting the harness opening portion 253 is provided in each of the elevating duct 69 and the cartridge tray 30 is illustrated.

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In the present embodiment, the harness opening portion 252 is formed on the drum unit 23 side which is a photo-conductor unit. This is because a part of a duct serving as a flow path of an airflow (development cooling airflow) for cooling the developing unit 24 is provided on the developing unit 24 side by the developing unit 24 and the developing support member 301. Thus, the harness opening portion 252 is provided on the drum unit 23 side so as not to affect the development cooling airflow.

In this manner, by providing the harness opening portion 252 on the drum unit 23 side arranged in the vicinity of the exposure head 4, it is possible to prevent the influence on the airflow flowing along the developing unit 24.

As illustrated in FIGS. 52, 53, and 57, in the state in which the exposure head 4 is dropped into the elevating duct 69 and the state in which the exposure head 4 is mounted, the extra length of the FFC 58 protrudes from the harness opening portion 252 toward the drum support member 302 side, and interferes with the insertion and removal locus of the drum unit 23.

The extra length of the FFC 58 is slack (extra length) of the FFC 58 formed between the exposure head 4 and the elevating duct 69 in a state where the exposure head 4 is dropped into the elevating duct and in a state where the exposure head 4 is mounted.

In this manner, the exposure head 4 can be mounted on the image forming apparatus in a state where the extra length of the FFC 58 is drawn out from the harness opening portion 252 including the first opening forming portion 55g and the second opening forming portions 69g and 30c.

After the exposure head 4 is mounted on the elevating duct 69, the extra length processing of the FFC 58 is performed. The FFC 58 has a bent portion 58a which is bent at least at one place in advance. The FFC 58 stores the extra length from the harness opening portion 252 while folding along the bent portion 58a.

FIGS. 63 and 64 illustrate a state where the extra length processing of the FFC 58 is performed. FIG. 63 is a right perspective view in a state where the extra length processing of the FFC 58 is performed. FIG. 64 is a front cross-sectional view of a state where the extra length processing of the FFC 58 is performed. FIG. 64 is a cross-sectional view in the front direction at the position of the FFC 58 in a state where the exposure head 4 approaches the photosensitive drum 2 by the moving mechanism of the exposure head 4 (hereinafter, the exposure head proximity state). By bending and storing the FFC 58 a plurality of times in the optical axis direction, an extra length of the FFC 58 can be stored in a space-saving manner particularly on the upper side close to the FFC connector 57.

As illustrated in FIG. 64, the length of the FFC 58 from the end portion on the connection side with the FFC connector 57 to the first bent portion 58a as viewed from the end portion is longer than the distance from the harness opening portion 252 to the FFC connector 57. As a result, the bent portion 58a of the FFC 58 is prevented from being caught by the harness opening portion 252, and the FFC 58 can be prevented from protruding from the harness opening portion 252.

FIG. 65 is a cross-sectional view in the front direction at the position of the FFC 58 in a state where the exposure head 4 is retracted from the photosensitive drum 2 by the moving mechanism of the exposure head 4 (hereinafter, the exposure head retracted state). It can be understood that the opening area of the harness opening portion 252 in the exposure head retracted state illustrated in FIG. 65 is smaller than that in the exposure head proximity state illustrated in FIG. 64. Accord-

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ingly, even when the exposure head **4** is moved a plurality of times between the exposure head proximity state and the retracted state by the moving mechanism, it is possible to prevent the FFC **58** from protruding from the harness opening portion **252**.

Finally, by assembling the positioning member **250** to the cartridge tray **30** as illustrated in FIG. **47**, the mounting of the exposure head **4** is completed as illustrated in FIG. **46**.

Next, a procedure for removing the exposure head **4** will be described with reference to FIGS. **49** to **65**.

The removal of the exposure head **4** is also performed in the exposure head proximity state (FIGS. **52** and **54**) similarly to the mounting described above, and is basically performed in the reverse order of the mounting.

First, the claw portion **250e** of the positioning member **250** is deformed from the state of FIG. **46** to release the engagement from the claw engagement portion **30g** of the cartridge tray **30**, and the positioning member **250** is removed from the cartridge tray **30** as illustrated in FIG. **25**.

In the case of removing the exposure head **4**, unlike the case of the mounting, it is not necessary to perform the extra length processing of the FFC **58**, for example, work of pulling out the FFC **58** from the harness opening portion **252**.

Next, the exposure head **4** is slid and moved in a direction opposite to the direction of the arrow B illustrated in FIG. **52** to release the engagement between the engagement claws **55b** and **55c** and the engagement holes **69b** and **69c** so as to be separable from the elevating duct **69**.

Finally, the exposure head **4** is lifted in a direction opposite to the direction of the arrow D illustrated in FIG. **49**, and the engagement claws **55b** and **55c** are pulled out from the engagement holes **69b** and **69c**. At the same time as the exposure head **4** is lifted, the FFC **58** that has been folded and stored extends, so that the exposure head **4** can be pulled out to the outside of the image forming apparatus **100**. By pulling out the FFC **58** from the FFC connector **57** in this state, the removal of the exposure head **4** is completed.

OTHER EMBODIMENTS

The configuration according to the present invention is not limited to the above-described embodiment.

In the above-described embodiment, the four-color full-color printer of the tandem type—intermediate transfer system has been described as an example, but for example, a direct transfer type in which a toner image is transferred from the photosensitive drum **2** to the recording sheet P without using the intermediate transfer belt **9** may be used. Furthermore, a mono-color or full color printer of five or more colors using a spot color toner may be used. In that case, a configuration including the exposure heads **4** corresponding to the number of colors may be used.

In the above-described embodiment, an elastic body such as sponge or rubber using urethane, silicone, or the like as a material is used as the sealing member **207**, but a gap of each opening portion may be closed by elastically deforming a resin sheet such as PET, modified PPE, or PE.

Further, although a configuration in which the sealing member **207** is disposed in the duct unit **60** is used, for example, a configuration in which the sealing member is disposed in the cartridge tray **30** or the elevating duct **69** may be used, or a configuration in which the sealing member is disposed in a plurality of parts may be selected.

Further, the third opening portion **201** and the fourth opening portion **202** of the duct unit **60** are connected to the

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opening portion **64** (the first opening portion **73** and the second opening portion **74**) formed by the cartridge tray **30** and the elevating duct **69**, but the present invention is not limited thereto. For example, the opening portion **64** may be eliminated, the first opening portion **73** and the second opening portion **74** may be provided in the elevating duct **69**, and the third opening portion **201** and the fourth opening portion **202** of the duct unit **60** may be directly connected to the first opening portion **73** and the second opening portion **74**, respectively.

Further, in the above-described embodiment, the duct unit **60** is configured to have one opening portion including both the third opening portion **201** and the fourth opening portion **202**, but only one of the openings may be provided. In that case, either the first opening portion **73** or the second opening portion **74** on the image forming apparatus side may be brought into close contact with the third opening portion **201** or the fourth opening portion **202** via the sealing member **207**. At this time, one of the first opening portion **73** and the second opening portion **74** on the image forming apparatus side may be brought into close contact, and the other opening portion may be extended to a space where there is no possibility of toner scattering.

Furthermore, in the above-described embodiment, the configuration in which the cooling duct **75** (see FIG. **26**) is formed between the duct unit **60** and the cartridge tray **30** or the elevating duct **69** has been exemplified, but the cooling duct is not necessarily formed by the duct unit **60**. In this case, the cooling duct **75** may be formed only by the cartridge tray **30** or the elevating duct **69**.

In the embodiment described above, the intake port **203** is configured to directly intake air from the outside of the image forming apparatus **100** and directly exhaust air from the exhaust port **204** to the outside of the apparatus, but such a configuration is not necessarily required. For example, the intake port **203** may be configured to intake relatively low-temperature air from a space without a heat source such as the sheet cassette **12**. In addition, the exhaust port **204** may also be configured to exhaust air to a space not affected by heat inside the image forming apparatus **100**.

In addition, the intake fan **62** and the exhaust fan **63** are not necessarily required, and one or both of them may not be disposed, and the airflow may be circulated by the pressure difference between the exposure cooling airflow and the outside air.

In addition, the vertical direction of the unit and the component has been described according to the arrangement of each unit in the cross-sectional view of the image forming apparatus **100** illustrated in FIG. **2**. However, a unit arrangement in which the photosensitive drum **2** is disposed above the intermediate transfer belt **9** and the exposure head **4** is further disposed above the photosensitive drum **2** as in an upper surface exposure method in which the photosensitive drum **2** is exposed from substantially above may be adopted. In this case, the up and down directions in the description of the embodiment are all reversed, and the duct unit **60** is configured to descend immediately before the assembling position.

In addition, although the guide portion **103** is a curved surface and the guided portion **208** is an inclined surface, the relationship may be reversed, or a combination of curved surfaces or inclined surfaces may be selected.

In the above-described embodiment, the configuration in which the harness opening portion **252** is provided on the drum unit **23** side has been exemplified, but the present invention is not limited thereto, and the harness opening portion may be provided on the developing unit **24** side. In

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this manner, by providing the harness opening portion **252** on the side of the developing unit **24** arranged in the vicinity of the exposure head, it is possible to prevent the influence on the airflow flowing along the drum unit **23**.

In the above-described embodiment, the configuration in which the engagement claws **55b** and **55c** are provided in the housing support member **55** constituting the exposure head **4** has been exemplified, but the engagement claws may be provided in a component on the image forming apparatus **100** side. In this case, an L-shaped engagement claw may be provided in the elevating duct **69**, and the engagement hole to be engaged with the engagement claw may be provided in the housing support member **55**.

Further, in the above-described embodiment, the configuration in which the concave portion **55f** is provided in the housing support member **55** constituting the exposure head **4** has been exemplified, but the concave portion may be provided in a component on the image forming apparatus **100** side. In this case, the concave portion may be provided in the elevating duct **69**, and the convex portion engaged with the concave portion may be provided in the housing support member **55**.

Further, in the above-described embodiment, the sliding operation for mounting the exposure head **4** is performed from the front side to the back side of the image forming apparatus **100**, but may be configured to slide from the back side to the front side. In this case, the shapes of the engagement claws and the engagement holes may be reversed in the front-back direction.

In the above-described embodiment, the exposure head **4** is positioned in the axial direction of the photosensitive drum **2** and grounded by the positioning pin **45F** on the front side, but may be positioned by the positioning pin **45B** on the back side. Furthermore, in the earth connection, both the front and back positioning pins **45F** and **45B** may be used.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-037622, filed Mar. 10, 2022, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An image forming apparatus comprising:

a rotatable photoconductor;

an exposure portion including:

a substrate on which a light-emitting chip, including a plurality of light-emitting elements arranged along a rotational axis direction of the photoconductor, is mounted; and

a support portion configured to support the substrate; a fan configured to generate an airflow for cooling the exposure portion; and

a duct connecting with the fan,

wherein the support portion directs the airflow generated by the fan along a longitudinal direction of the substrate,

wherein the exposure portion is movable between an exposed position where the photoconductor is exposed and a retracted position where the photoconductor is retracted from the exposed position,

wherein, in a state where the exposure portion is at the exposure position:

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the duct is connected to the support portion, the fan-generated airflow is directed through the duct, and

a gap between the support portion and the duct is sealed by a seal member, and the support portion and the duct communicate with each other.

2. The image forming apparatus according to claim 1, further comprising:

a charging portion configured to charge a surface of the photoconductor; and

a developing portion configured to develop a latent image formed on the surface of the photoconductor with toner, wherein the exposure portion and the duct are located between the charging portion and the developing portion in a direction orthogonal to a rotation axis direction of the photoconductor and a vertical direction.

3. The image forming apparatus according to claim 1, wherein:

the duct guides the airflow to a second surface of the substrate opposite to a first surface on which the plurality of chips is mounted, and

the airflow guided to the second surface is guided from one side to the other side in a rotation axis direction of the photoconductor.

4. The image forming apparatus according to claim 1, wherein, in a state where the exposure portion is located at the retracted position:

the gap exists between the support portion and the duct, and

the support portion and the duct do not communicate with each other.

5. The image forming apparatus according to claim 1, wherein:

the support portion includes a first opening portion communicating with the duct and provided on one side in a rotation axis direction of the photoconductor and a second opening portion provided on the other side in the rotation axis direction of the photoconductor, and the first opening portion and the second opening portion communicate with each other.

6. The image forming apparatus according to claim 5, wherein:

the duct includes a third opening portion provided on one side in the rotation axis direction of the photoconductor and a fourth opening portion provided on the other side in the rotation axis direction of the photoconductor, and the first opening portion and the third opening portion communicate with each other, and the second opening portion and the fourth opening portion communicate with each other, so that the support portion and the duct communicate with each other.

7. The image forming apparatus according to claim 1, wherein:

an exterior cover of the image forming apparatus includes a first opening at a position facing the fan, and the fan generates an airflow for cooling the exposure portion through the first opening.

8. The image forming apparatus according to claim 1, wherein:

the fan is a first fan,

the image forming apparatus includes a second fan configured to discharge the airflow generated by the first fan to the outside of the image forming apparatus, and the duct allows the first fan and the second fan to communicate with each other.

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9. The image forming apparatus according to claim 1, further comprising:

a fixing portion that fixes a toner image on a recording medium onto the recording medium,

wherein the fan is located on the opposite side of the fixing portion with respect to the center of the image forming apparatus in a left-right direction. 5

10. The image forming apparatus according to claim 1, wherein the plurality of light emitting portions are organic EL. 10

11. An image forming apparatus comprising:

a rotatable photoconductor;

an exposure portion including:

a substrate on which a light-emitting chip, including a plurality of light-emitting elements arranged along a rotational axis direction of the photoconductor, is mounted; and 15

a support portion configured to support the substrate; a fan configured to generate an airflow for cooling the exposure portion; and 20

a duct connecting with the fan,

wherein the support portion directs the airflow generated by the fan along a longitudinal direction of the substrate, 25

wherein the exposure portion is movable between an exposed position where the photoconductor is exposed and a retracted position where the photoconductor is retracted from the exposed position,

wherein, in a state where the exposure portion is at the exposure position:

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the duct is connected to the support portion, and the fan-generated airflow is directed through the duct, and

wherein, in a state where the exposure portion is located at the retracted position:

a gap exists between the support portion and the duct, and

the support portion and the duct do not communicate with each other.

12. The image forming apparatus according to claim 11, further comprising:

a charging portion configured to charge a surface of the photoconductor; and

a developing portion configured to develop a latent image formed on the surface of the photoconductor with toner, wherein the exposure portion and the duct are located between the charging portion and the developing portion in a direction orthogonal to a rotation axis direction of the photoconductor and a vertical direction.

13. The image forming apparatus according to claim 11, wherein:

the duct guides the airflow to a second surface of the substrate opposite to a first surface on which the plurality of chips is mounted, and

the airflow guided to the second surface is guided from one side to the other side in a rotation axis direction of the photoconductor.

14. The image forming apparatus according to claim 11, wherein the plurality of light emitting portions are organic EL.

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